



HARDWARE TUTORIAL 1

Intro to Hardware & Schematics

START

MORE





TABLE OF CONTENT

1. Logistics
2. What Is Hardware & Your Expectation
3. Design Overview
4. Basics of a Circuit & Passive Components
5. Schematic Design Flow & Function of Each Parts
6. How to Implement ICs + KiCAD guidelines
7. External Resources
8. MCU Control/Communication Signals
 - (if time allows)



ITEM 1

LOGISTICS



MAIN COMMUNICATION CHANNEL:

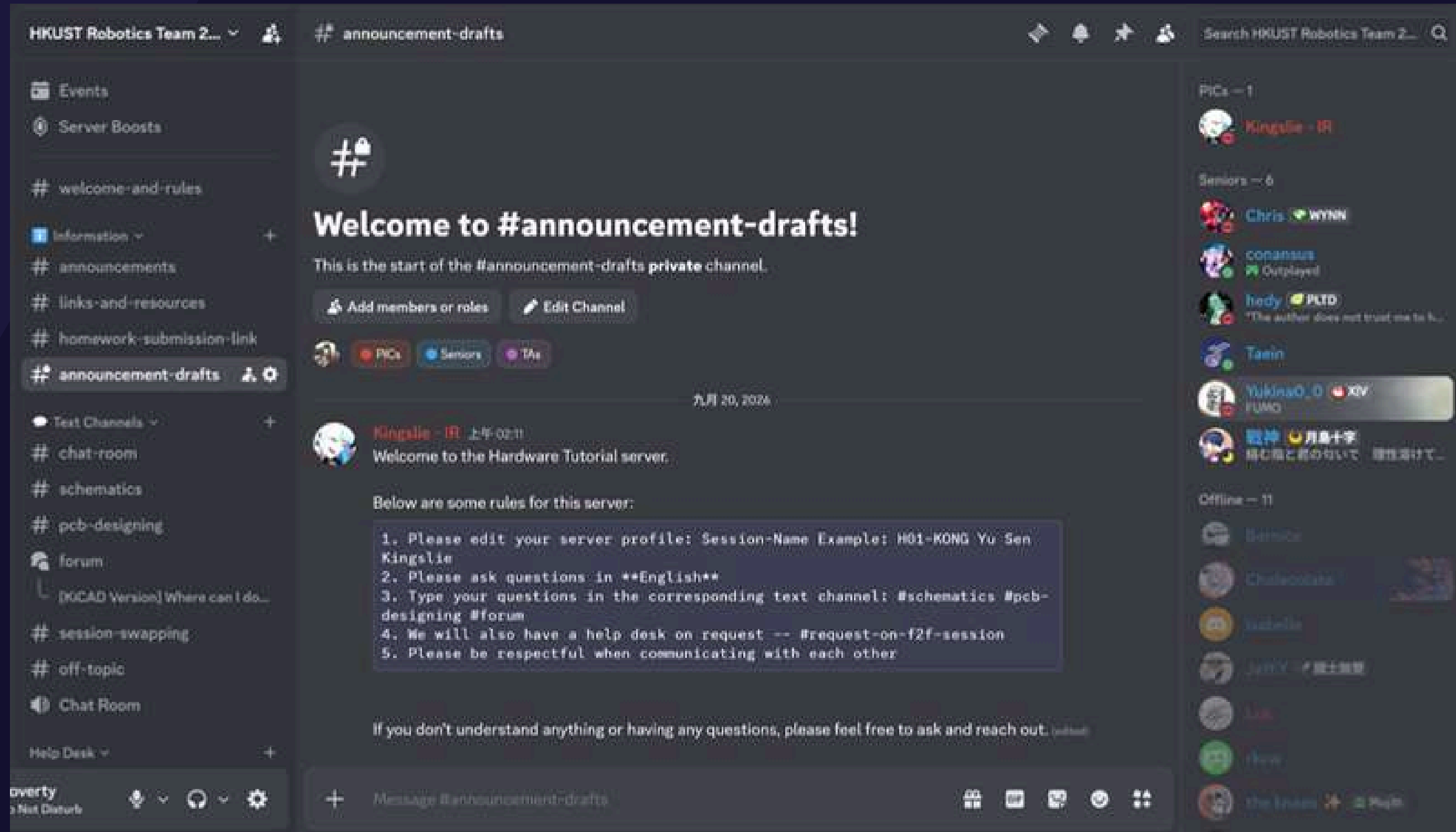




!!! BE SURE TO TURN ON YOUR NOTIFICATION FOR DISCORD !!!



LOGISTICS⁷



HKUST Robotics Team 2...  # announcement-drafts     Search HKUST Robotics Team 2... 

Events
Server Boosts

welcome-and-rules

Information 

announcements

links-and-resources

homework-submission-link

announcement-drafts  

Text Channels 

chat-room

schematics

pcb-designing

forum

[XCAD Version] Where can I do...

session-swapping

off-topic

Chat Room

Help Desk 

overly   

Message Banannouncement-drafts     

 **Welcome to #announcement-drafts!**

This is the start of the #announcement-drafts **private** channel.

 Add members or roles  Edit Channel

九月 20, 2024

 **Kingslie - IR** 上午 02:11
Welcome to the Hardware Tutorial server.

Below are some rules for this server:

1. Please edit your server profile: Session-Name Example: H01-KONG Yu-Sen Kingslie
2. Please ask questions in ****English****
3. Type your questions in the corresponding text channel: #schematics #pcb-designing #forum
4. We will also have a help desk on request -- #request-on-f2f-session
5. Please be respectful when communicating with each other

If you don't understand anything or having any questions, please feel free to ask and reach out. (edited)

Pics — 1

 **Kingslie - IR**

Seniors — 6

 **Chris WYNN**

 **conansur** 

 **hedy**  *"The author does not trust me to h..."*

 **Tamin**

 **Yukina0.0**  FUMO

 **戰神**  掃心術と術の匂いで、理性消けて...

Offline — 11

 **Barnack**

 **Chaleontaka**

 **Isabelle**

 **Jenny** 

 **Lia**

 **Alex**

 **The Enam**  



LOGISTICS ⁷

Useful
Info

Information

announcements

links-and-resources

homework-submission-link

Open for
questions &
chatting

Text Channels

chat-room

schematics

pcb-designing

forum

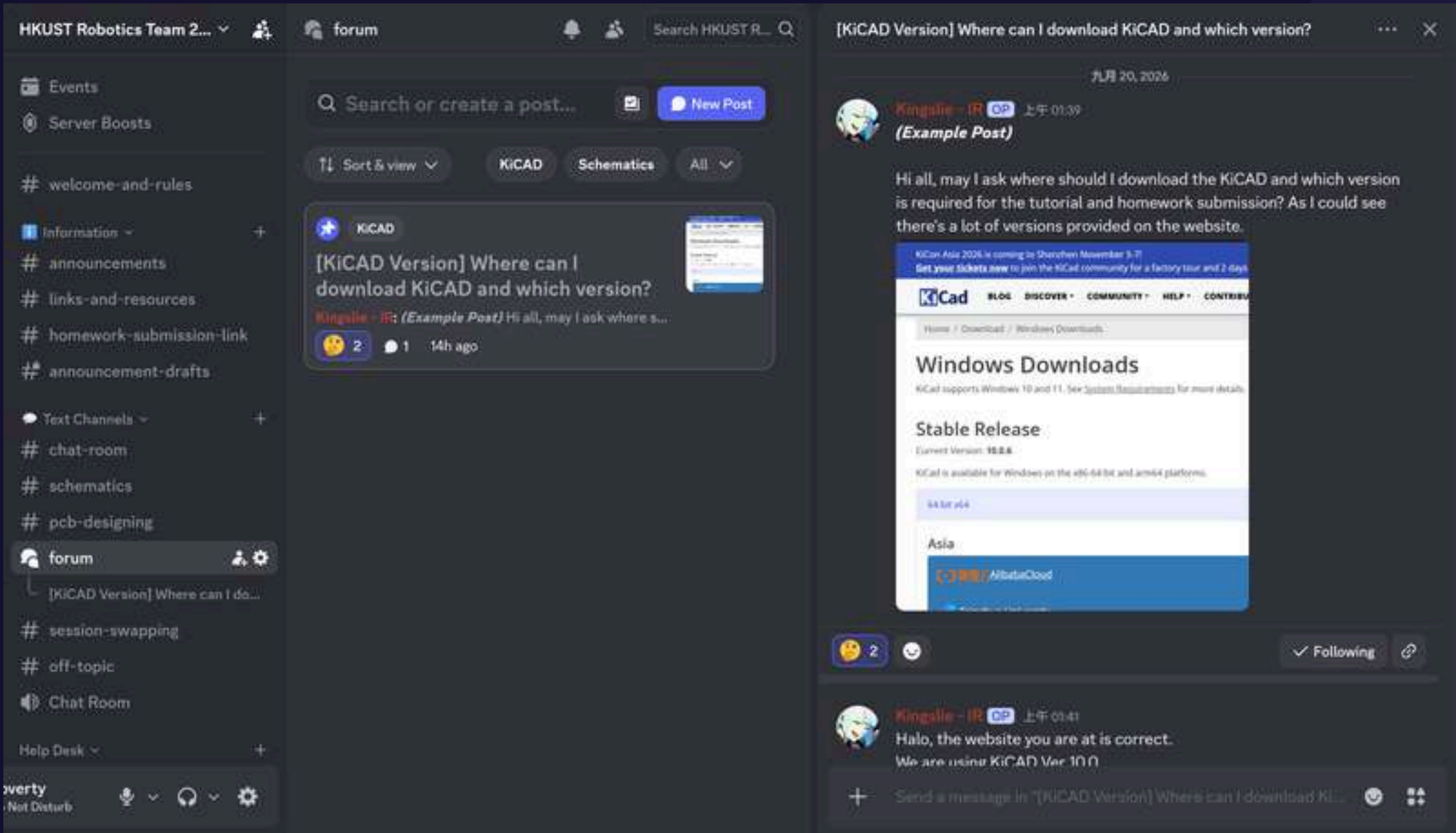
[KiCAD Version] Where can I do...

session-swapping

off-topic

Chat Room

For Quick Address of Questions & Discussion



TUTORIAL NOTES & OTHER RESOURCES

HKUST Robotics Team 2...

links-and-resources

Link to Github: <https://github.com/YuanL-013/2026-Hardware-Tutorial/tree/main>

Events

Server Boosts

welcome-and-rules

Information

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announcement-drafts

Text Channels

chat-room

schematics

pcb-designing

forum

[KiCAD Version] Where can I do...

session-swapping

off-topic

Chat Room

Help Desk

#

Welcome to #links-and-resources!

This is the start of the #links-and-resources channel. Link to Github: <https://github.com/YuanL-013/2026-Hardware-Tutorial/tree/main>

Edit Channel

2026-Hardware-Tutorial

Public

Watch 0

Fork 0

Star 0

main

1 Branch

0 Tags

Go to file

Add file

Code

thomas2204 changed a03400a footprint 55T4324 · 15 hours ago 9 Commits

RDC2026_Libraries Update libraries 2 days ago

LIBRARIES.md Update libraries, guides and readme last month

RDC2026_Libraries.zip changed a03400a footprint 15 hours ago

README.md change 9.0 to 10.0 15 hours ago

README

Hardware-Tutorial

Welcome to the HKUST Robotics Team Hardware Tutorial!

Below will be a brief overview of our schedule: (Tutorial Arrangement)[]
(Homework Arrangement)[]

Installation of KiCAD 10.0

Please install this before the first tutorial!

Please install this before the first tutorial!

Installation guides for different operating systems:

- Windows
- MacOS
- Linux
 - Install from terminal via flatpak: Flatpak install-from <https://flathub.org/repo/appstream/org.kicad.KiCad.flatpakref>

We will be using a customized components library for RDC to ease the designing process for many. Below is where the library can be downloaded and steps on how to import the library:

(RDC2026-Libraries)[]

(Importing libraries into KiCAD)[]

If you have any questions/facing issues, feel free to reach out the the seniors!

Self Learn Module

About

Materials for HKUST Robotics Team recruitment hardware tutorial

Readme

Activity

0 stars

0 watching

0 forks

Report repository

Releases

No releases published

Contributors 2

YuanL-013 Yuan

thomas2204



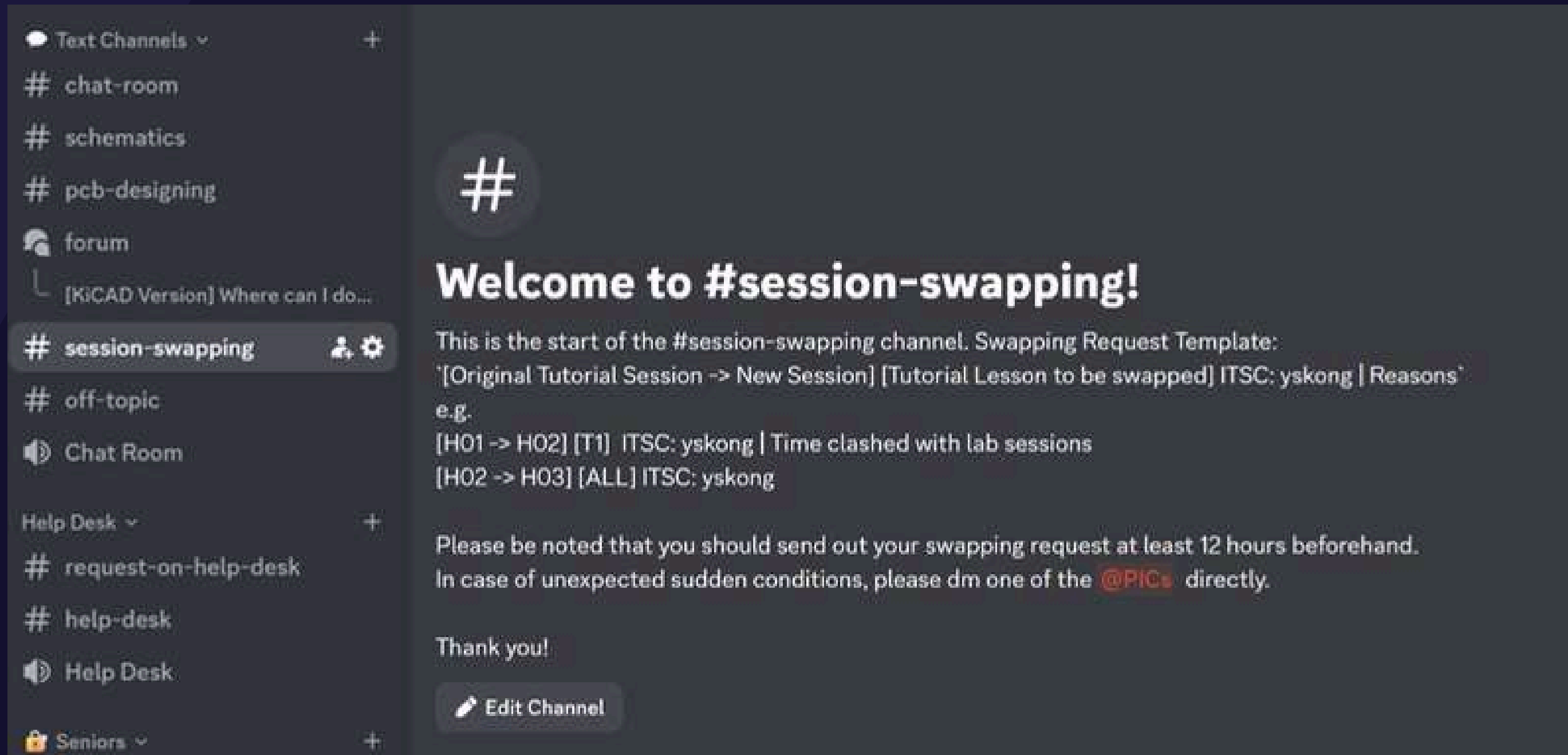
TUTORIAL SCHEDULE

- Time for each tutorial: 19:00 - 22:00

	Sun	Mon	Tue	Wed	Thurs	Fri	Sat
Sep	6	7	8	9	10	11	12
	13	14	15	16	17 RDC Interview	18 RDC Interview	19 RDC Interview
	20	21	22	23 T1_H01 (Rm1103) Release of HW1(H01)	24 T1_H02 (Rm4472) Release of HW1(H02)	25 Mid-Autumn Festival	26
	27	28 T1_H03 (Rm4219) Release of HW1(H03)	29 HW1 DDL(H01)	30 T2_H01 (Rm1103) Release of HW2(H01)	1 National Day HW1 DDL(H02)	2 T2_H02 (Rm1103) Release of HW2(H02)	3
Oct	4 HW1 DDL(H03)	5 T2_H03 (Rm1103) Release of HW2(H03)	6 HW2 DDL(H01)	7 T3_H01 (Hall IX) HW2 DDL(H02)	8 T3_H02 (Hall IX)	9	10
	11 HW2 DDL(H03)	12 T3_H03 (Hall IX) Release general mistakes according to trainees' PCBs	13	14 T4_H01 (Hall IX)	15 T4_H02 (Hall IX)	16	17
	18 PCB Re- submissio n DDL	19 The Day following the Chung Yeung Festival	20 T4_H03 (Hall IX)	21 Decide groupings for RDC	22	23	24



SWAPPING TUTORIAL SESSIONS



Text Channels ▾

- # chat-room
- # schematics
- # pcb-designing
- forum
- [KiCAD Version] Where can I do...
- # session-swapping**  
- # off-topic
- Chat Room
- Help Desk ▾
- # request-on-help-desk
- # help-desk
- Help Desk
- Seniors ▾

#

Welcome to #session-swapping!

This is the start of the #session-swapping channel. Swapping Request Template:
 '[Original Tutorial Session -> New Session] [Tutorial Lesson to be swapped] ITSC: yskong | Reasons'
 e.g.
 [H01 -> H02] [T1] ITSC: yskong | Time clashed with lab sessions
 [H02 -> H03] [ALL] ITSC: yskong

Please be noted that you should send out your swapping request at least 12 hours beforehand.
 In case of unexpected sudden conditions, please dm one of the [@PICs](#) directly.

Thank you!

 Edit Channel

HELP DESKS

announcements

links-and-resources

homework-submission-link

announcement-drafts

Text Channels

+

chat-room

schematics

pcb-designing

forum

[KiCAD Version] Where can I do...

session-swapping

off-topic

Chat Room

Help Desk

+

request-on-help-desk

help-desk

Help Desk

#

Welcome to #request-on-help-desk!

This is the start of the #request-on-help-desk channel. You may request a help desk with our senior members. Please type out the preferred meeting time and venue (if any).

Template:

'[Item]'

'[Time]'

'[Venue]'

For exmaple:

[PCB Routing]

Time: 26/9 3pm

Venue: Library LG3 - 17

If no senior has replied for 12 hours, sorry for that and please @ us again.

Thank you!

Edit Channel



REMINDERS

- The main announcement will be shared via Discord and email.
- YOU are responsible for any issues that arise from not checking these messages regularly.
- Please submit your homework on time.
- Information provided by seniors is not guaranteed to be 100% correct.
 - We are all learning throughout this process, so please verify and ask questions when in doubt.



DISCLAIMER

- **Tutorial Content**
 - **This tutorial includes both mandatory and optional content (the latter is in hidden slides or extra documentation posted on Github).**
 - **Mathematics is not required to follow this tutorial.**
 - **Due to time constraints, we will cover only the mandatory content during the session.**
 - **The mandatory material is sufficient for completing the first homework assignment.**
- **Feedback & Alternatives**
 - **If you feel that this hardware tutorial does not meet your needs and you would prefer to attend a tutorial offered by another department, please contact us after the session.**



KICAD ⁰
LIBRARIES

IMPORT KICAD LIBRARIES



KICAD⁰ LIBRARIES

DOWNLOAD IT

2026-Hardware-Tutorial / KiCAD-Materials / 

Add file ▾ 

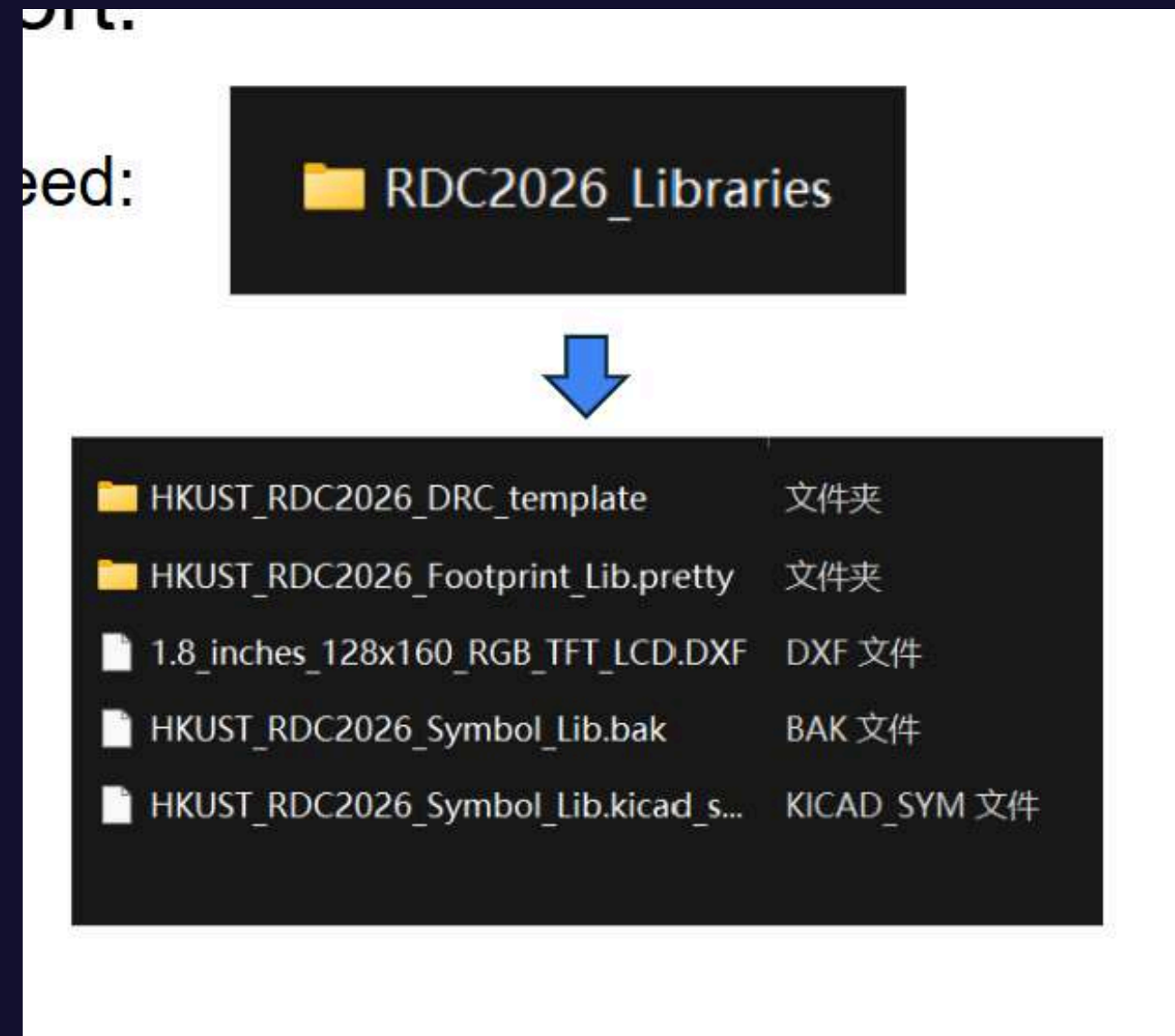
 k-kys Added Installation guidelines (PDF ver) b4f88bd · 4 minutes ago  History

Name	Last commit message	Last commit date
 ..		
 2026RDC-Libraries	update-libraries	6 hours ago
 2026RDC-Libraries.zip	update-libraries	6 hours ago
 KiCAD Installation Set-up.pdf	Added Installation guidelines (PDF ver)	4 minutes ago
 KiCAD-Installation-Setup-Plugins.md	Fix formatting for download instructions in plugin setup	7 minutes ago

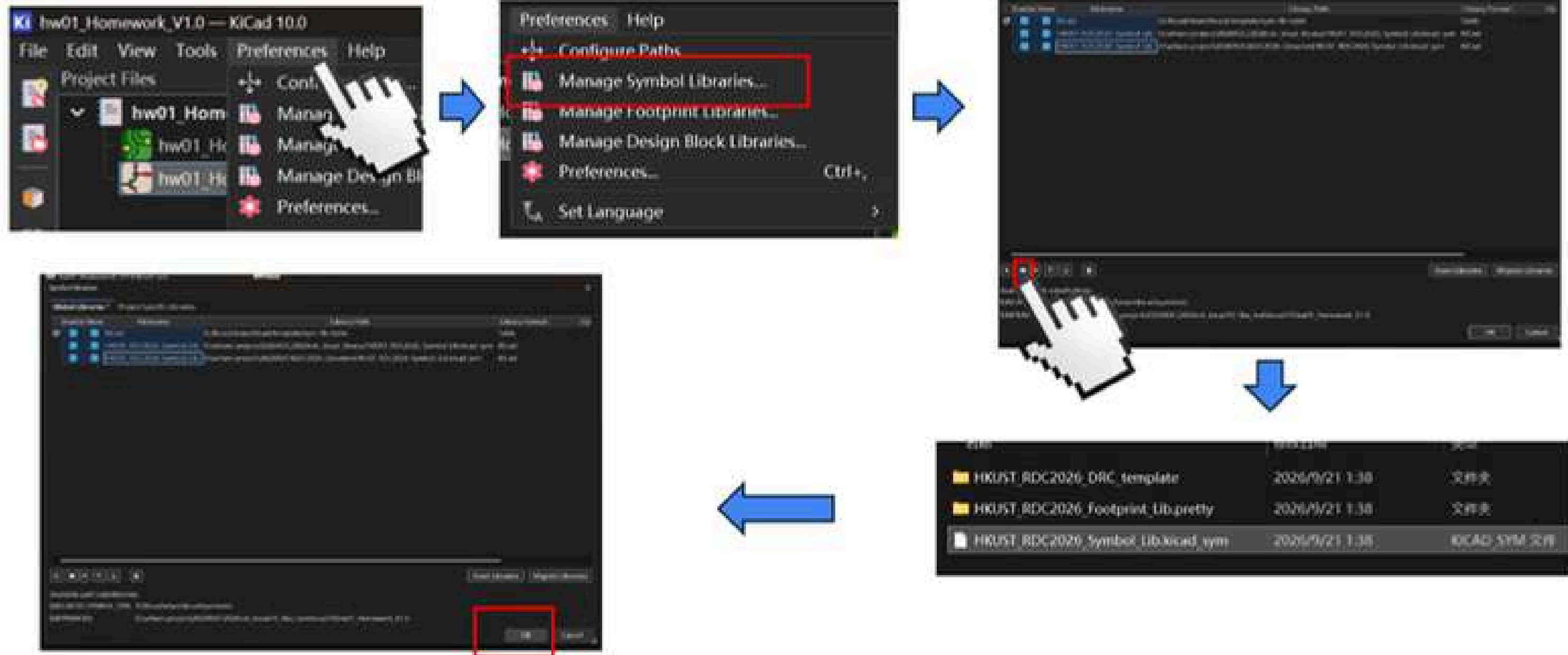


Just download the ZIP file is okay

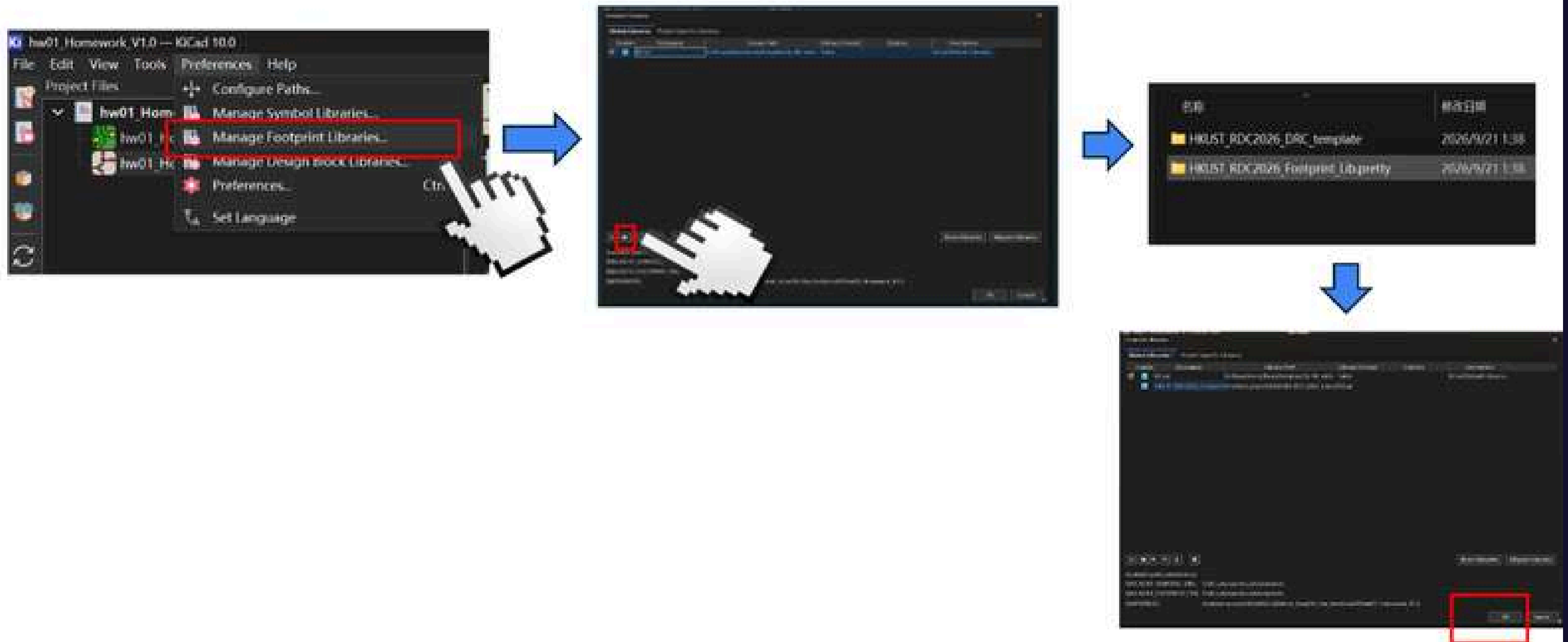
PUT IT IN A GOOD PLACE (DO NOT JUST PUT IT IN YOUR “DOWNLOAD” FOLDER, THEN UNZIP IT



To add symbol library :



To add footprint library :



XXXX

ITEM 2

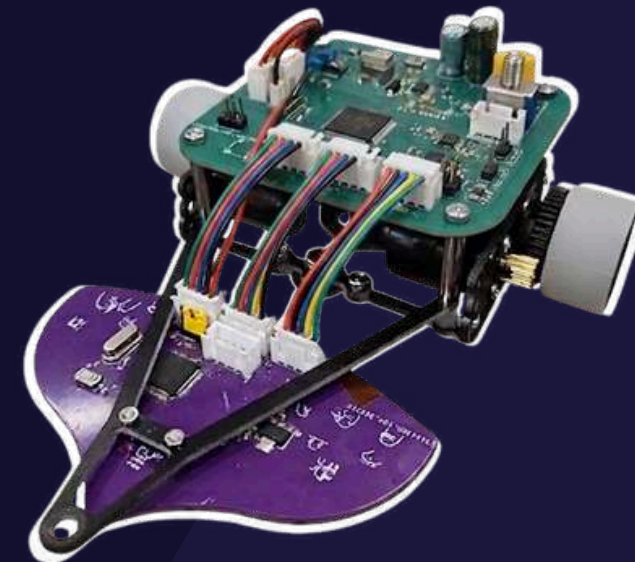
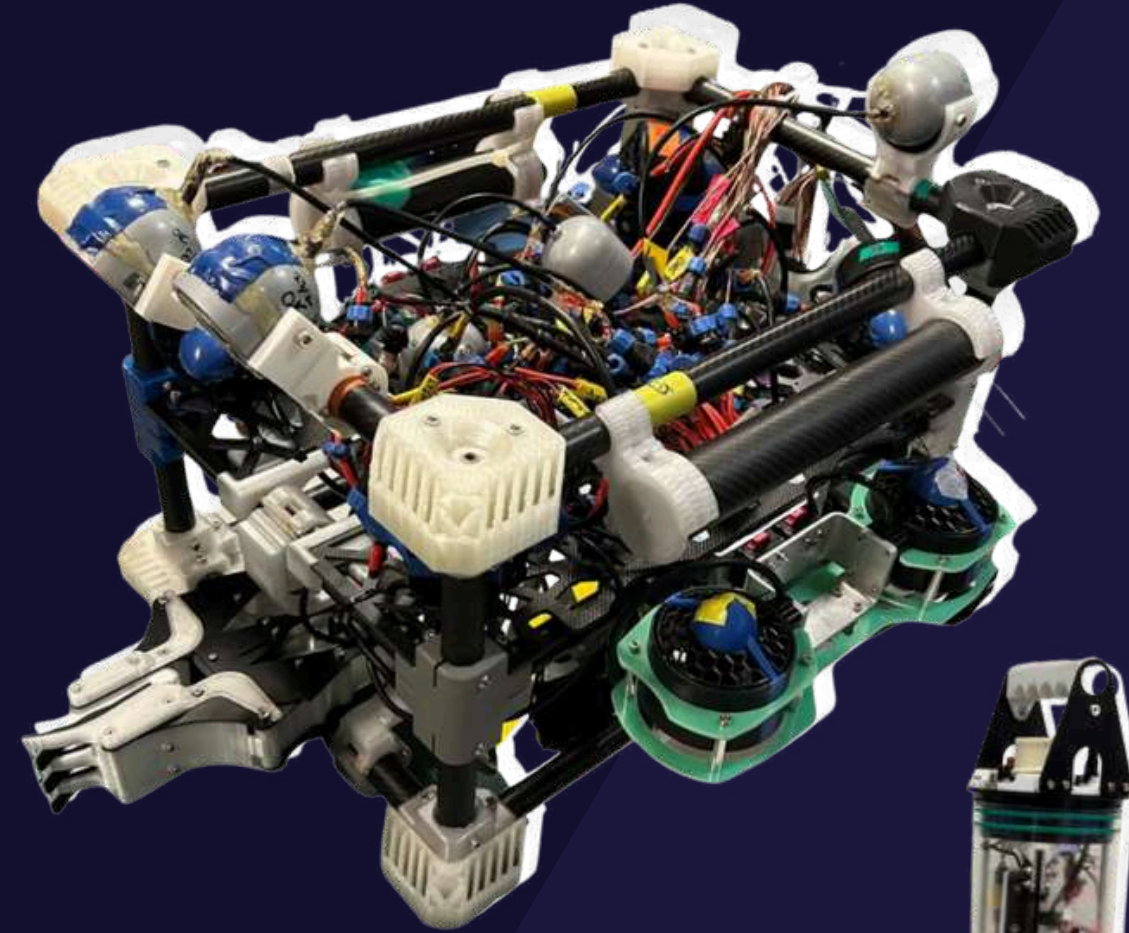
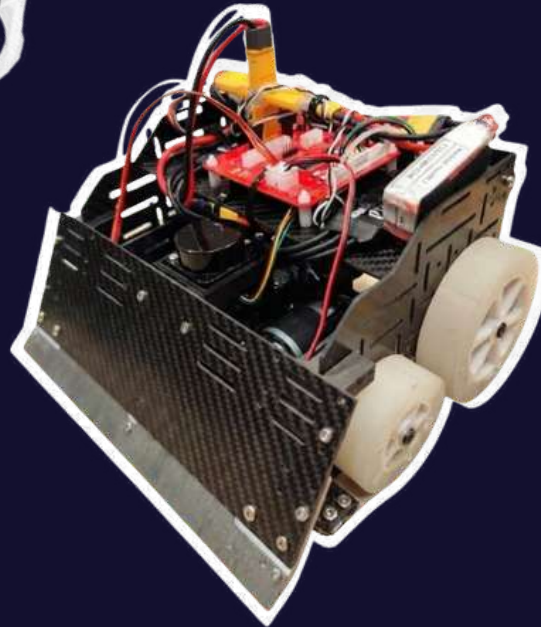
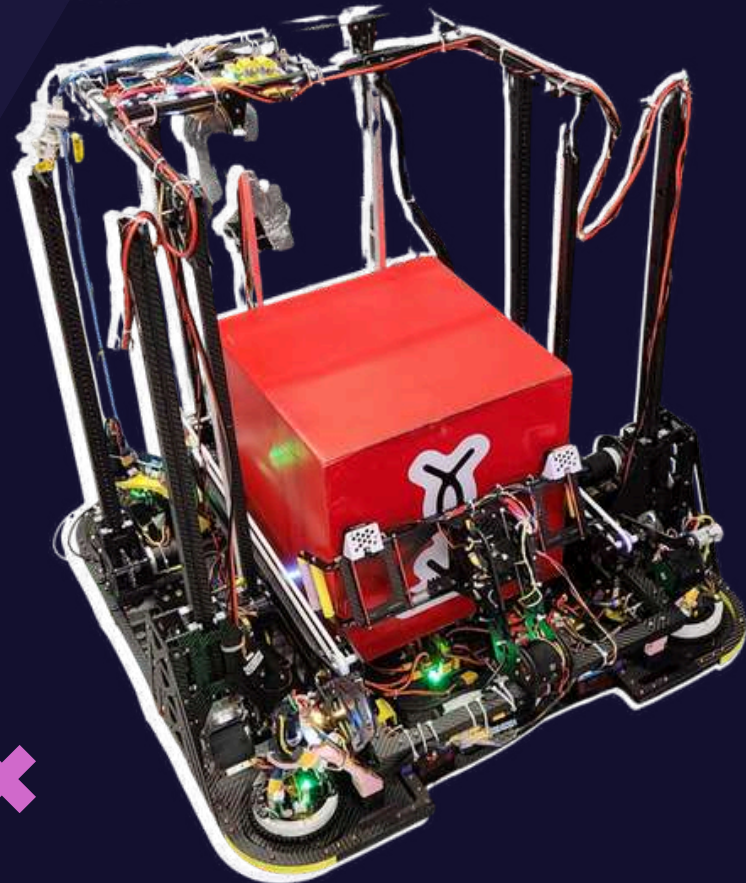
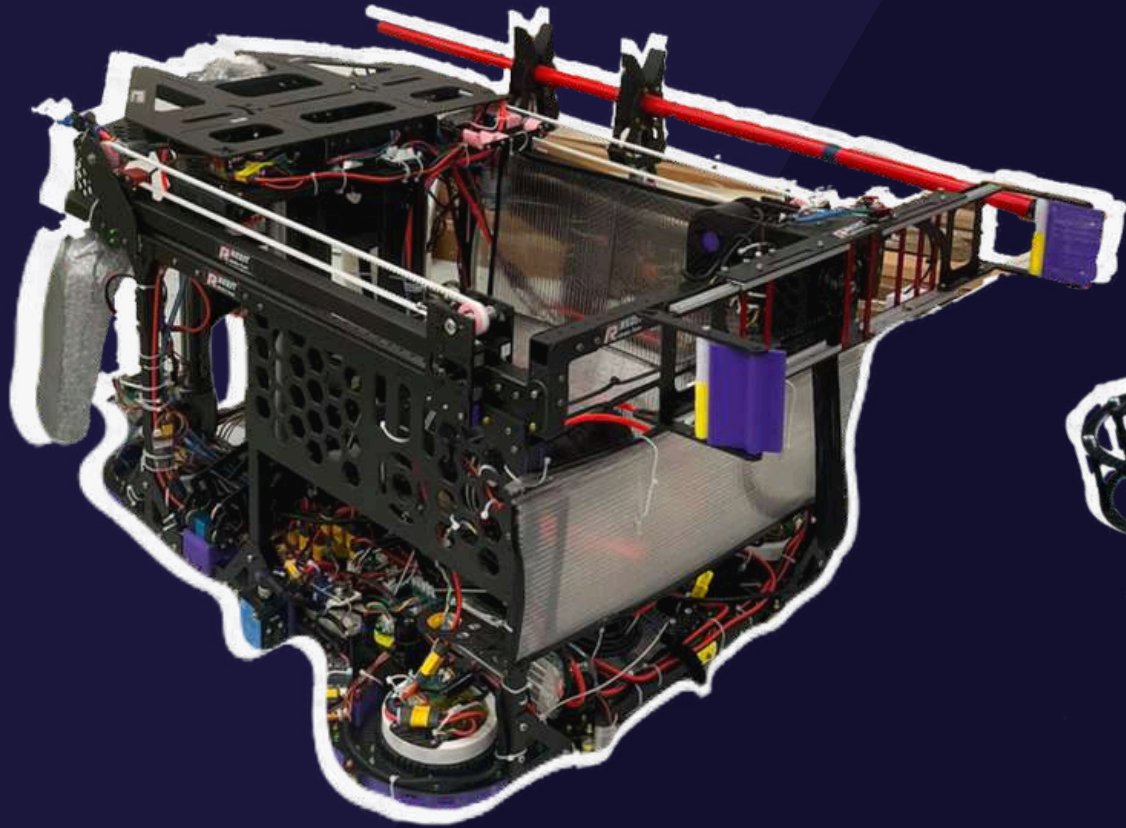
WHAT IS HARDWARE & YOUR EXPECTATION

A white text box with a purple border containing the text 'PCB? MCU?'. The background of the slide features a close-up, angled view of a printed circuit board (PCB) with various electronic components like chips, capacitors, and resistors, overlaid with a purple and blue geometric design.

XXXX

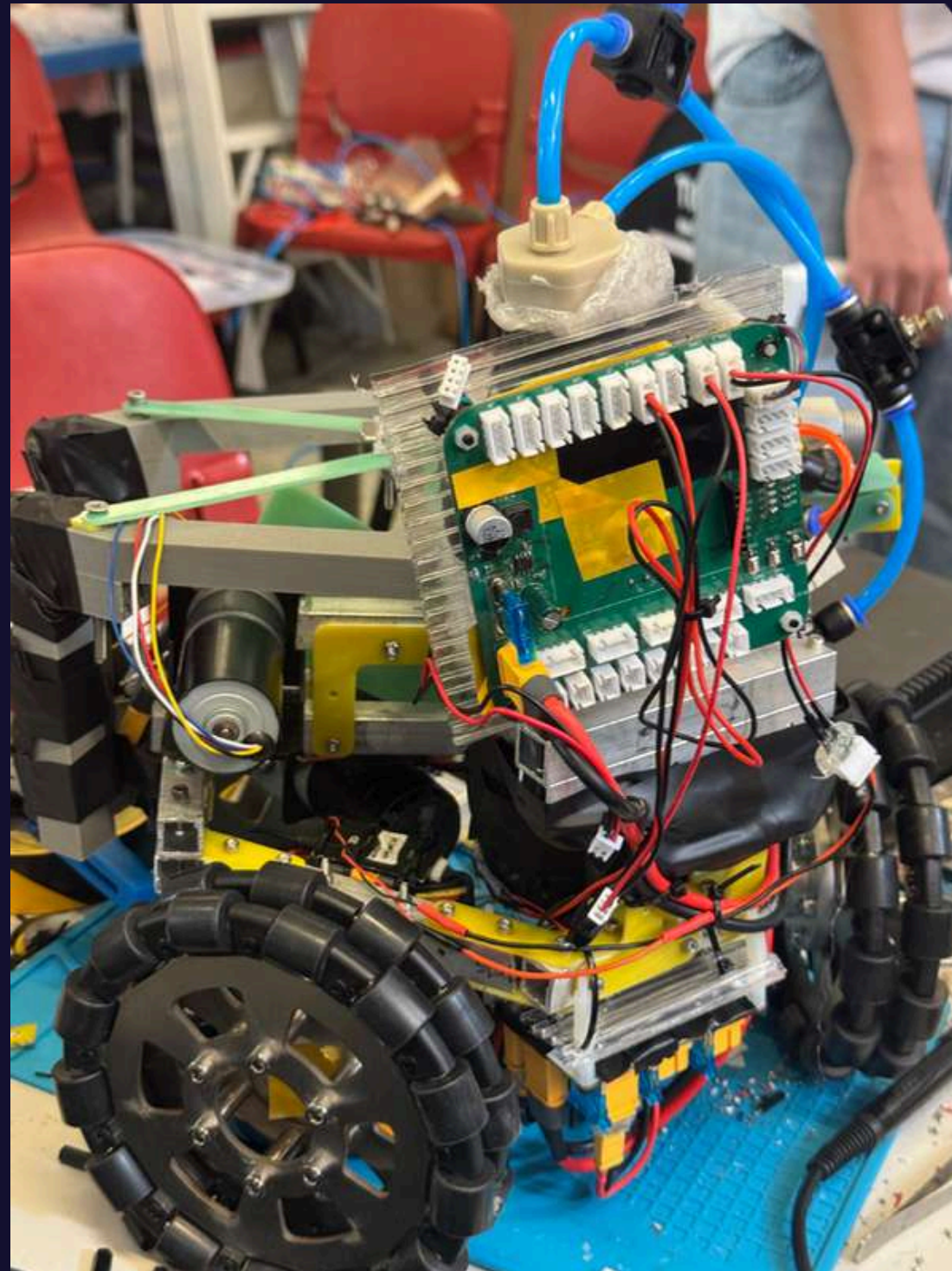
INTRODUCTION²

YUP, A ROBOT.



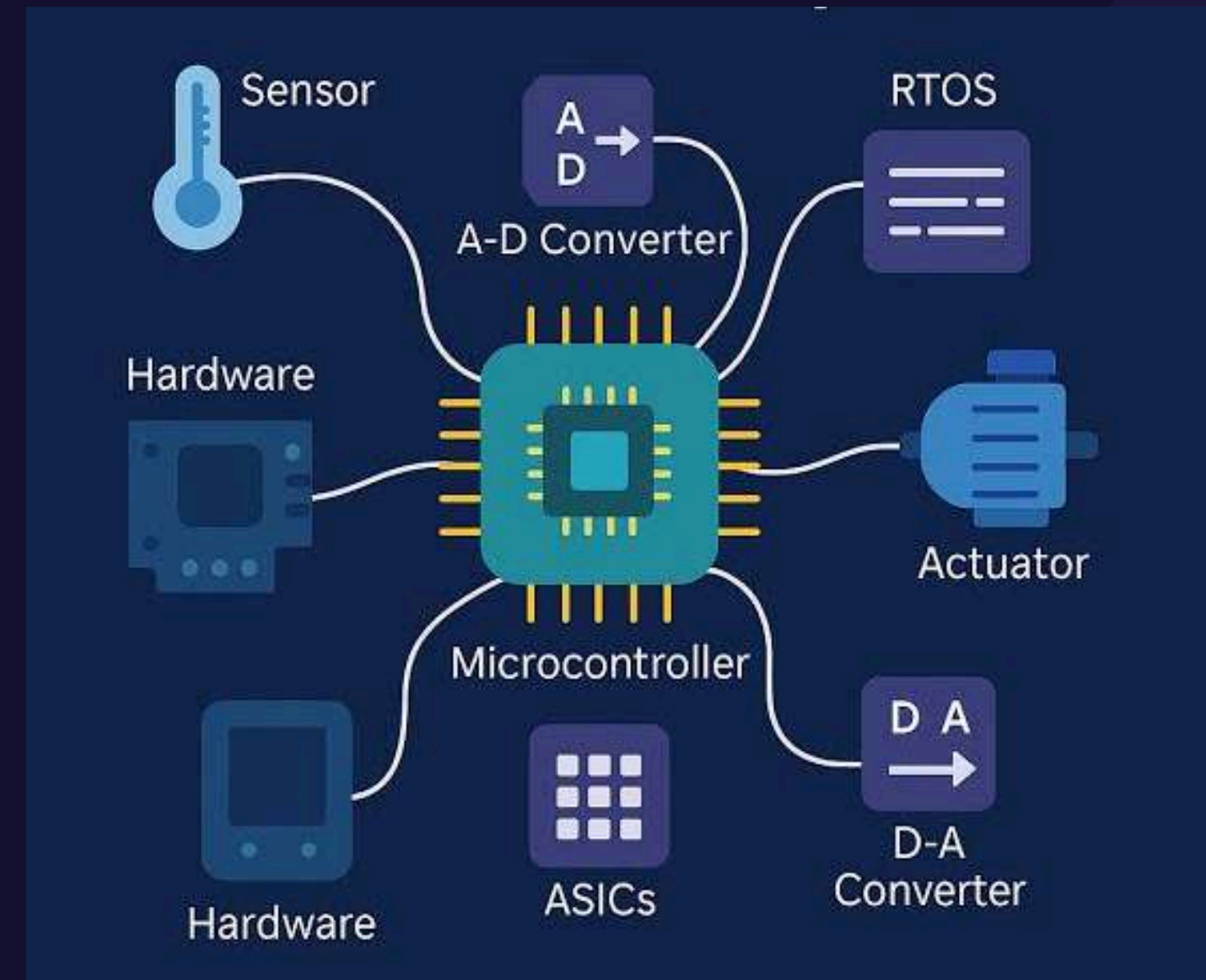
INTRODUCTION²

ALSO A ROBOT...



EMBEDDED SYSTEM

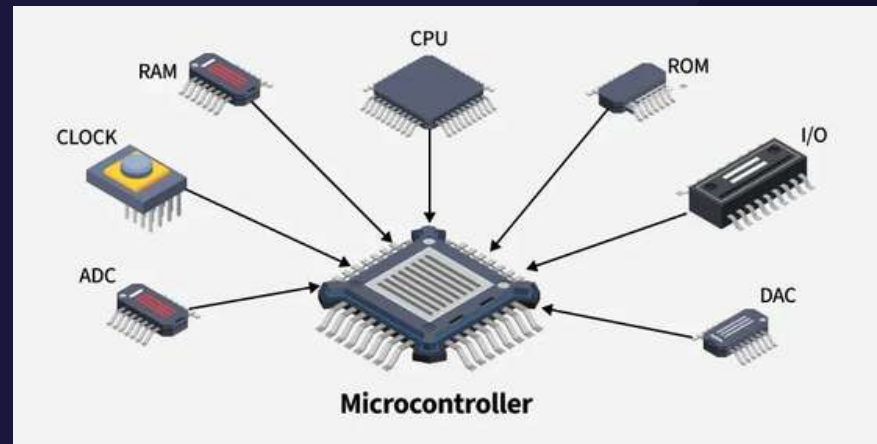
- **Specialized computer system for a dedicated function, comprising:**
 - **Software**
 - **Logic Operations**
 - **... (See software tutorial)**
 - **Hardware**
 - **Power management**
 - **Micro-controllers**
 - **Integrated Circuits**
 - **Motors**
 - **Much more**



INTRODUCTION²

ROBOT COMPONENTS V.S. HUMAN

Micro-controller (MCU)



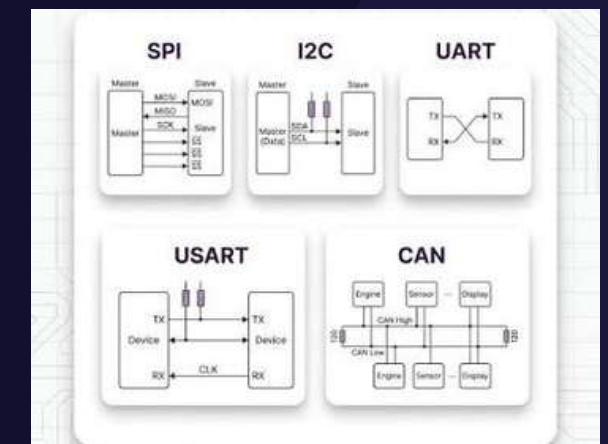
Sensors



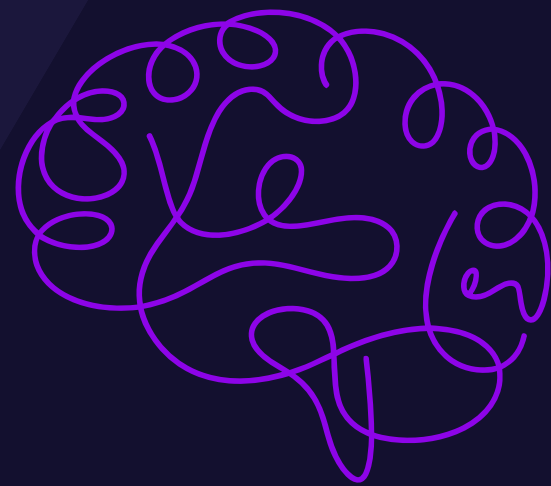
Actuators



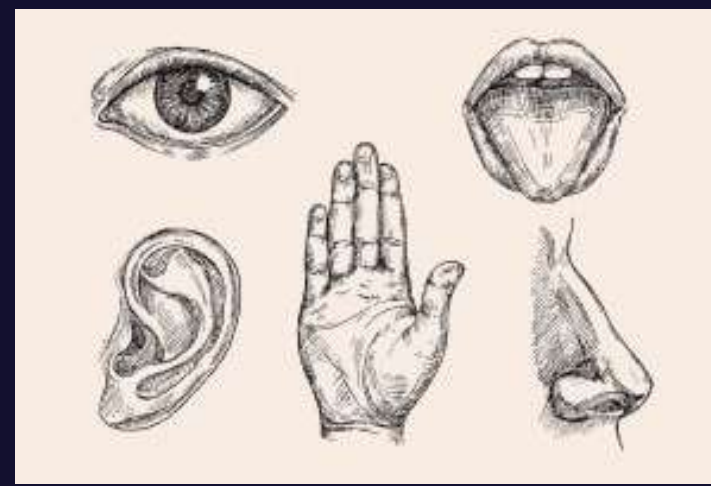
Communication Protocols



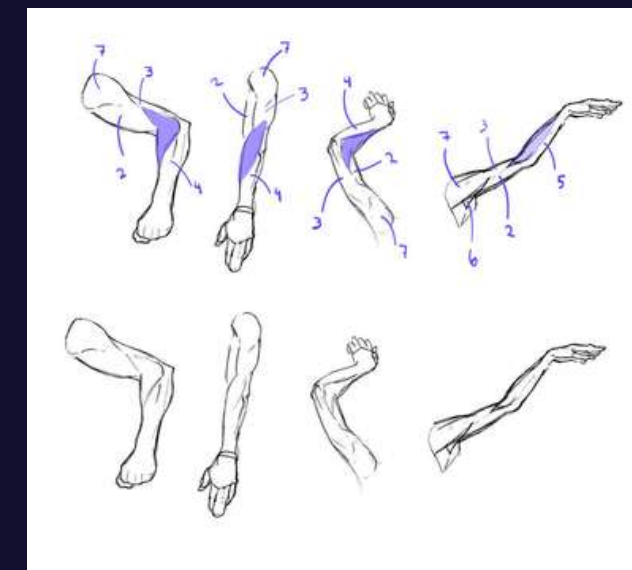
Brain



Sensations



Limbs

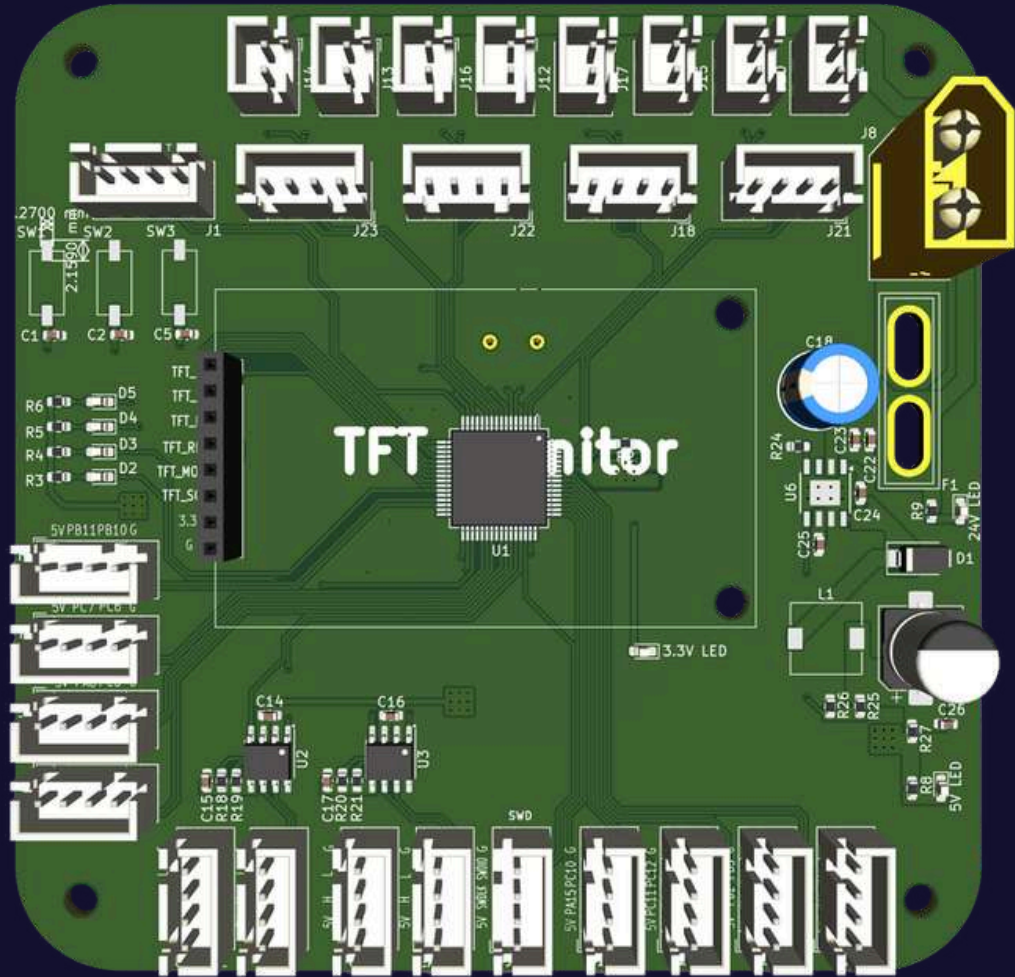
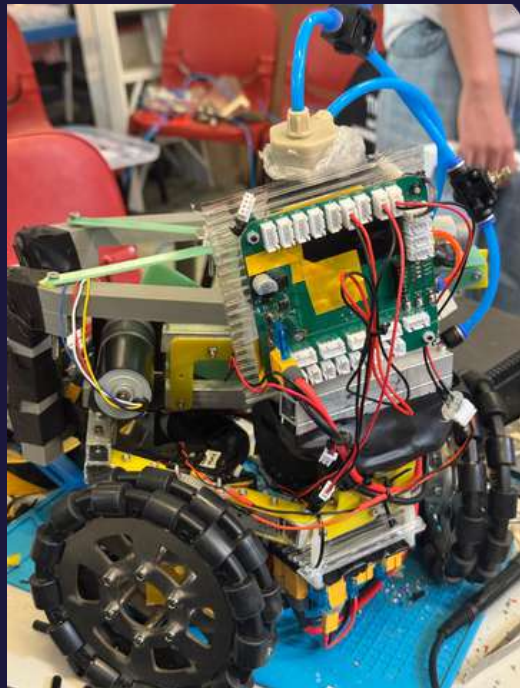


Languages



INTRODUCTION ²

ROBOT HARDWARE

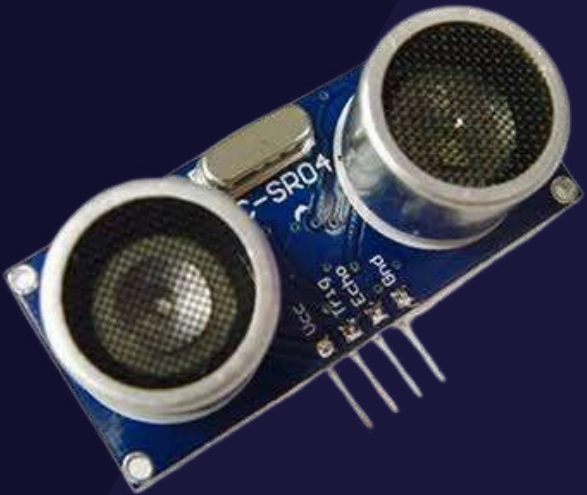


Control Board

Motors



Sensors



Battery



Pneumatics

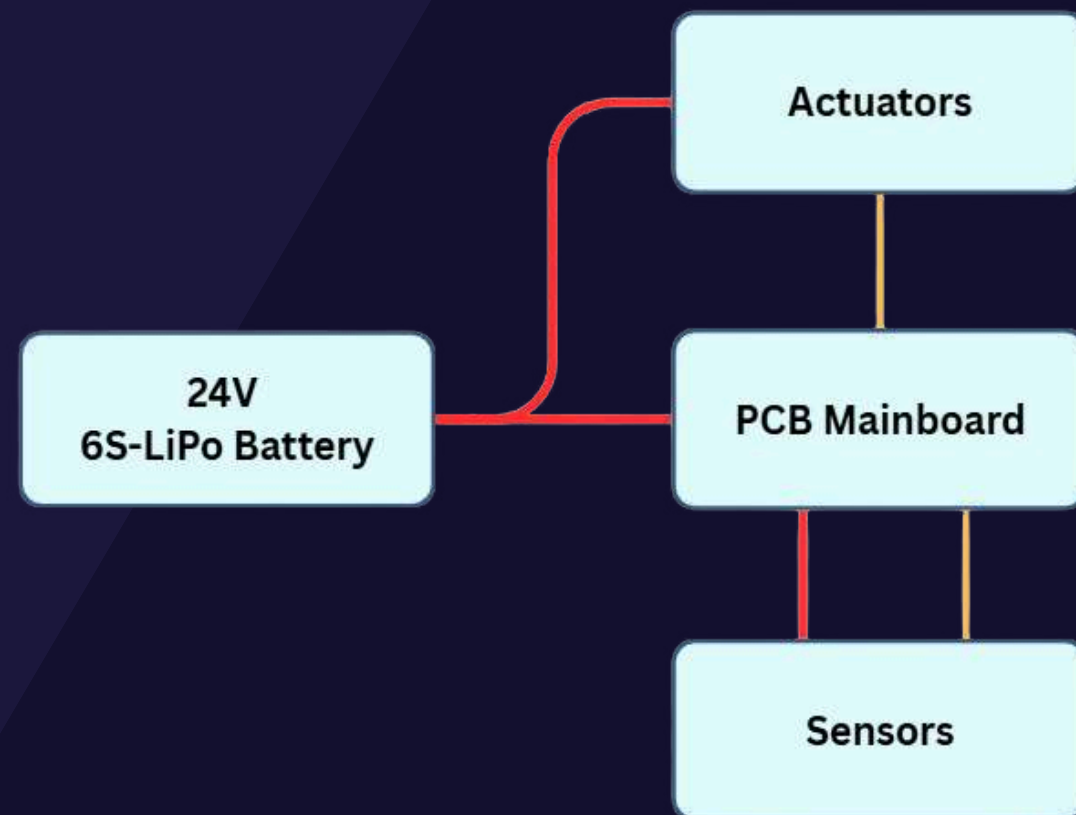


Design Robot Layout



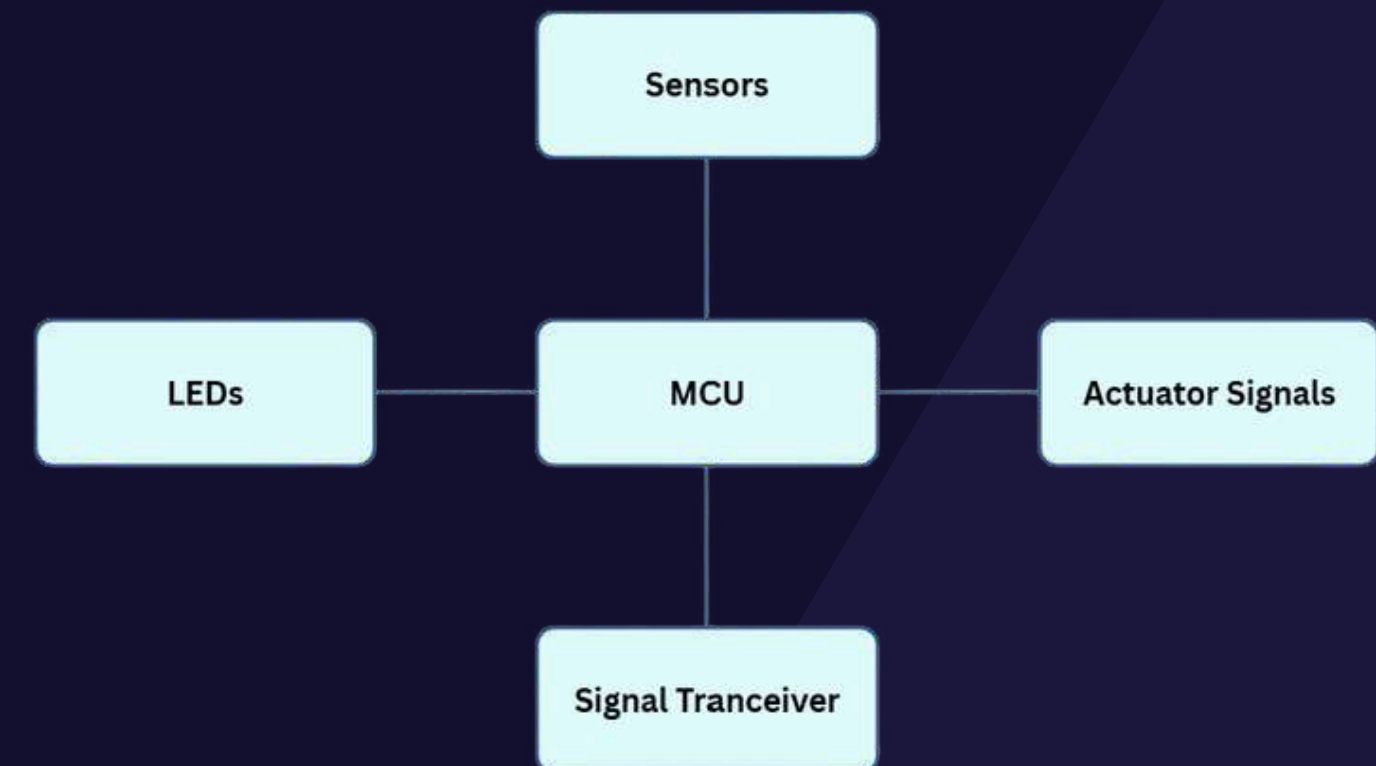
SIMPLIFIED LAYOUT

Grand View



***No worries**
it won't be so easy in the actual case

Mainboard Grand View



***No worries *2**
it won't be so easy in the actual case, as well



We will go through them briefly in this tutorial session together

AFTER THIS TUTORIAL...

1. How hardware operates in a robot
2. Breakdown of circuit board structure
3. How to draw an STM32-based control board circuit
4. Draw the schematics

HW1: SCHEMATIC OF YOUR RDC MAINBOARD



XXXX

ITEM 3

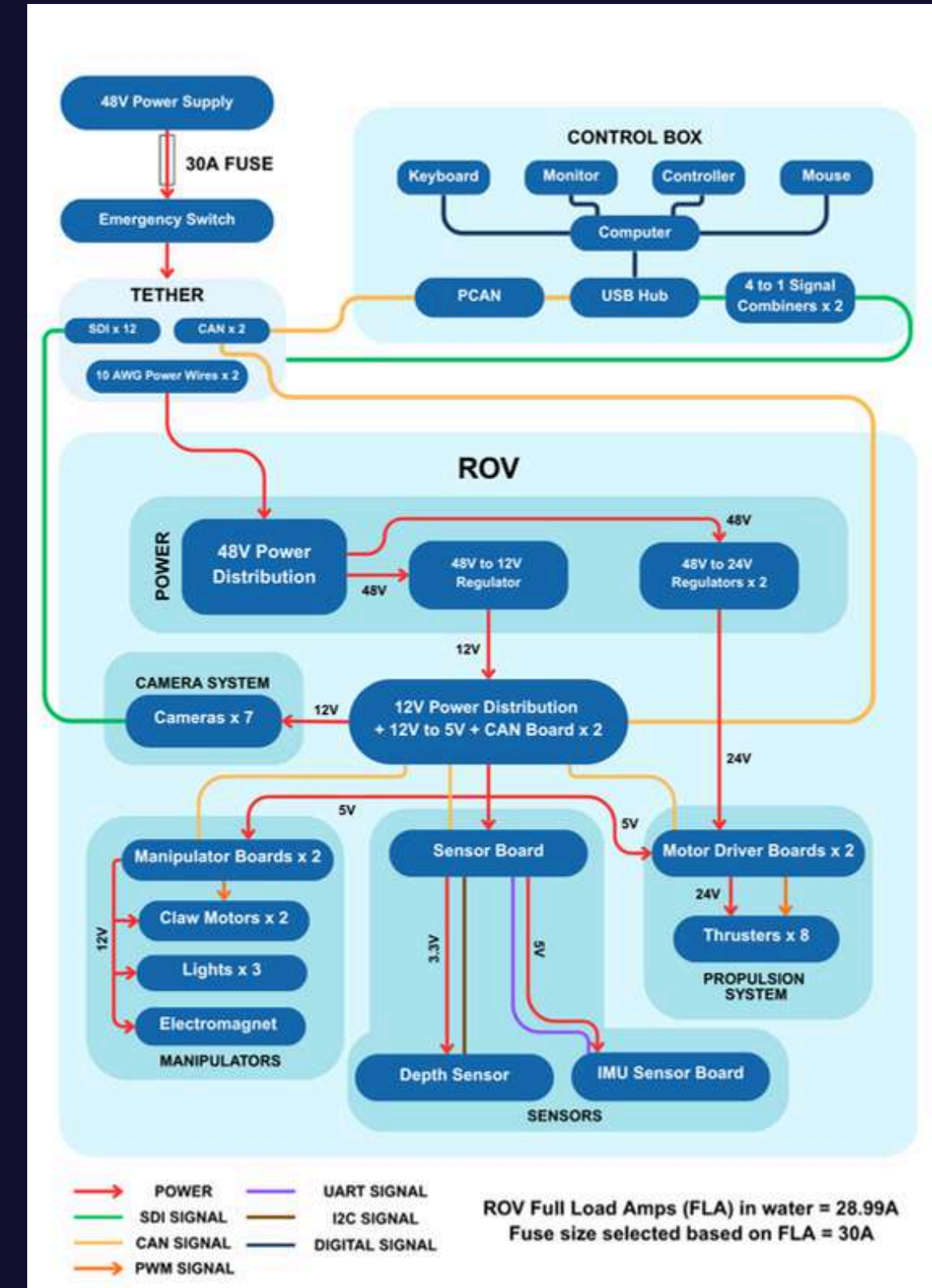
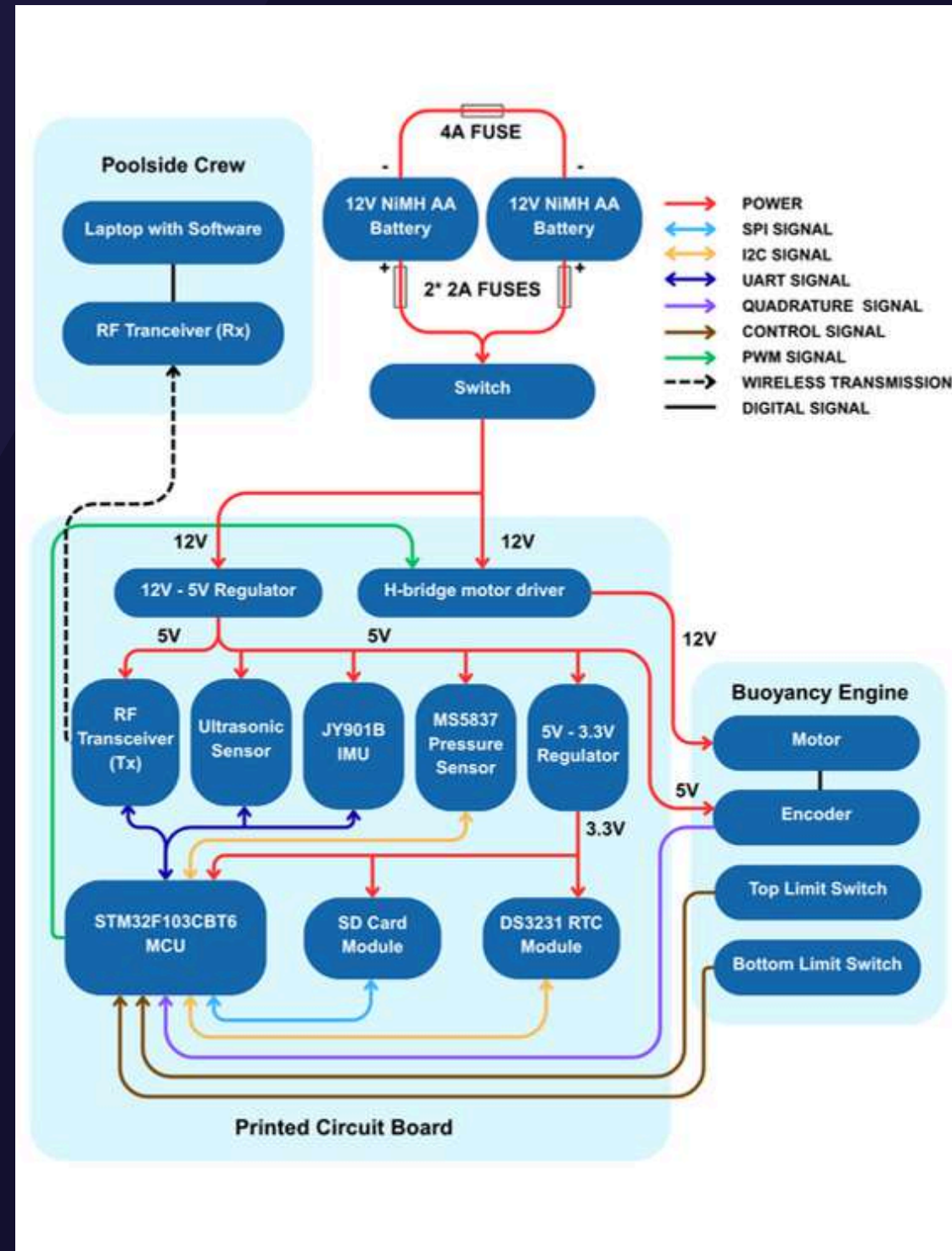
DESIGN OVERVIEW

XXXX



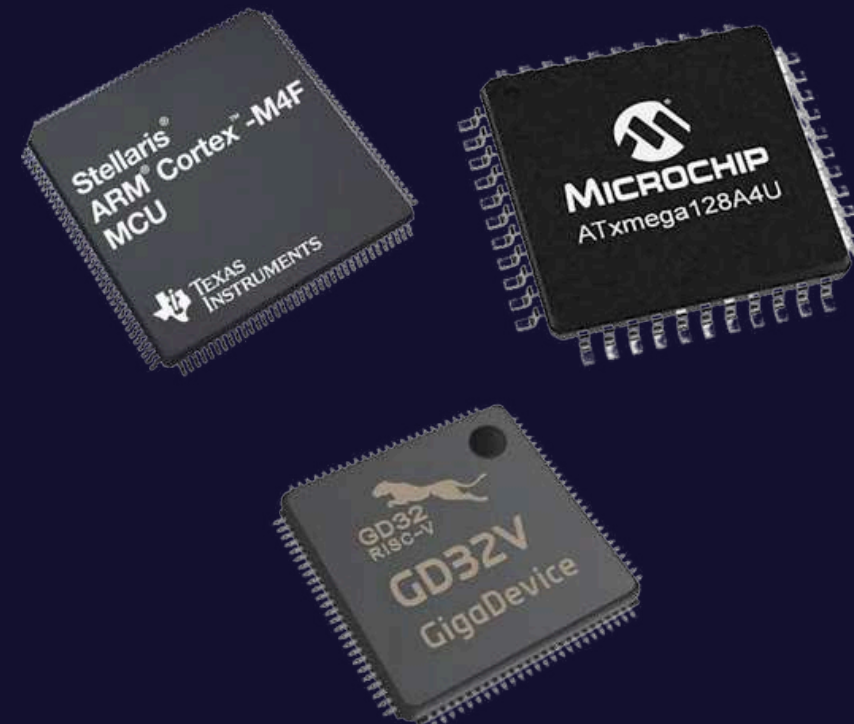
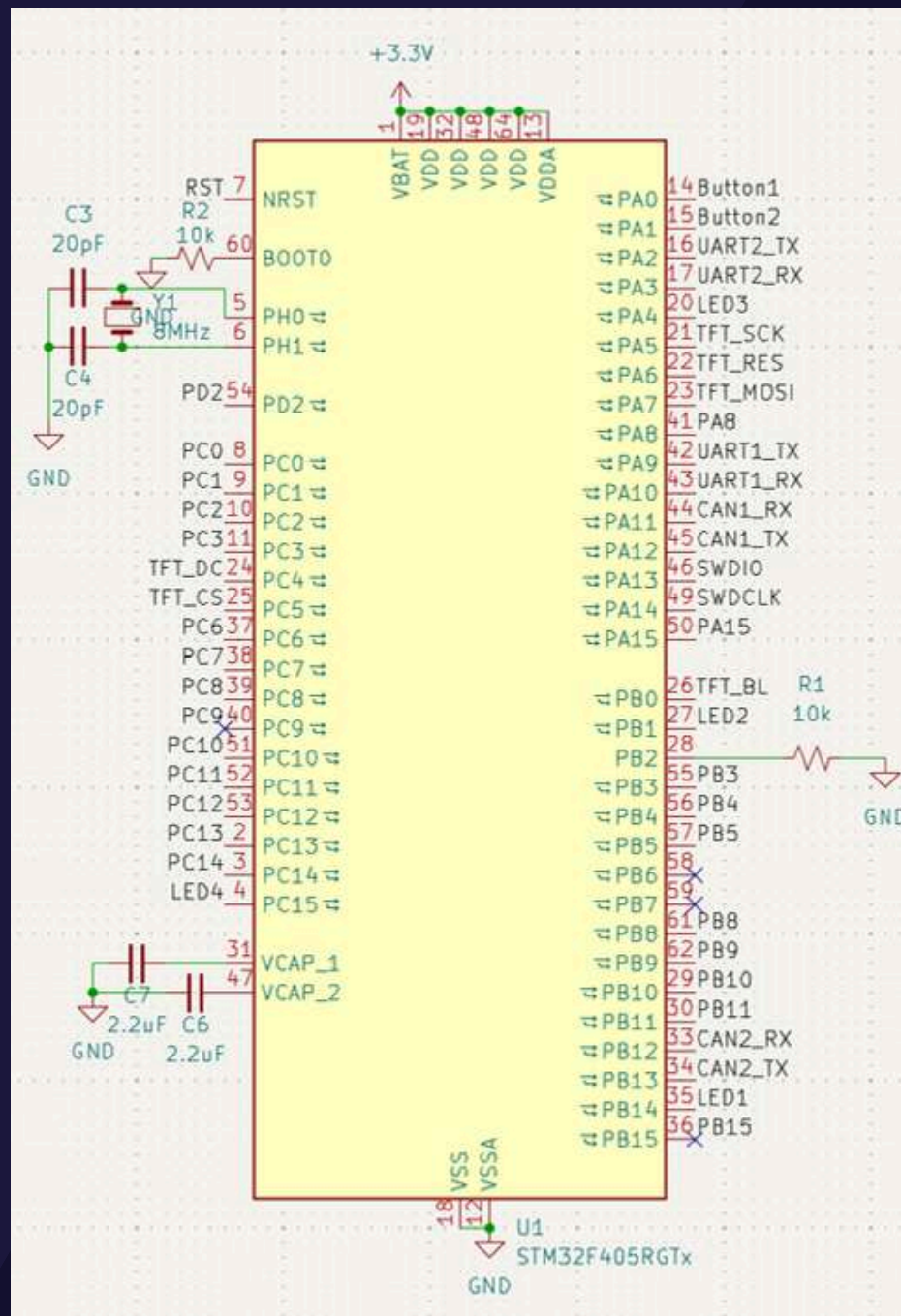
DESIGN ³ OVERVIEW

EXAMPLES



DESIGN ³ OVERVIEW

HOW TO CONTROL STUFF? - MCU



- Usually 3.3V/5V
 - Run command
 - Send command to another device
 - Read sensor values
 - Much more...
-
- Considering factors when choosing a microcontroller unit:
 - Pins available
 - Computational speed
 - Compatibility with different communication protocols
 - RAM size (capability in handling requests)



HOW TO CONTROL STUFF? - MCU

STM32

F

051

R

8

T

6

X

XX

Family

STM32 32-bit MCUs

STM8 8-bit MCUs

Product type

A Automotive

F Foundation

L Ultra-low power

S Standard

T Touch sensing

W Wireless

xP Fastrom

Pin count (pins for STM8 and STM32)

D 14 pins

Y 20 pins (STM8)

F 20 pins (STM32)

E 24 & 25 pins

G 28 pins

K 32 pins

T 36 pins

H 40 pins

S 44 pins

C 48 & 49 pins

U 63 pins

R 64 & 66 pins

J 72 pins

M 80 pins

O 90 pins

V 100 pins

Q 132 pins

Z 144 pins

A 169 pins

I 176 & 201 (176+25) pins

B 208 pins

N 216 pins

X 256 pins

Auto

8 48

9 64

A 80

Specific features (3 digits)

(Depends of product series

None exhaustive list)

STM32x ...

051 Entry-level

103 STM32 foundation

303 103 upgraded with DSP and Analog

407 High-performance and DSP with FPU

152 Ultra-low-power

STM8x .../STM8Ax...

103 Mainstream access line

F52 Automotive CAN

L31 Automotive low-end

Code size (Kbytes)

0	1
1	2
2	4
3	8
4	16
5	24
6	32
7	48
8	64
9	72
A	96 or 128*
B	128
Z	192
C	256
D	384
E	512
F	768
G	1024
H	1536
I	2048

Note:

* For STM8A only

Package

B Plastic DIP*

D Ceramic DIP*

G Ceramic QFP

H LFBGA /TFBGA

I UFBGA Pitch 0.5**

J UFBGA Pitch 0.8**

K UFBGA Pitch 0.65**

M Plastic SO

P TSSOP

Q Plastic QFP

T QFP

U UQFPN

V VQFPN

Y WLCSP

* Dual-in-Line package

** For new product serie only for existing product marketing serie please use H letter

Firmware Royalties

U Universal Not for production (Sampling and tools)

V MP3 decoder

W MP3 Codec

J 0.80 mm

D IS2T JAVA

Option

xxx Fastrom code

or

xTR Tape and Reel

Dxx No RTC (STM8L)

Dxx BOR OFF with Special bonding + Boot standard

Dxx BOR OFF with Boot I2CS (Special)

Sxx BOR OFF

bxx BOR ON

No Letter BOR ON + Boot standard

or

Yxx Die rev (Y)

Temperature range (°C)

6 and A - 40 to + 85

7 and B - 40 to + 105

3 and C - 40 to + 125

D - 40 to + 150

●SON @MEYOU_Cc

FOR EXAMPLE, STM32F405RGT6

(1) 32-BIT MCU

(2) 66 IO PIN

(3) STORAGE: 1024 K FLASH

(4) QFP PACKAGE

(5) TEMP RANGE -40 TO +85



****SUPPLEMENTARY****

MCU PINOUTS AND PIN DESCRIPTION

1. THE PINS

2. PINS NAME

3. TYPE

S -> POWER PIN

I -> INPUT PIN

I/O -> INPUT/OUTPUT PIN

4. I/O STRUCTURE

FT -> ALLOW 5V

TTA -> ONLY ALLOW 3V3, AND
CONNECT TO ADC

B -> BOOT PIN

RST -> RST PIN

5. MAIN FUNCTION

6. ALTERNATIVE FUNCTION

****SUPPLEMENTARY****

Table 5. High-density STM32F103xx pin definitions

Pins ¹						²	³	⁴	⁵	Alternate functions ⁽⁴⁾	
LFPGA144	LFPGA100	WLCSP64	LQFP64	LQFP100	LQFP144	Pin name	Type ⁽¹⁾	I/O Level	Main function ⁽³⁾ (after reset)	Default	⁷ Remap
A3	A3	-	-	1	1	PE2	I/O	FT	PE2	TRACECK/ FSMC_A23	
A2	B3	-	-	2	2	PE3	I/O	FT	PE3	TRACED0/FSMC_A19	
B2	C3	-	-	3	3	PE4	I/O	FT	PE4	TRACED1/FSMC_A20	
B3	D3	-	-	4	4	PE5	I/O	FT	PE5	TRACED2/FSMC_A21	
B4	E3	-	-	5	5	PE6	I/O	FT	PE6	TRACED3/FSMC_A22	
C2	B2	C6	1	6	6	V _{BAT}	S		V _{BAT}		CSDN@MEYOU.CC



MCU - BOOT PIN

2.4

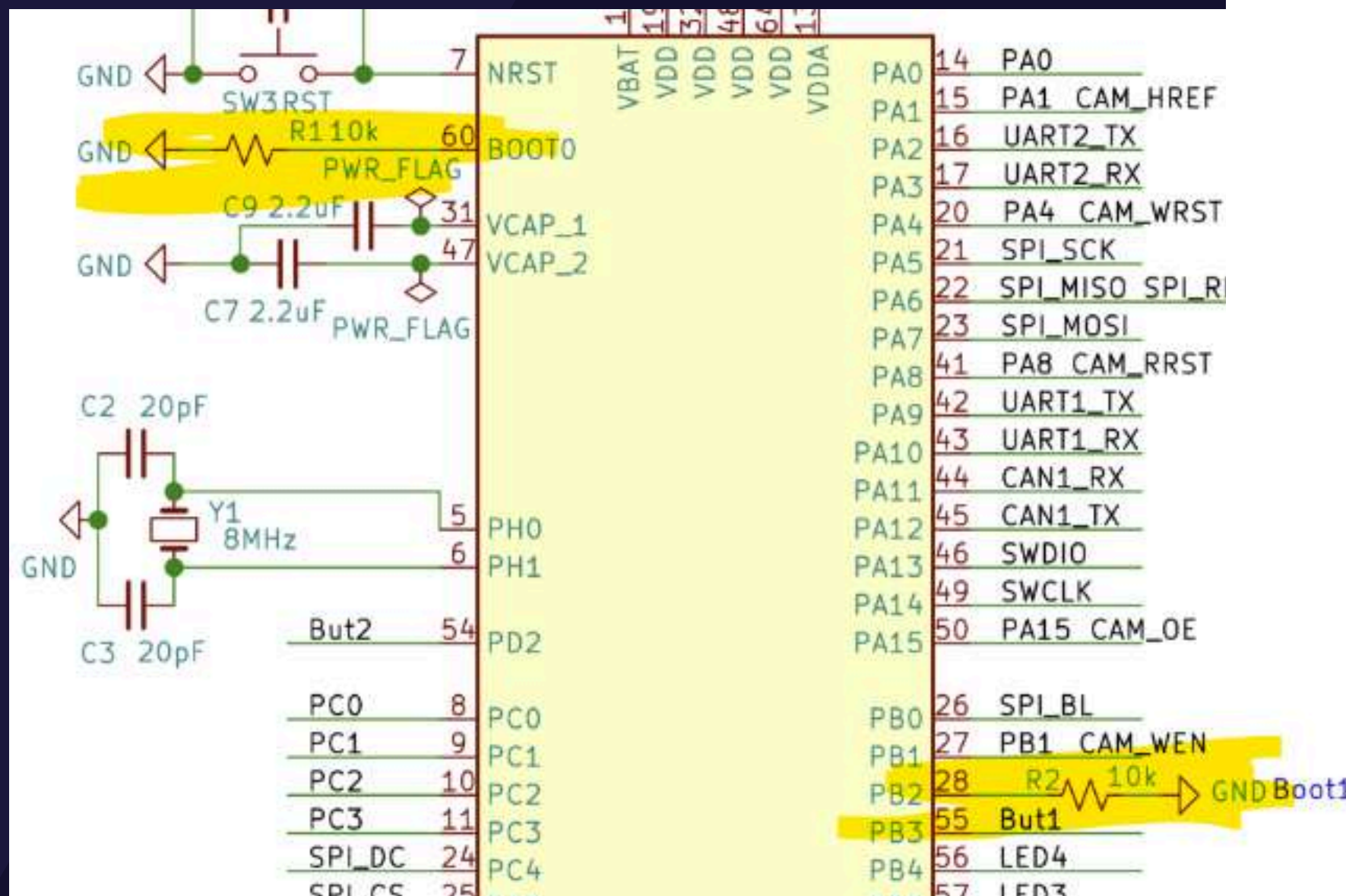
Boot configuration

Due to its fixed memory map, the code area starts from address 0x0000 0000 (accessed through the ICode/DCode buses) while the data area (SRAM) starts from address 0x2000 0000 (accessed through the system bus). The Cortex[®]-M4 with FPU CPU always fetches the reset vector on the ICode bus, which implies to have the boot space available only in the code area (typically, Flash memory). STM32F4xx microcontrollers implement a special mechanism to be able to boot from other memories (like the internal SRAM).

In the STM32F4xx, three different boot modes can be selected through the BOOT[1:0] pins as shown in [Table 2](#).

Table 2. Boot modes

Boot mode selection pins		Boot mode	Aliasing
BOOT1	BOOT0		
x	0	Main Flash memory	Main Flash memory is selected as the boot space
0	1	System memory	System memory is selected as the boot space
1	1	Embedded SRAM	Embedded SRAM is selected as the boot space



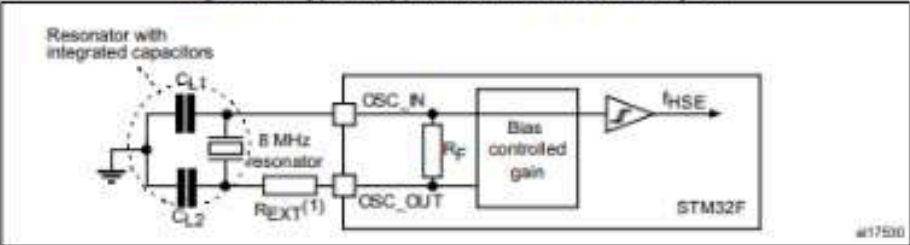
MCU - EXTERNAL CRYSTAL

Electrical characteristics

STM32F405xx, STM32F407xx

Note: For information on electing the crystal, refer to the application note AN2867 "Oscillator design guide for ST microcontrollers" available from the ST website www.st.com.

Figure 32. Typical application with an 8 MHz crystal



1. R_{EXT} value depends on the crystal characteristics.

Low-speed external clock generated from a crystal/ceramic resonator

The low-speed external (LSE) clock can be supplied with a 32.768 kHz crystal/ceramic resonator oscillator. All the information given in this paragraph are based on characterization results obtained with typical external components specified in Table 33. In the application, the resonator and the load capacitors have to be placed as close as possible to the oscillator pins in order to minimize output distortion and startup stabilization time. Refer to the crystal resonator manufacturer for more details on the resonator characteristics (frequency, package, accuracy).

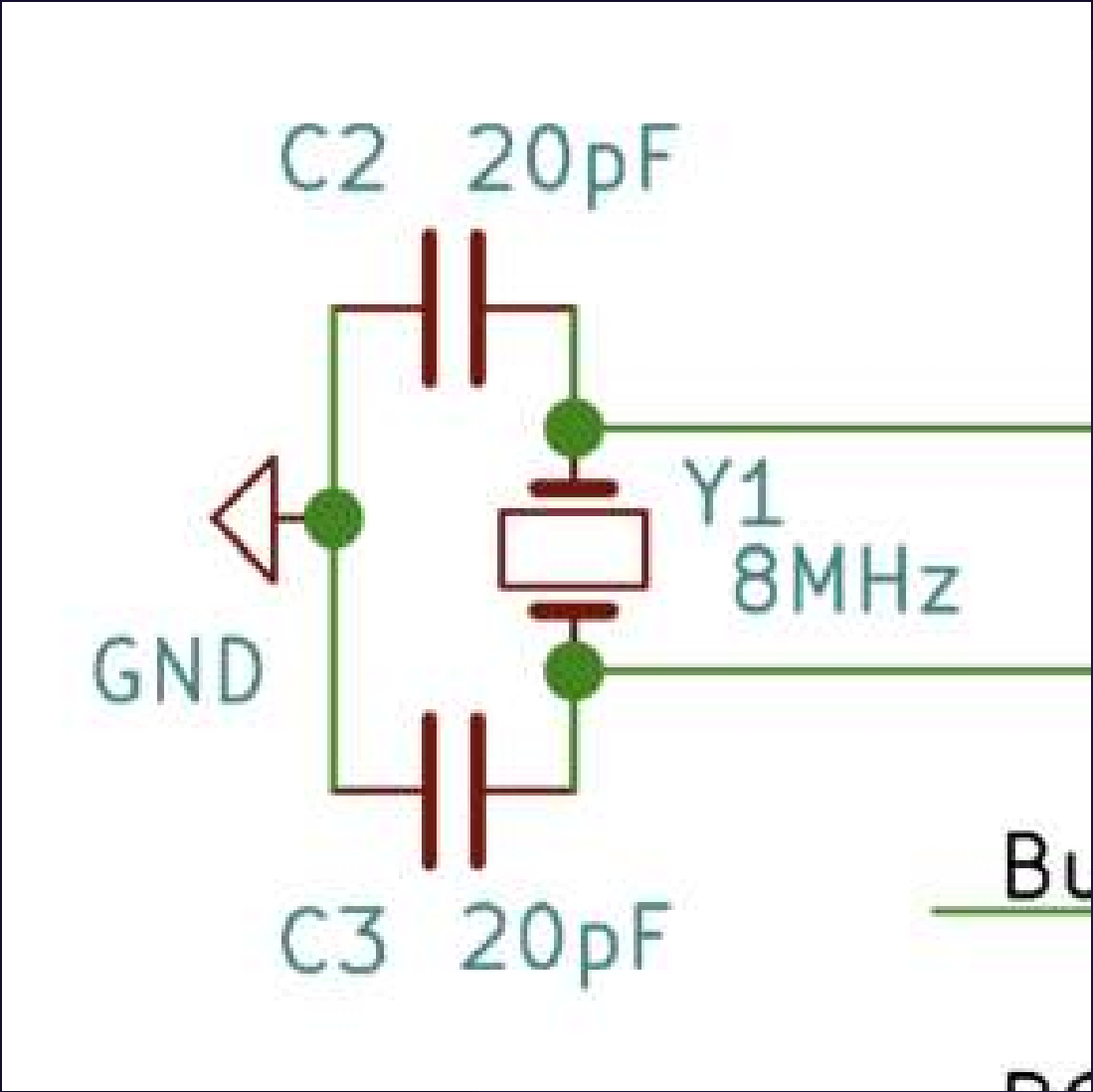
Table 33. LSE oscillator characteristics ($f_{LSE} = 32.768 \text{ kHz}$) ⁽¹⁾

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
f_{OSC_IN}	Oscillator frequency	-	-	32.768	-	MHz
R_F	Feedback resistor	-	-	18.4	-	M Ω
I_{DD}	LSE current consumption	-	-	-	1	μA
G_m	Oscillator transconductance	Startup	2.8	-	-	$\mu\text{A/V}$
$G_{m\text{critmax}}$	Maximum critical crystal G_m		-	-	0.56	
$t_{SU(LSE)}^{(2)}$	startup time	V_{DD} is stabilized	-	2	-	s

1. Guaranteed by design.

2. Guaranteed by characterization. $t_{SU(LSE)}$ is the startup time measured from the moment it is enabled (by software) to a stabilized 32.768 kHz oscillation is reached. This value is measured for a standard crystal resonator and it can vary significantly with the crystal manufacturer.

Note: For information on electing the crystal, refer to the application note AN2867 "Oscillator design guide for ST microcontrollers" available from the ST website www.st.com.



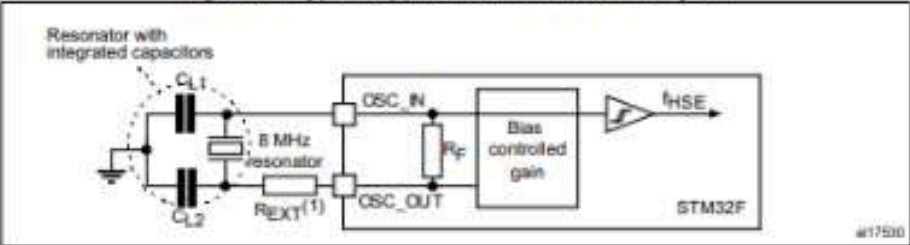
MCU - EXTERNAL CRYSTAL

Electrical characteristics

STM32F405xx, STM32F407xx

Note: For information on electing the crystal, refer to the application note AN2867 "Oscillator design guide for ST microcontrollers" available from the ST website www.st.com.

Figure 32. Typical application with an 8 MHz crystal



1. R_{EXT} value depends on the crystal characteristics.

Low-speed external clock generated from a crystal/ceramic resonator

The low-speed external (LSE) clock can be supplied with a 32.768 kHz crystal/ceramic resonator oscillator. All the information given in this paragraph are based on characterization results obtained with typical external components specified in Table 33. In the application, the resonator and the load capacitors have to be placed as close as possible to the oscillator pins in order to minimize output distortion and startup stabilization time. Refer to the crystal resonator manufacturer for more details on the resonator characteristics (frequency, package, accuracy).

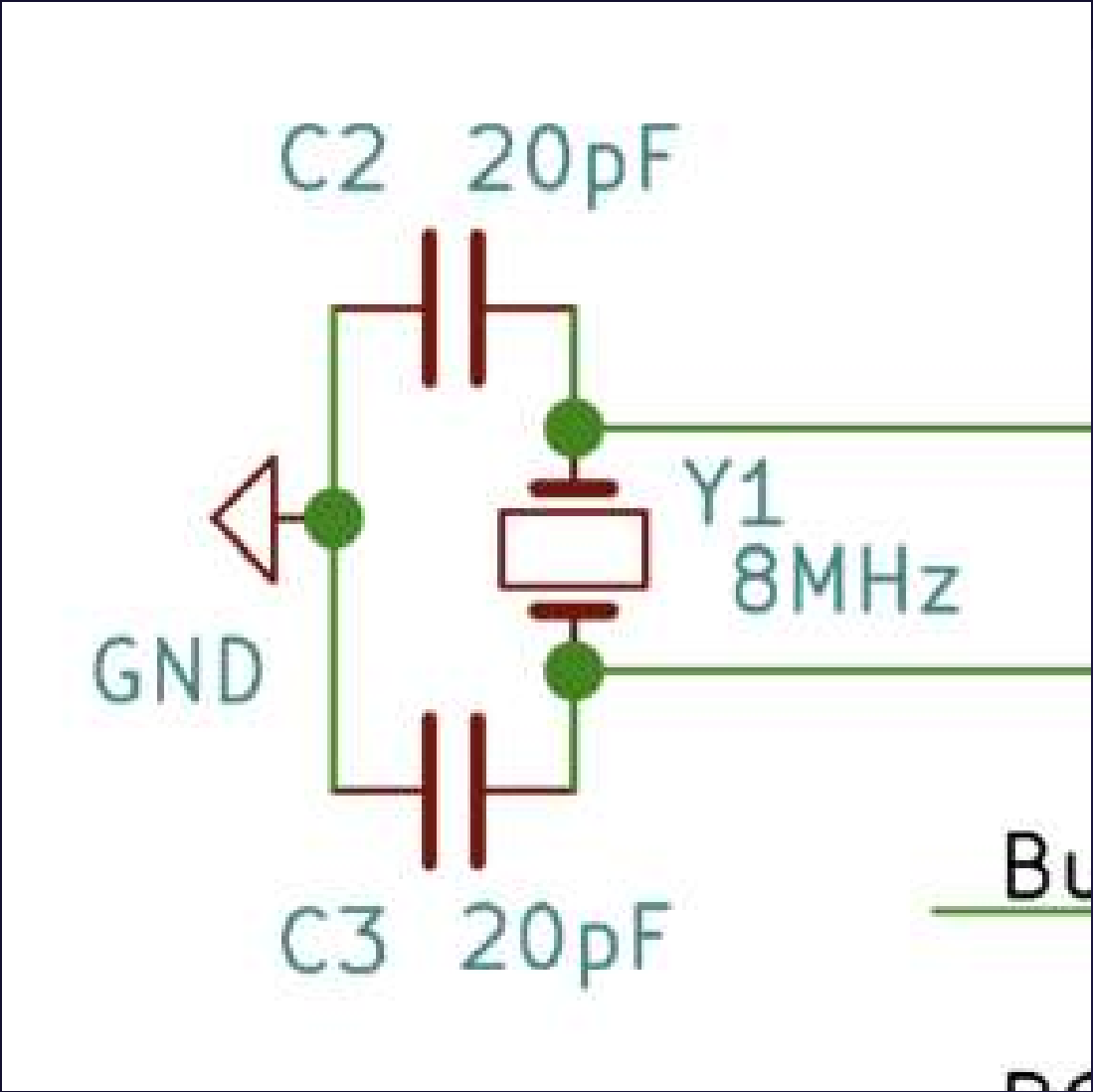
Table 33. LSE oscillator characteristics ($f_{LSE} = 32.768 \text{ kHz}$) ⁽¹⁾

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
f_{OSC_IN}	Oscillator frequency	-	-	32.768	-	MHz
R_F	Feedback resistor	-	-	18.4	-	k Ω
I_{DD}	LSE current consumption	-	-	-	1	μA
G_m	Oscillator transconductance	Startup	2.8	-	-	$\mu\text{A/V}$
$G_{m\text{critmax}}$	Maximum critical crystal G_m		-	-	0.56	
$t_{SU(LSE)}^{(2)}$	startup time	V_{DD} is stabilized	-	2	-	s

1. Guaranteed by design.

2. Guaranteed by characterization. $t_{SU(LSE)}$ is the startup time measured from the moment it is enabled (by software) to a stabilized 32.768 kHz oscillation is reached. This value is measured for a standard crystal resonator and it can vary significantly with the crystal manufacturer.

Note: For information on electing the crystal, refer to the application note AN2867 "Oscillator design guide for ST microcontrollers" available from the ST website www.st.com.

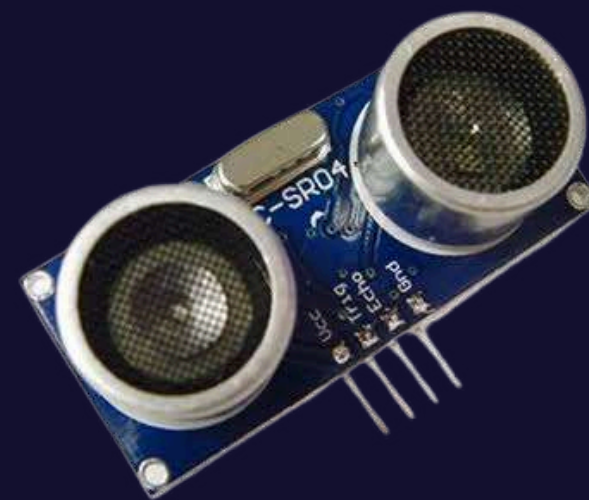


DESIGN ³ OVERVIEW

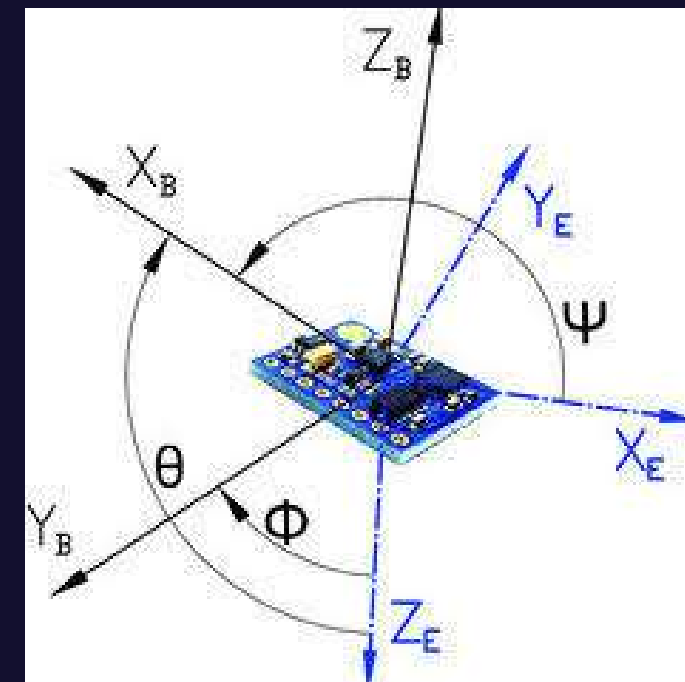
HOW TO SENSE/INDICATE/WRITE? - MODULES



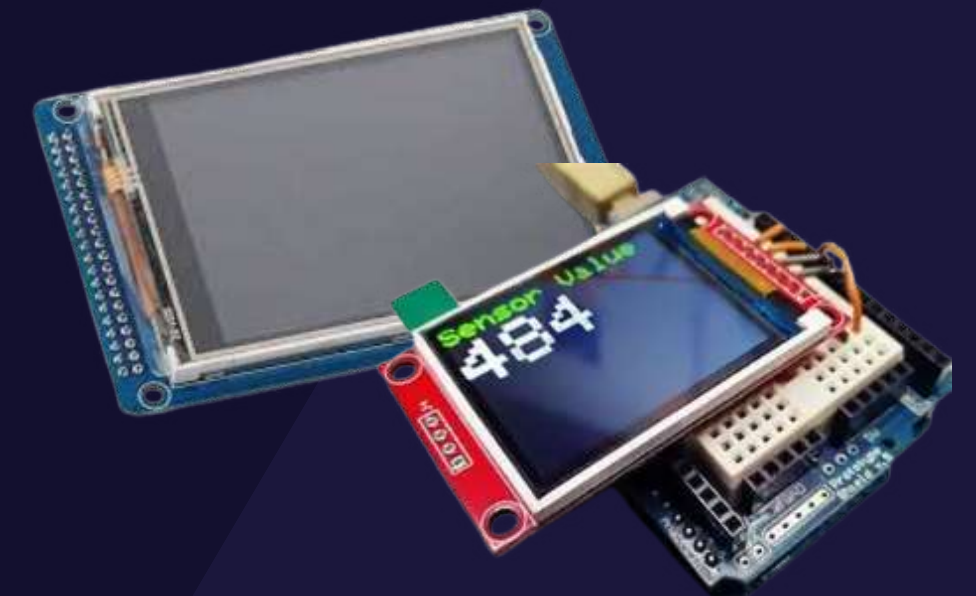
ST-Link
- SWD
(Flash code into MCU)



Ultrasonic Sensor
- UART/PWM



Inertial Measurement Unit (IMU)
- SPI/I2C/UART



TFT Monitors
- SPI/I2C

-are communication protocols which will be covered later

XXXX

Usually powered by 5V / 3.3V

HOW TO MOVE? - ACTUATORS



Motors
- PWM/CAN



Pneumatics
- IO



Usually powered by $>5V$, e.g. 12V/24V

SYSTEM POWER INPUT

Battery



- Considering factors when choosing batteries
 - Voltage rating
 - Current rating
 - Capacity
 - Stability



Power Supply



DESIGN ³ OVERVIEW

Battery

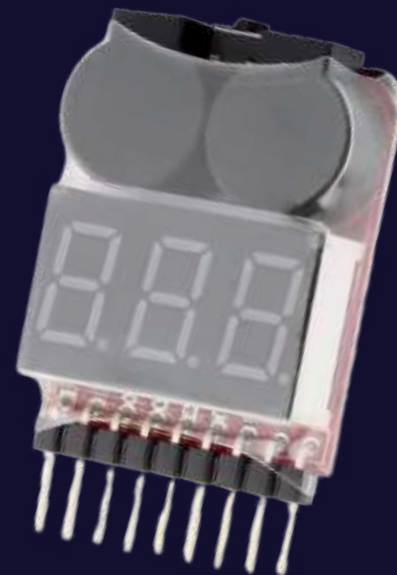


3s battery = 12V
6s battery = 24V
One cell = 3.8-4.2V

24V(6s) battery is used in RDC

The battery provides power to both mainboard and motors

Beeper



Check the voltage of the battery
Please go to charge if you see the battery is lower than 12.0V (for 3 cells) / 24.0V (for 6 cells)

Emergency Stop Button



Aim to switch off the circuit immediately when you see something wrong, such as explosion

Wireless E-stop



Allow people to shut down the robot remotely



MISCELLANEOUS

DESIGN ³
OVERVIEW

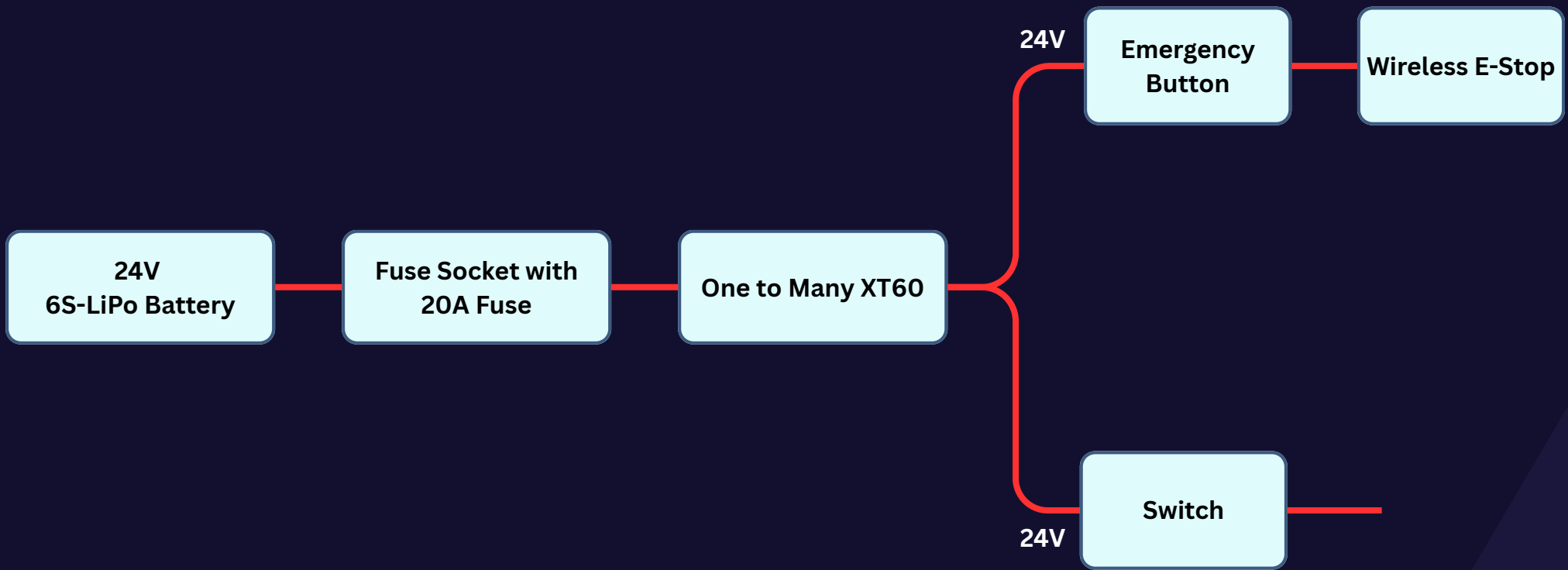
Fuse Socket



XT60 Headers



Switch



***You will learn more about safe power routing in RDC**



HOW???

1. How to choose an MCU?
2. How to choose sensors?
3. How to choose motors?
4. How to select power input specifications?
5. How to have 24V/12V/5V/3.3V at the same time?
6. How to let the MCU control/communicate with all the components?
7. Other concerns?



HOW....

1. How to choose an MCU?

- Ask & discuss with software
 - Depends on computational speed, specific sensors they want, and etc etc.

2. How to choose sensors?

- Ask & discuss with software & mech - depends on an agreed game rule approach

3. How to choose motors?

- Ask MECH
 - Need motors with specific strength
- Ask Software
 - Need to decide how to get feedback from motors

HOW....

1. How to select power input specifications?
2. How to have 24V/12V/5V/3.3V at the same time?
3. How to let the MCU control/communicate with all the components?
4. Other concerns?
 - Thermal breakdown | Noise → Inaccurate data

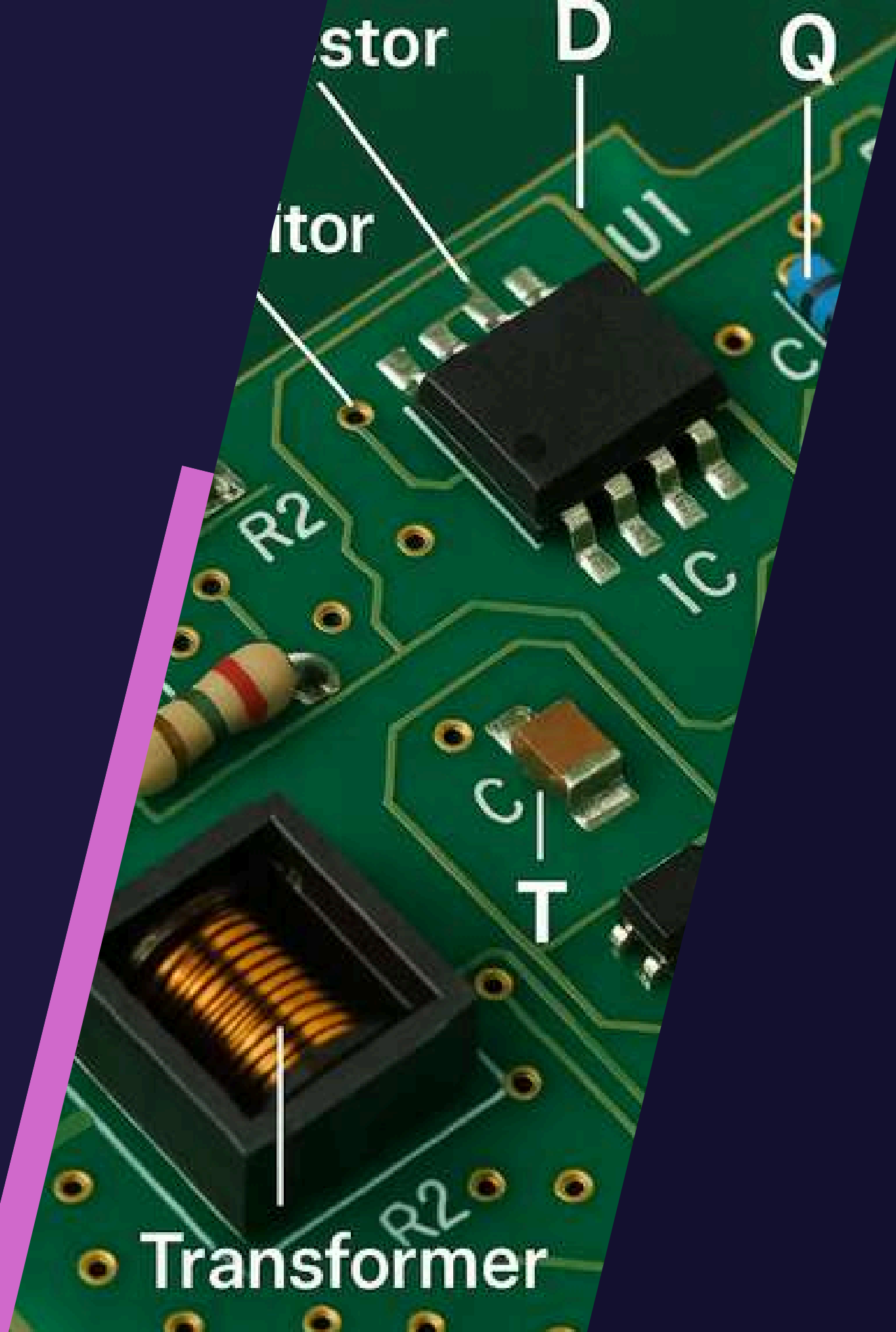
Depends on how WE design the PCB & route the robot

XXXX

ITEM 4

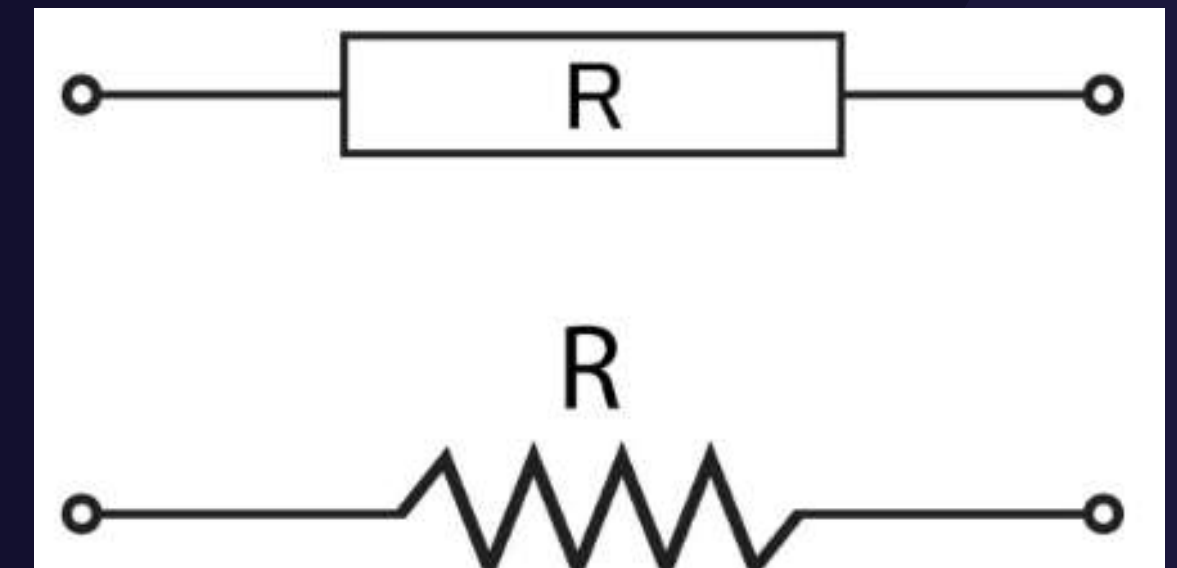
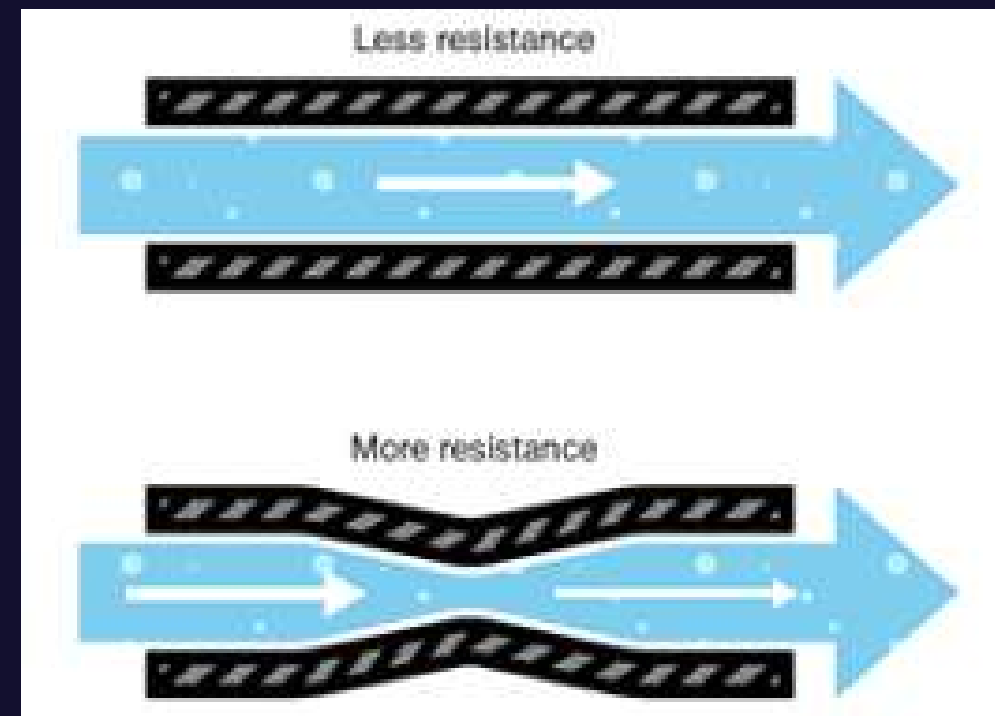
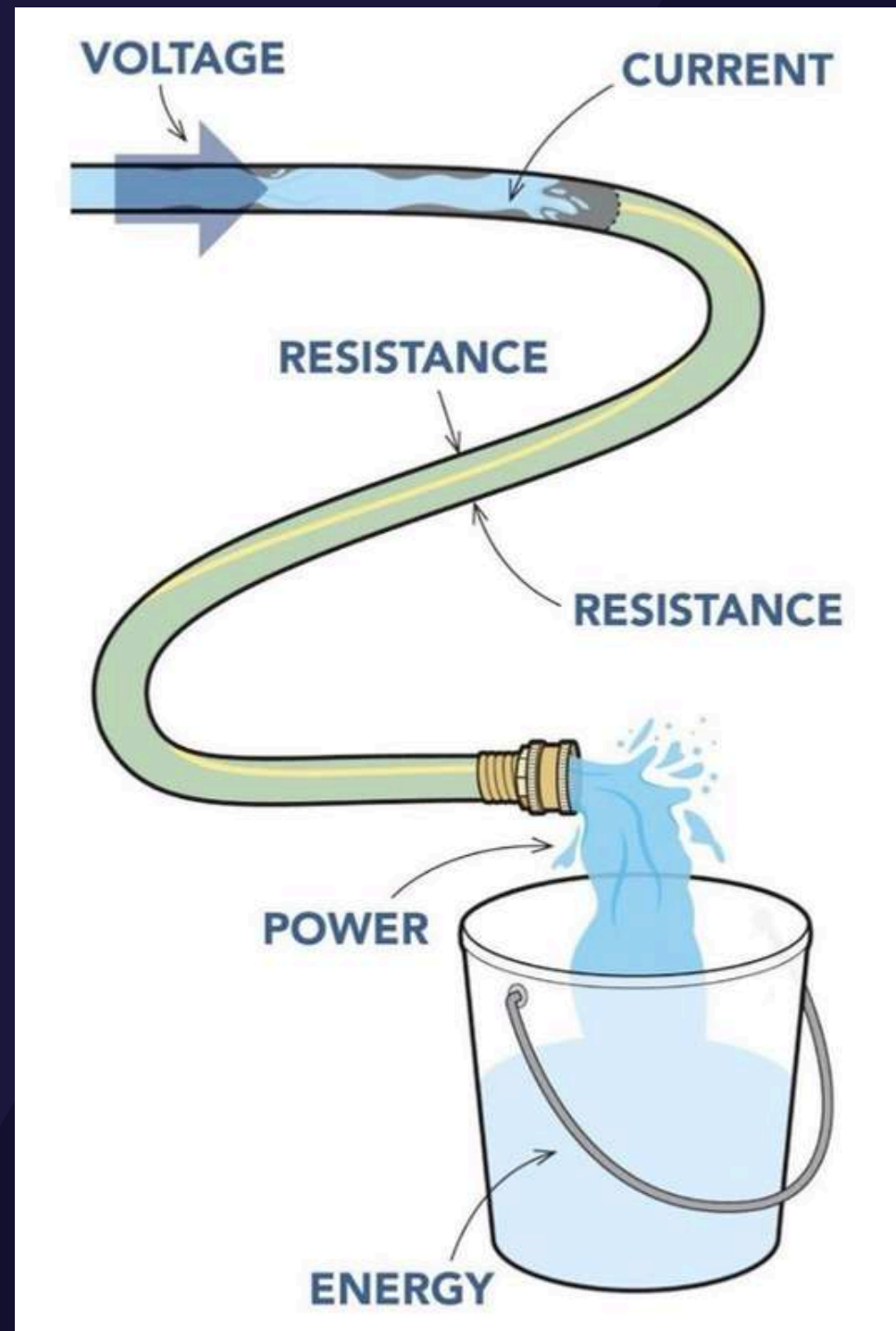
BASICS OF A CIRCUIT (PASSIVE COMPONENTS)

XXXX



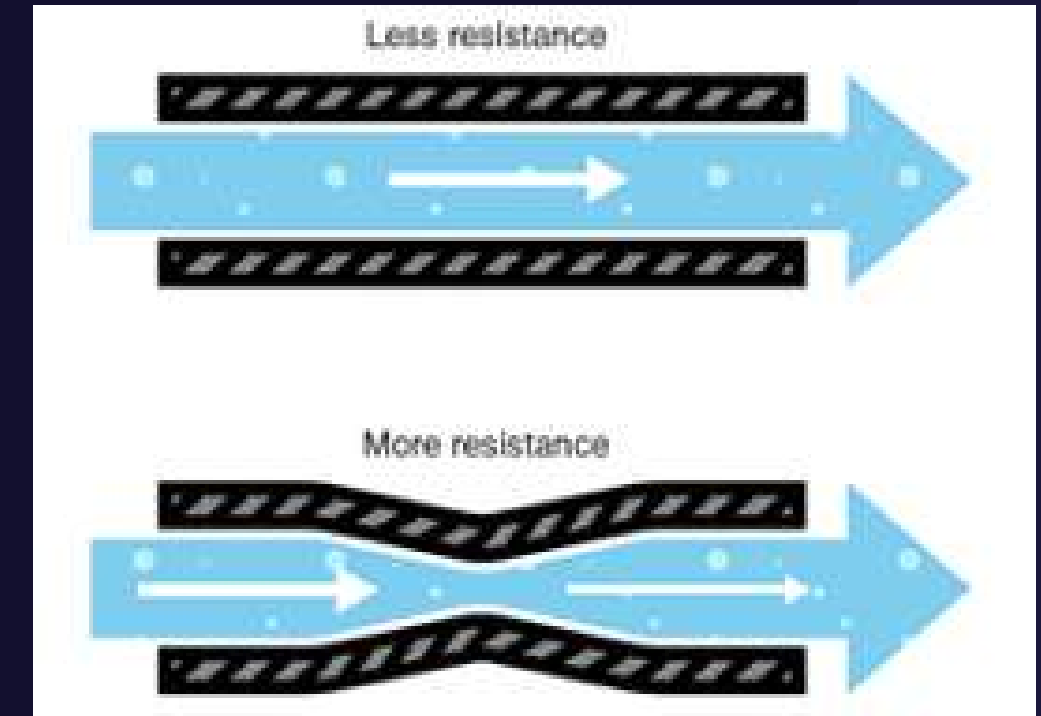
BASICS⁴ OF CIRCUIT

VOLTAGE, CURRENT, RESISTANCE



VOLTAGE, CURRENT, RESISTANCE

- Voltage (V, volts) $\approx$ water pressure
 - The “electrical push” or pressure that makes charge want to move between two points.
- Current (I, amperes or amps) $\approx$ flow rate of water
 - how much charge is actually flowing per second; the flow rate of electricity.
- Resistance (R, ohms, Ω) $\approx$ how narrow or rough the pipe is
 - how much something resists that flow; it makes it harder for current to pass.

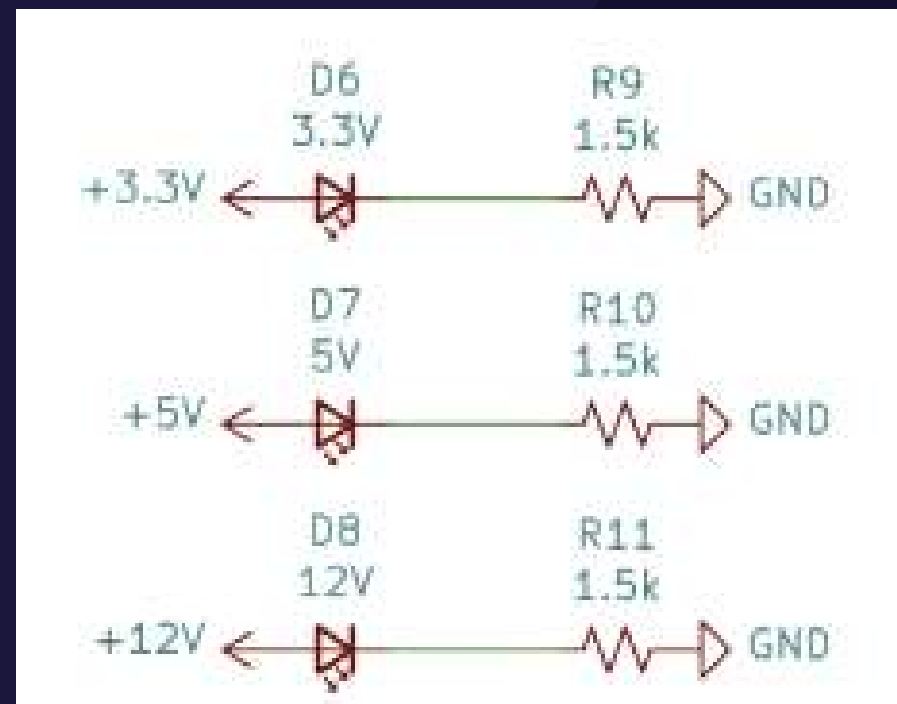


$$V = IR$$

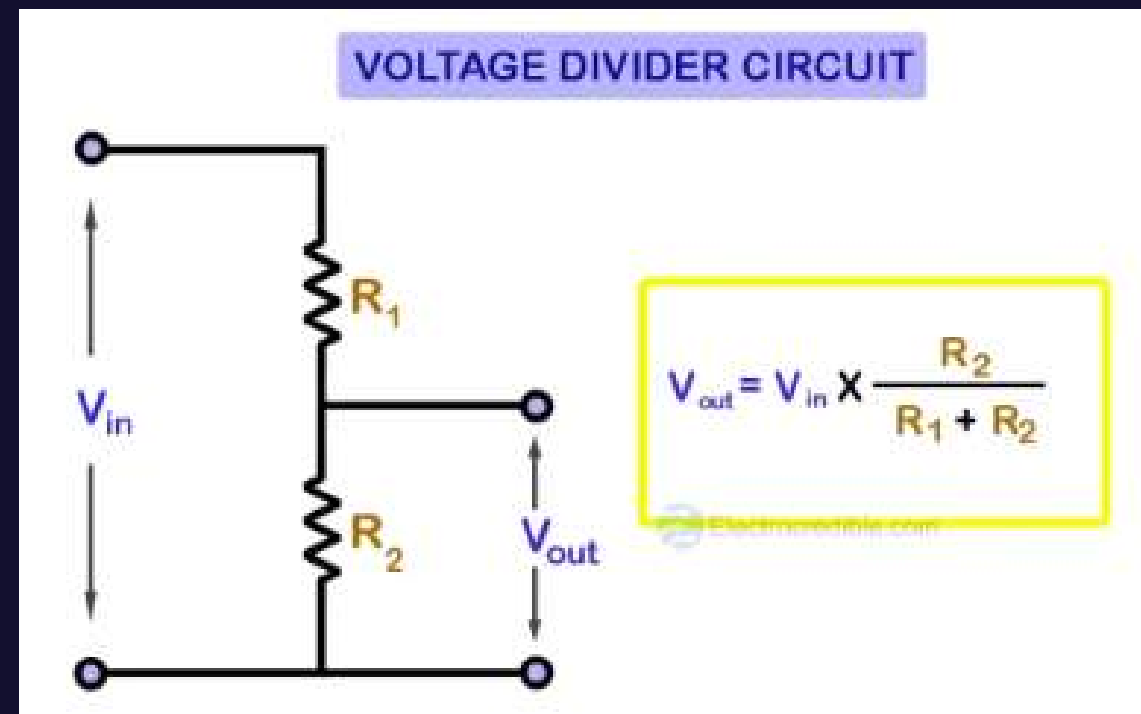
(Your favourite)

BASICS⁴ OF CIRCUIT

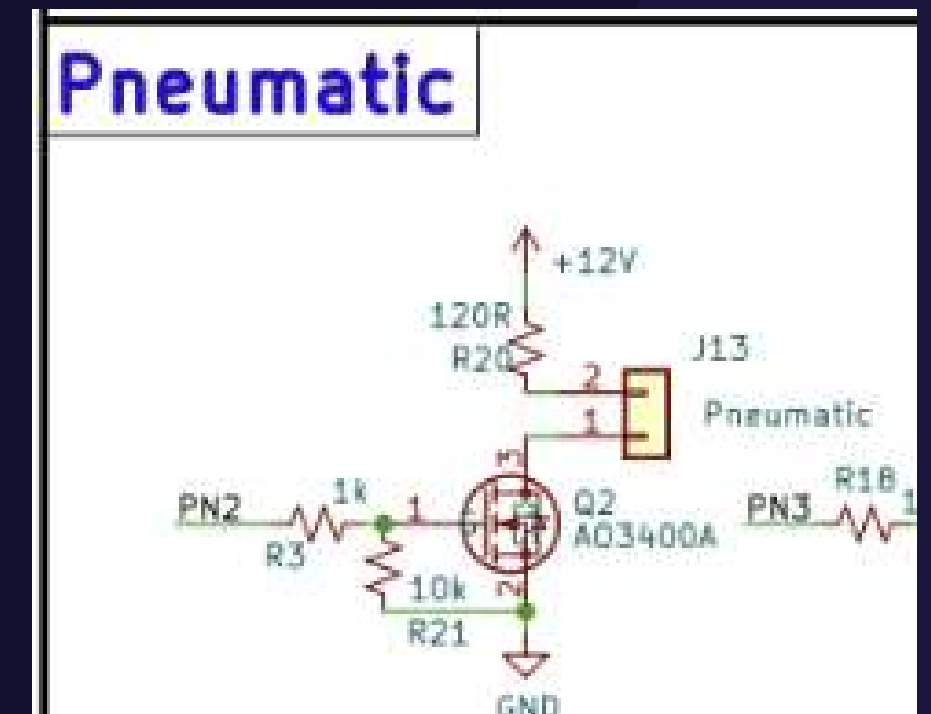
RESISTOR - APPLICATION



Current Limiting

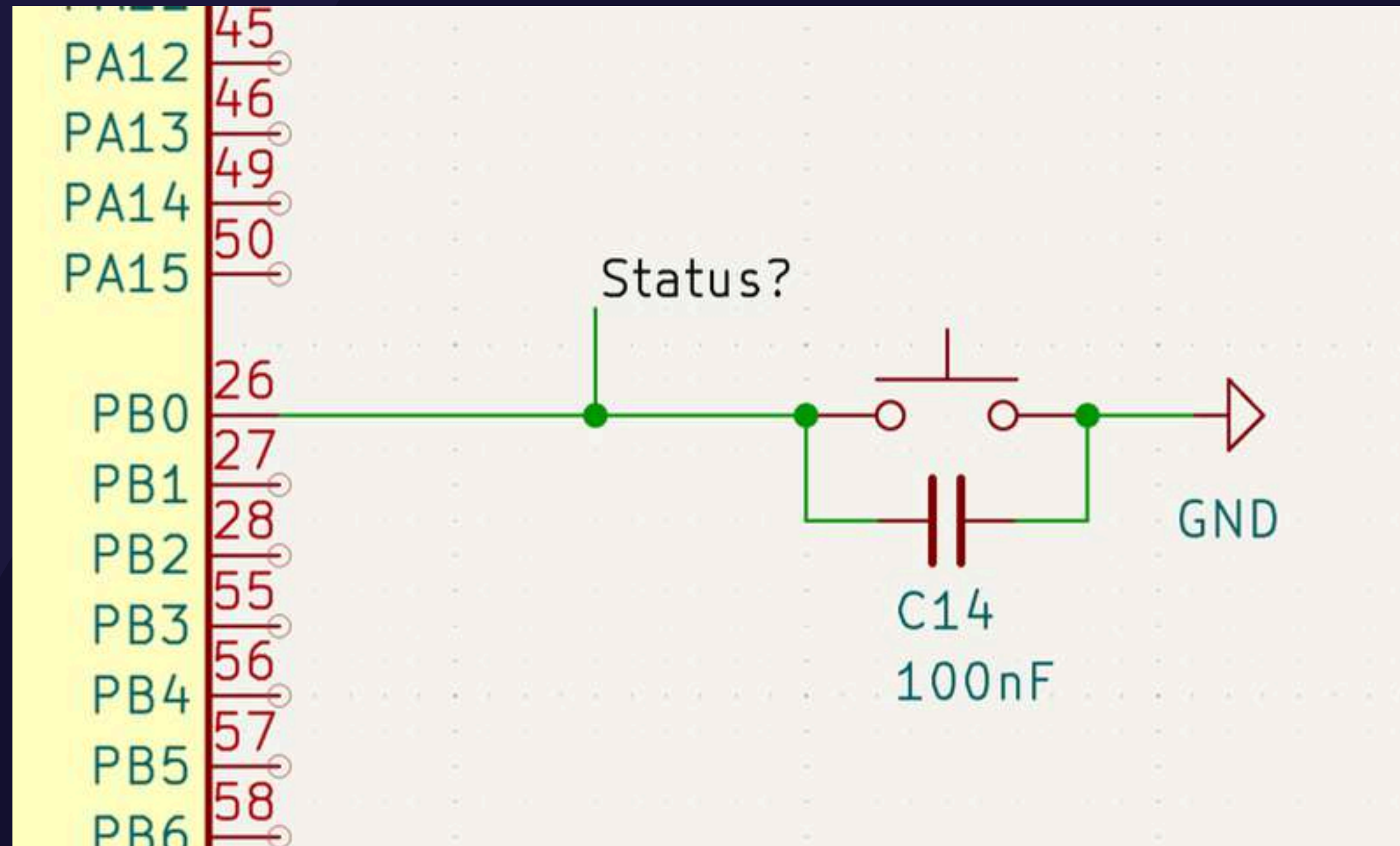


Voltage Divider



Pull up/down*

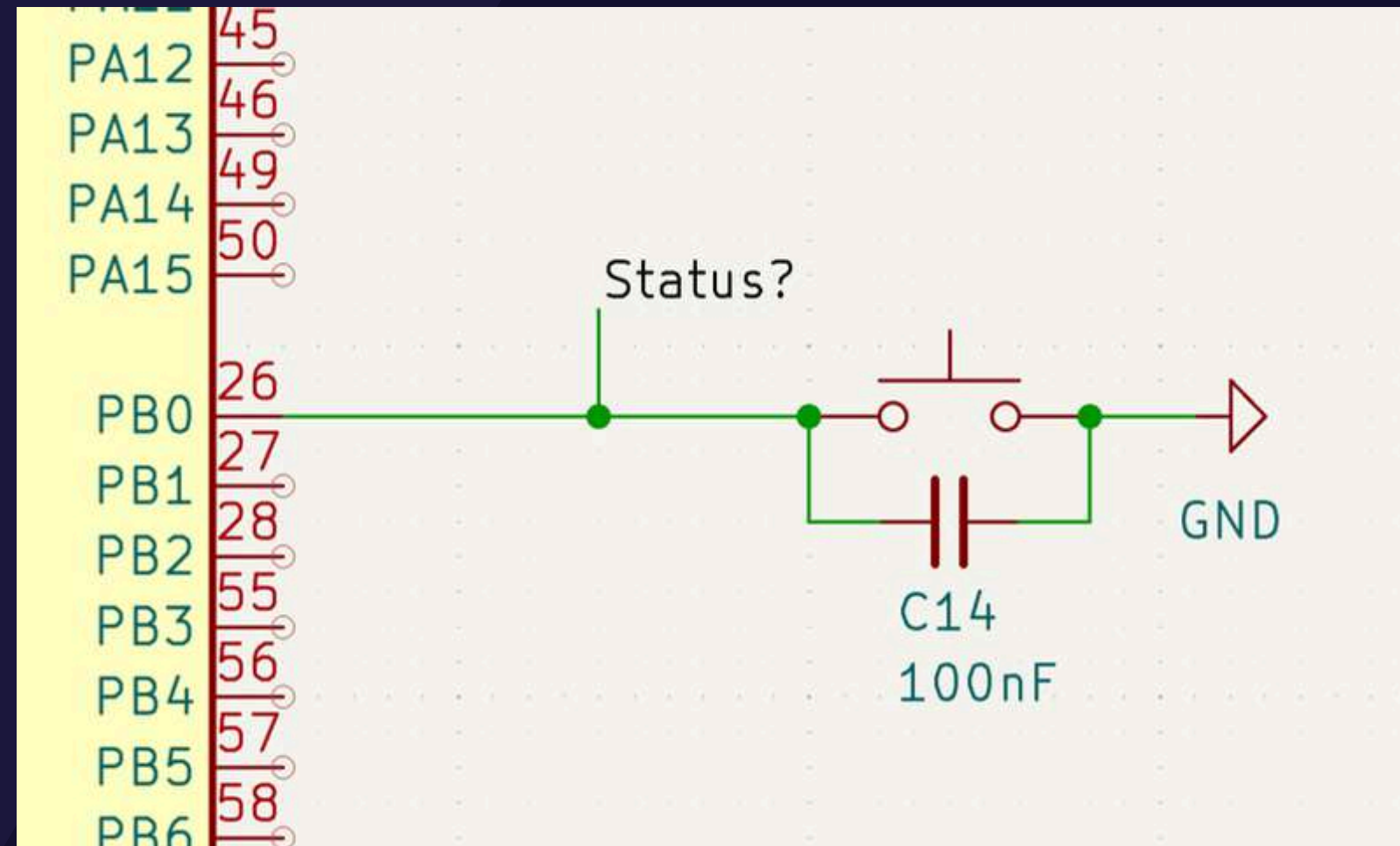
PULL-UP/DOWN RESISTORS - INCENTIVE



After pressing: LOW

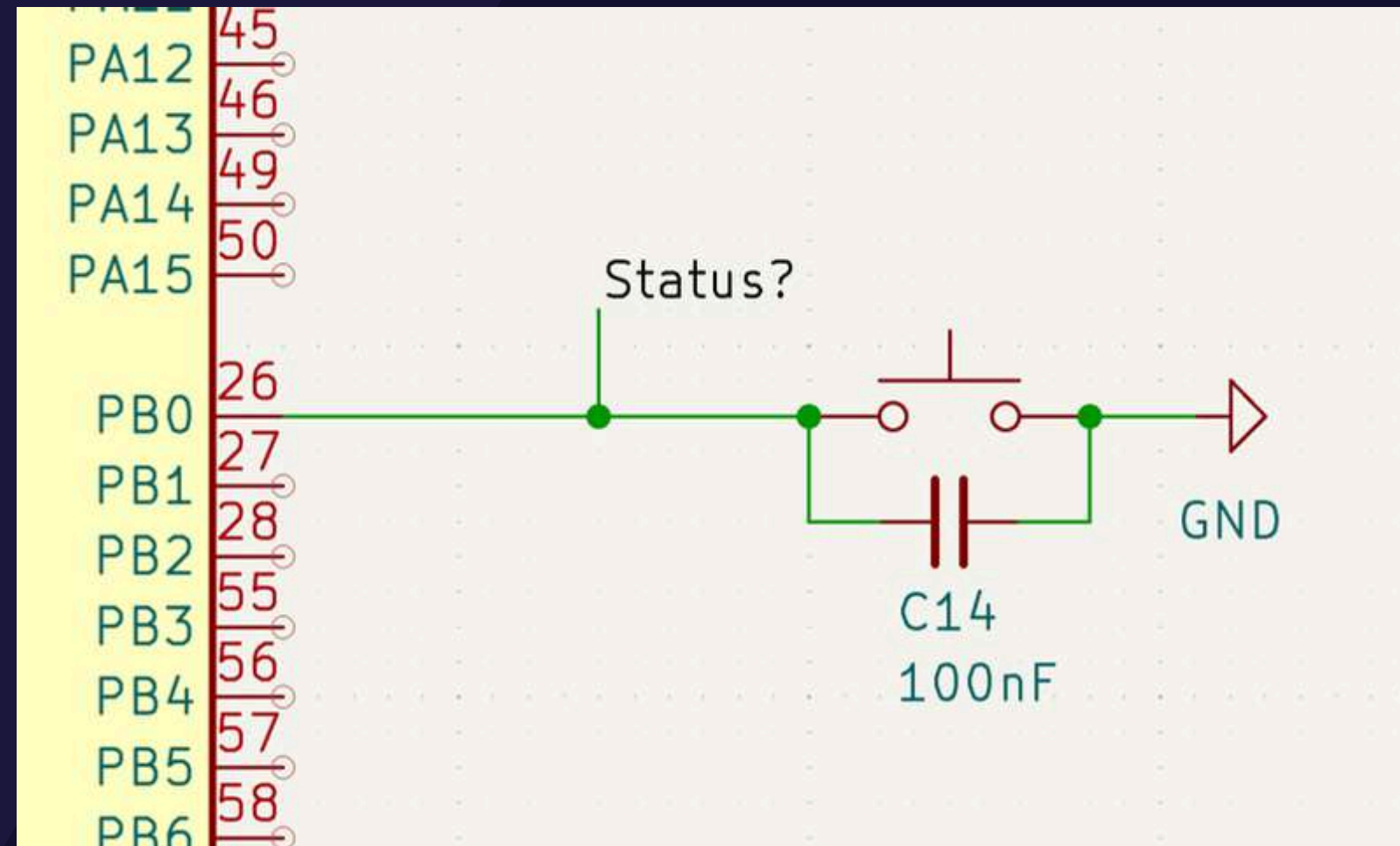
but is “status” high / low before pressing the button?

PULL-UP/DOWN RESISTORS - INCENTIVE



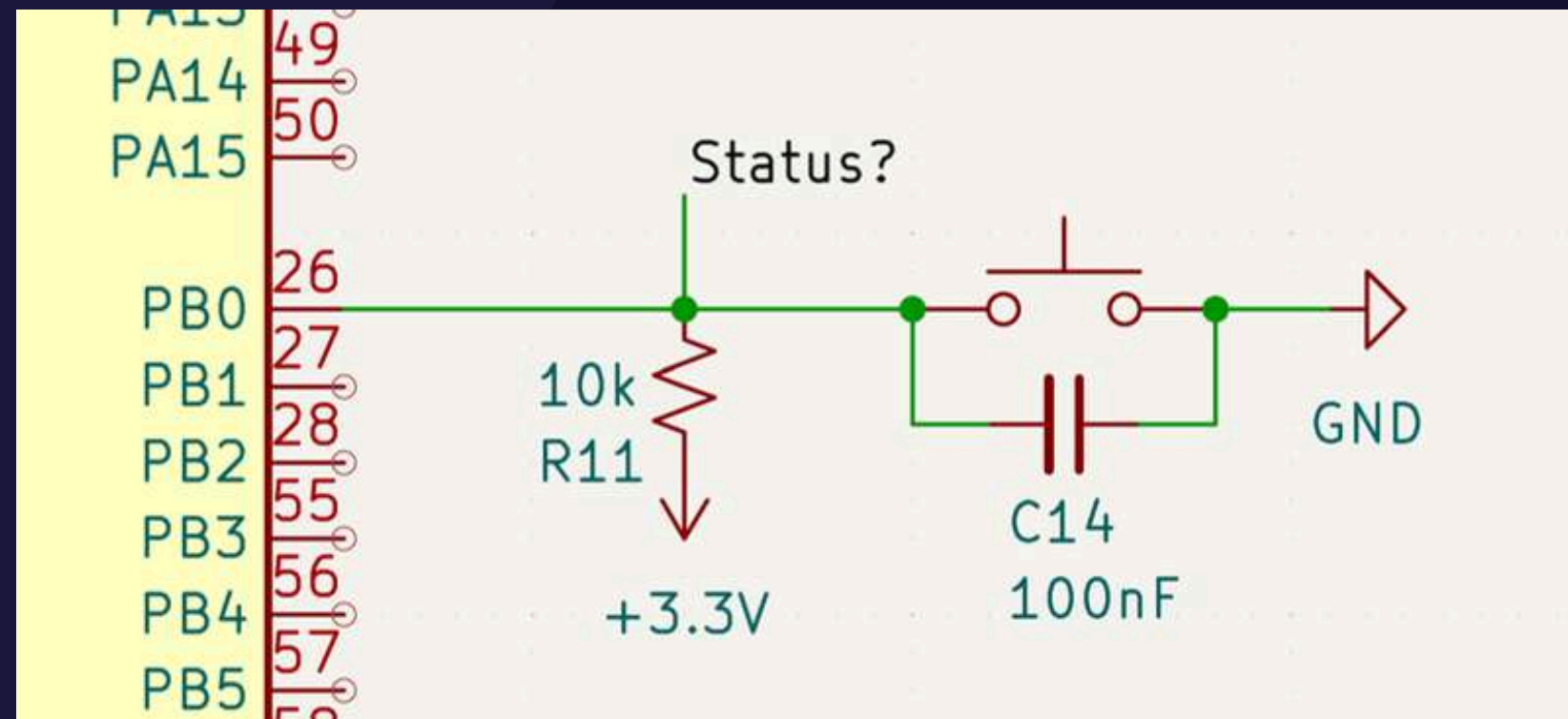
FLOATING: UNCLEAR HIGH / LOW

PULL-UP/DOWN RESISTORS - INCENTIVE



SOLUTION:
PROVIDE REFERENCE WHEN THE POINT IS UNCONNECTED

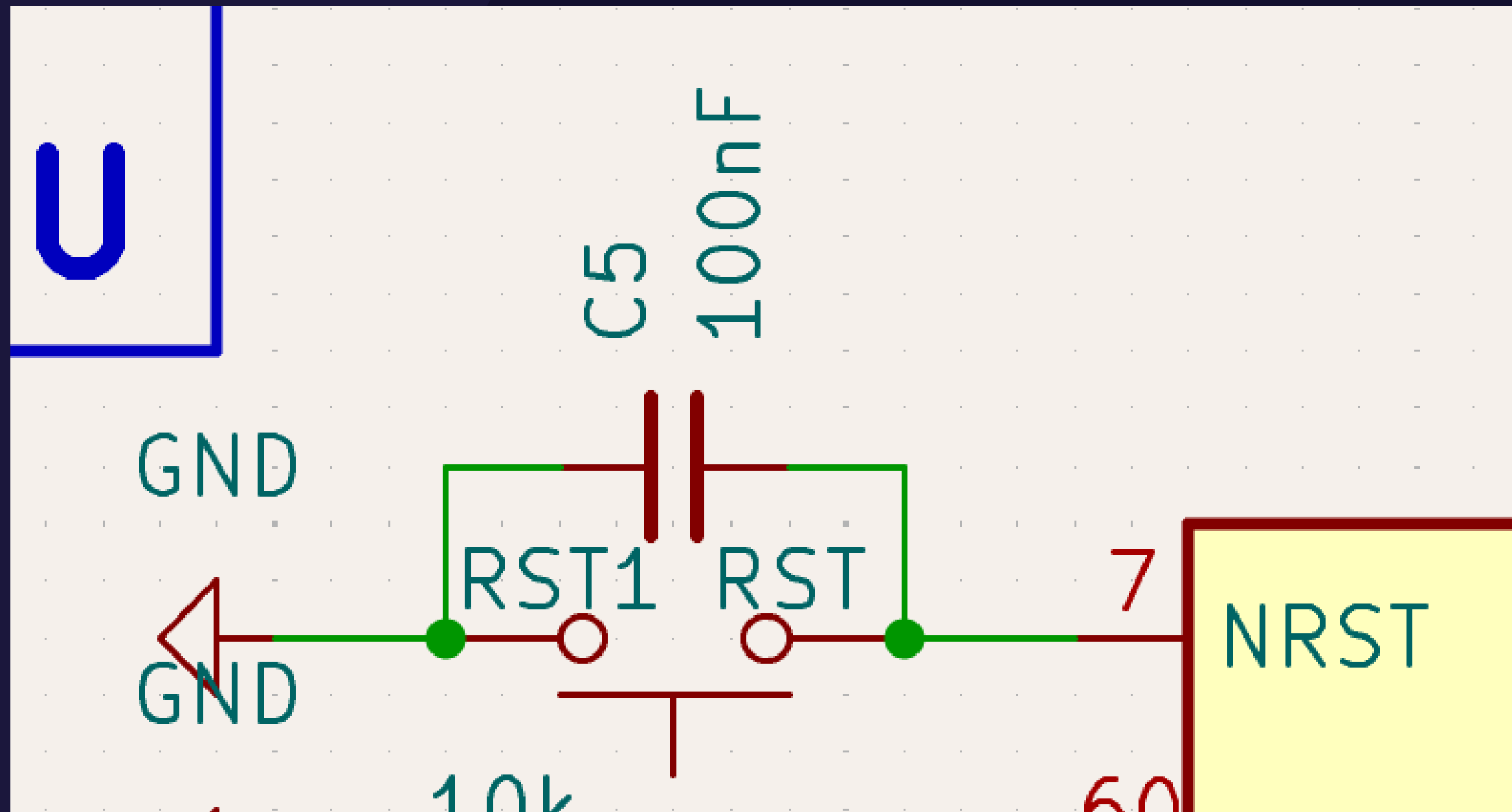
PULL-UP/DOWN RESISTORS - INCENTIVE



SOLUTION:
PROVIDE REFERENCE WHEN THE POINT IS UNCONNECTED

“status” = HIGH when the button is not pressed?

PULL-UP/DOWN RESISTORS - INCENTIVE



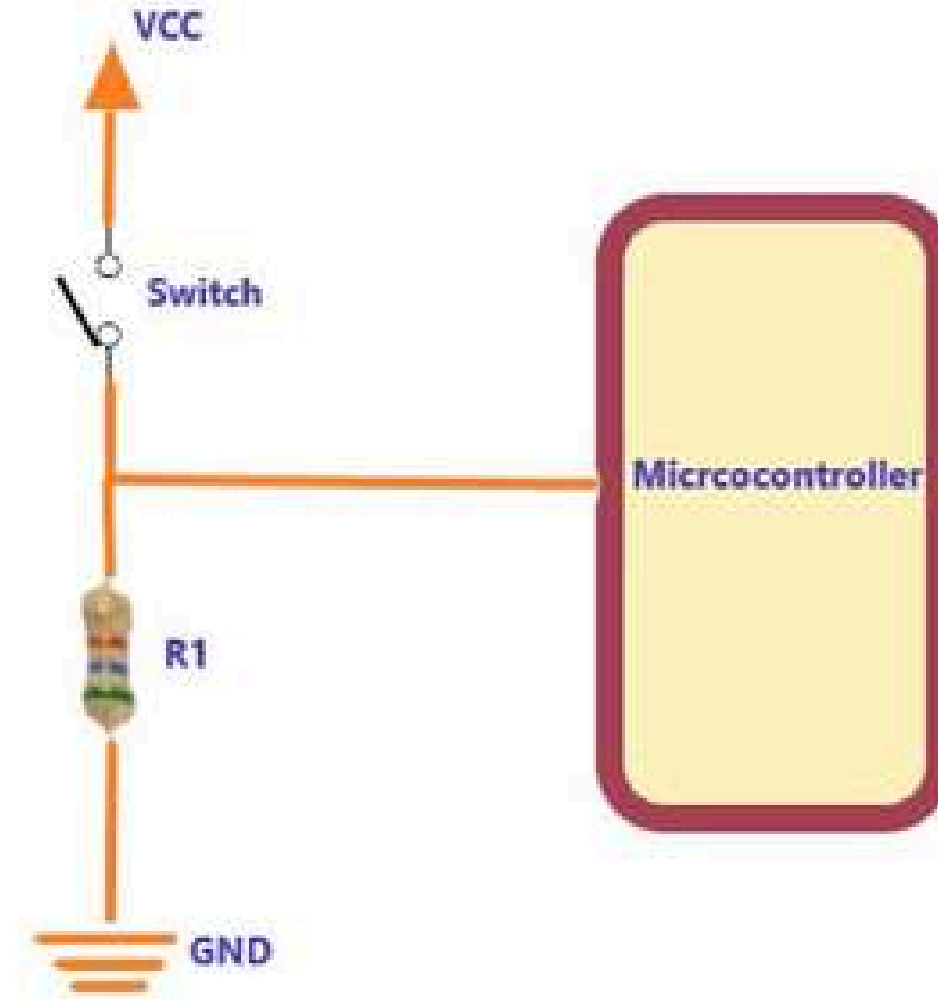
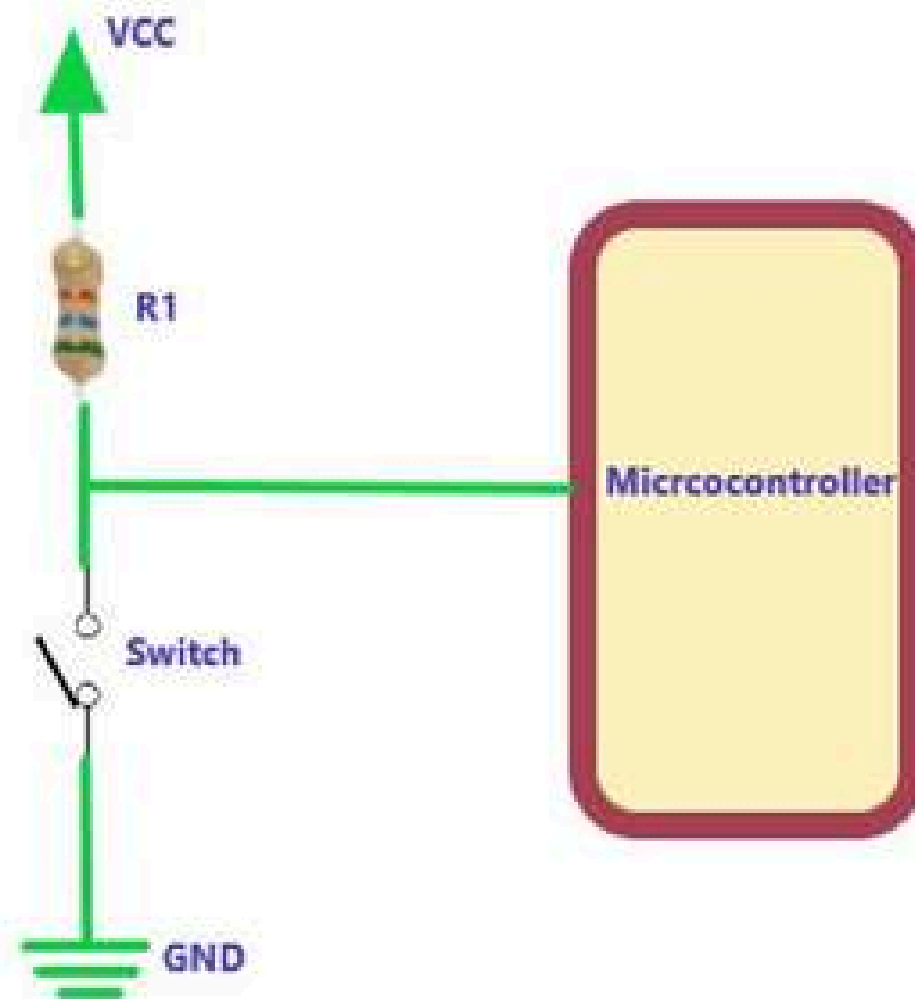
Some MCU pins have internal
pull-ups / downs

Read datasheet of the MCU / ask your software

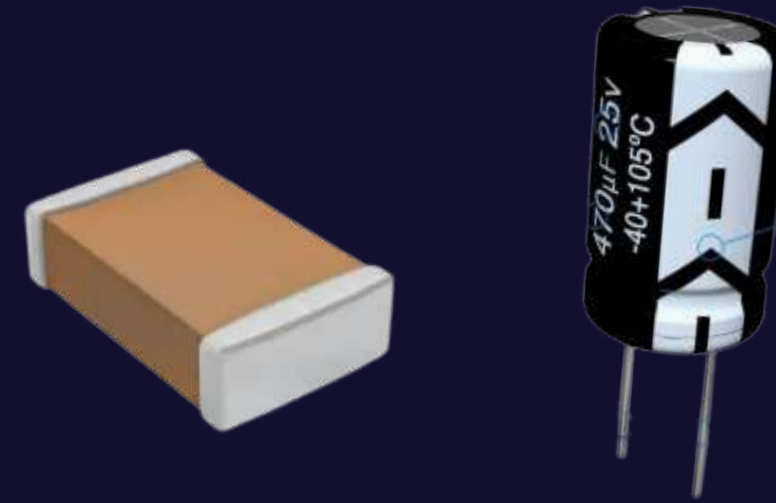
XXXX

****SUPPLEMENTARY****

Pull-up Pull-down Resistors



CAPACITOR



- Capacitor

- passive component that ****stores electrical energy**** in an electric field and releases it quickly when needed.

- Typical Applications:

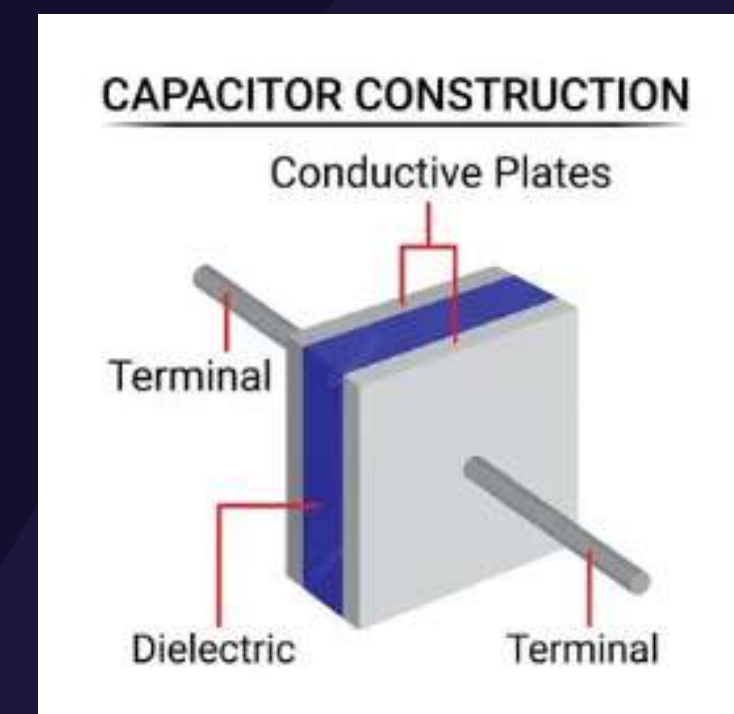
- Smoothing/filtering: reduce voltage ripple in power supplies.
- Decoupling: provide local, fast energy to ICs and suppress noise.
- Timing: with resistors, set delays and frequencies (RC circuits).
- Coupling: pass AC signals while blocking DC between stages

Capacitance = Charge/ Voltage

Current = (Capacitance) * (rate of change of voltage)

→ Opposing voltage change

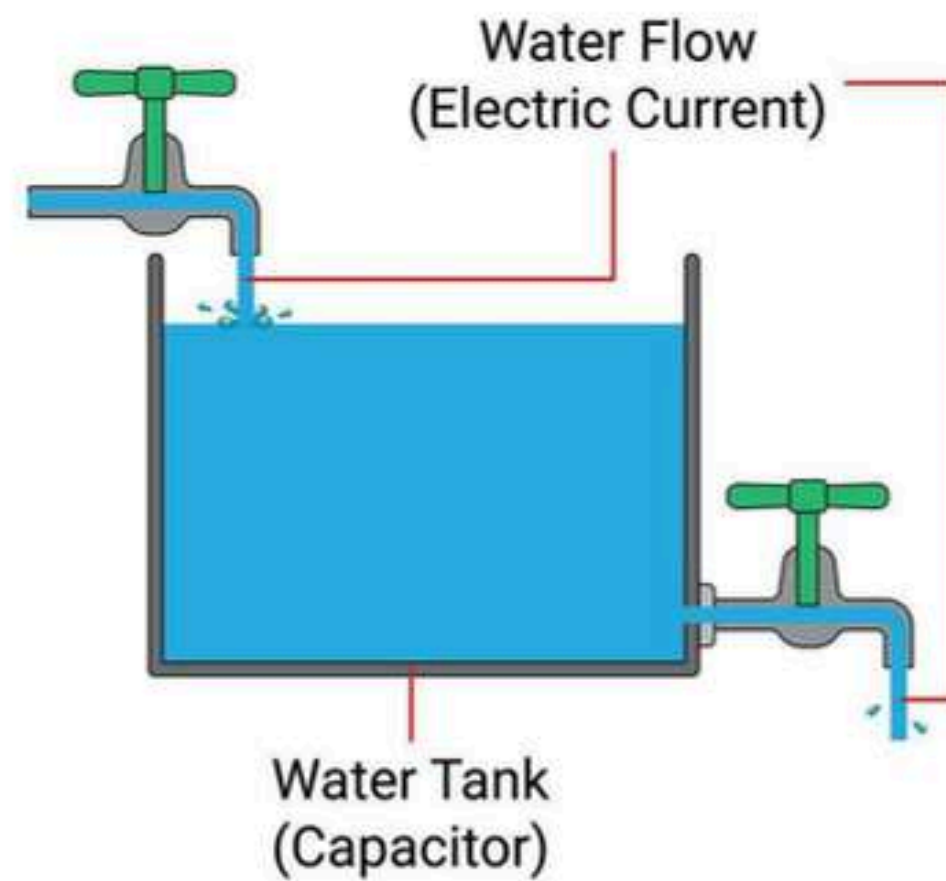
Energy Stored = $0.5 * C * V^2$



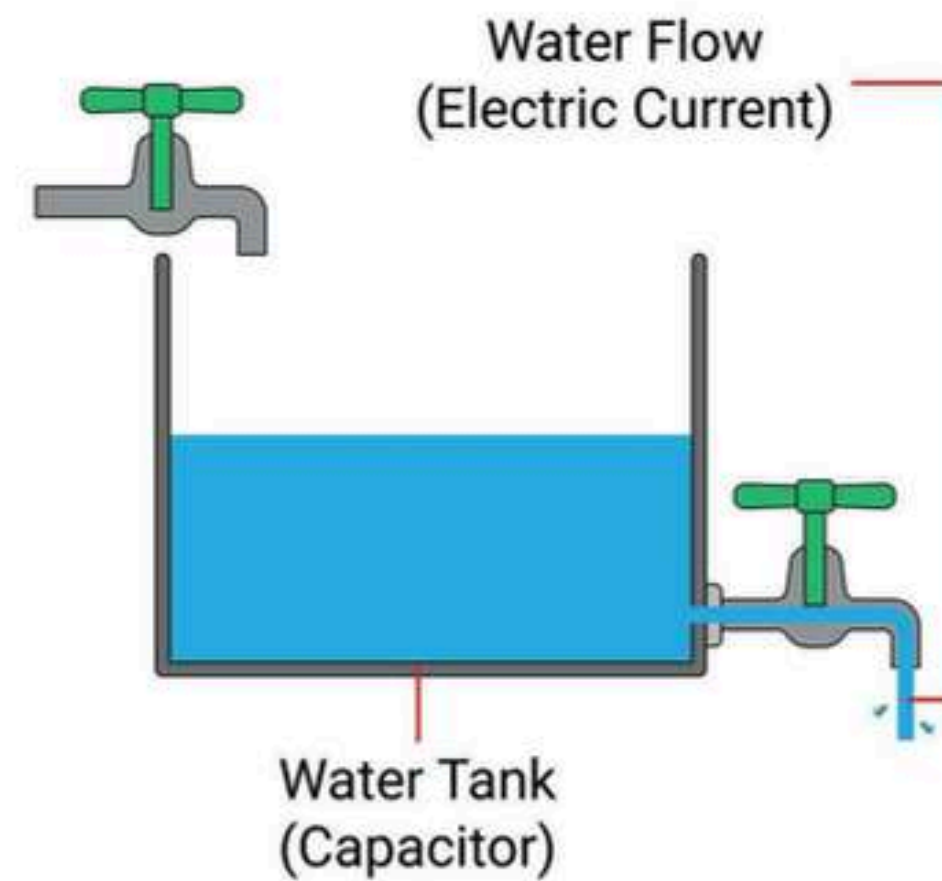
CAPACITOR

Water Tank Analogy for a Capacitor

Step 1



Step 2



CAPACITOR - APPLICATION 1 - TEMP ENERGY

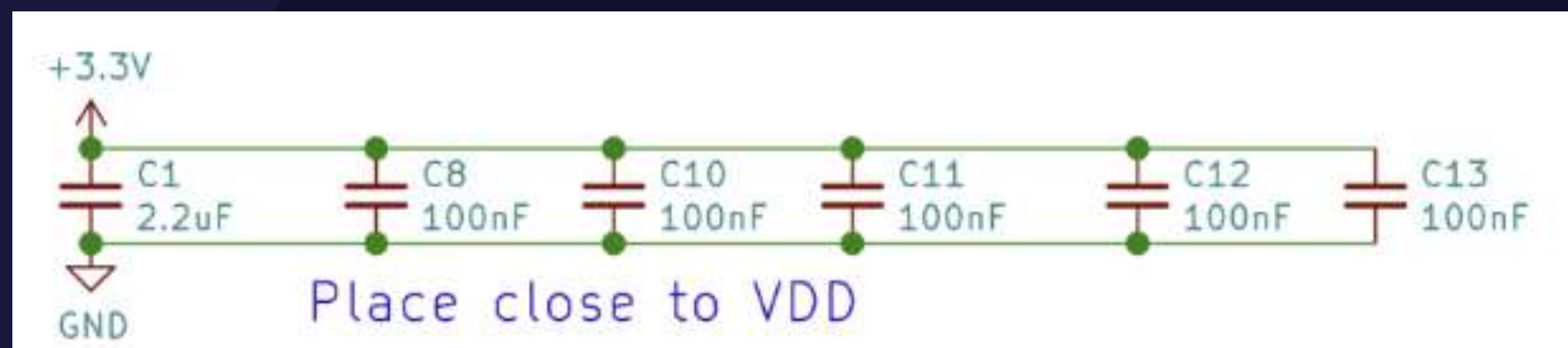
Energy Storage - temporary battery

Prevent voltage fluctuations

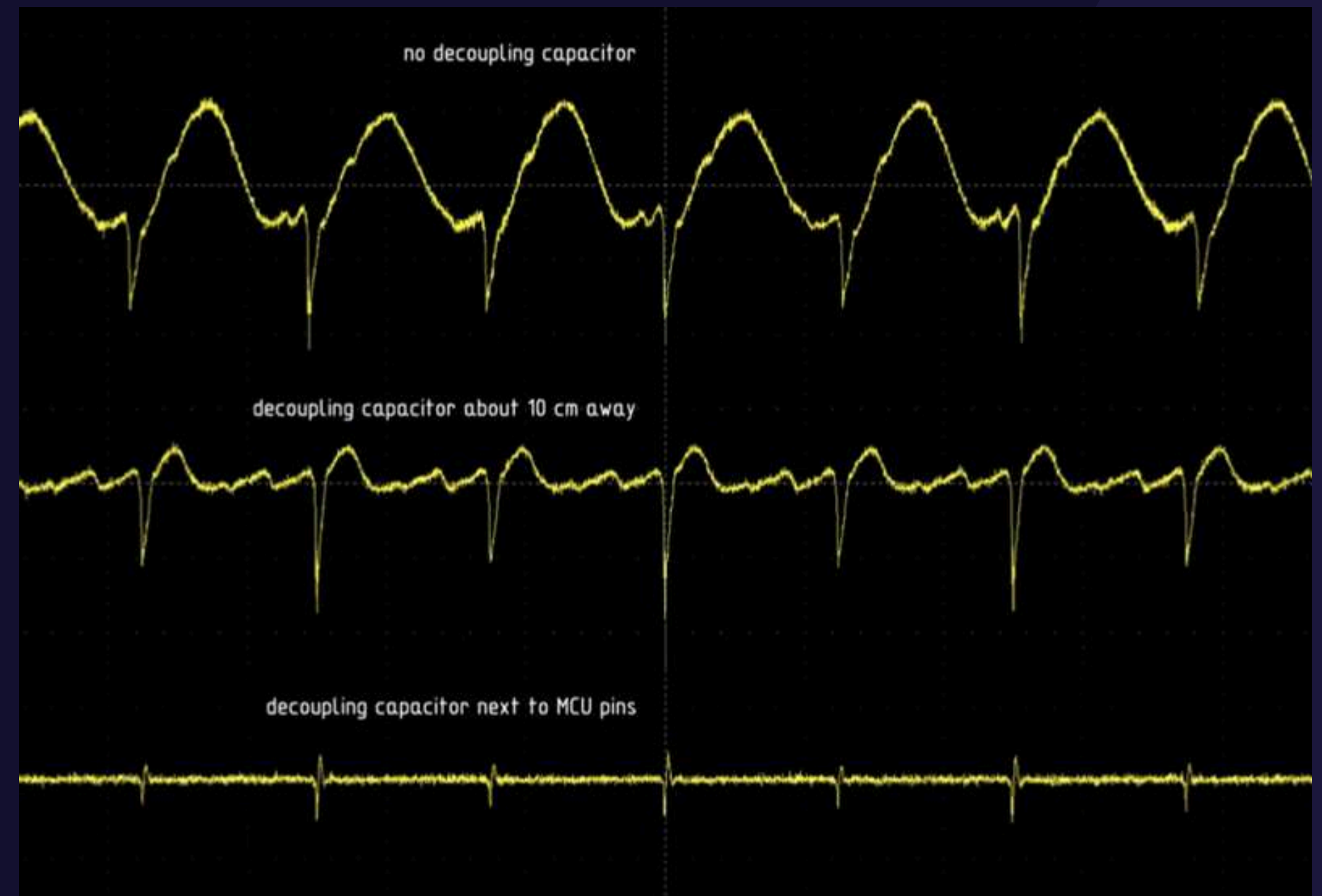
For Example: Decoupling Capacitors:

- filtering out noise
- stabilizing the power supply voltage for ICs

→ less voltage fluctuations



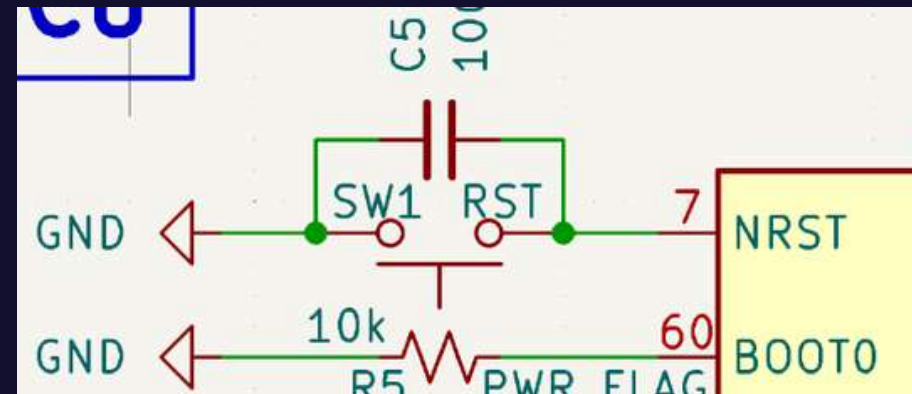
MCU Decoupling Capacitors at power inputs



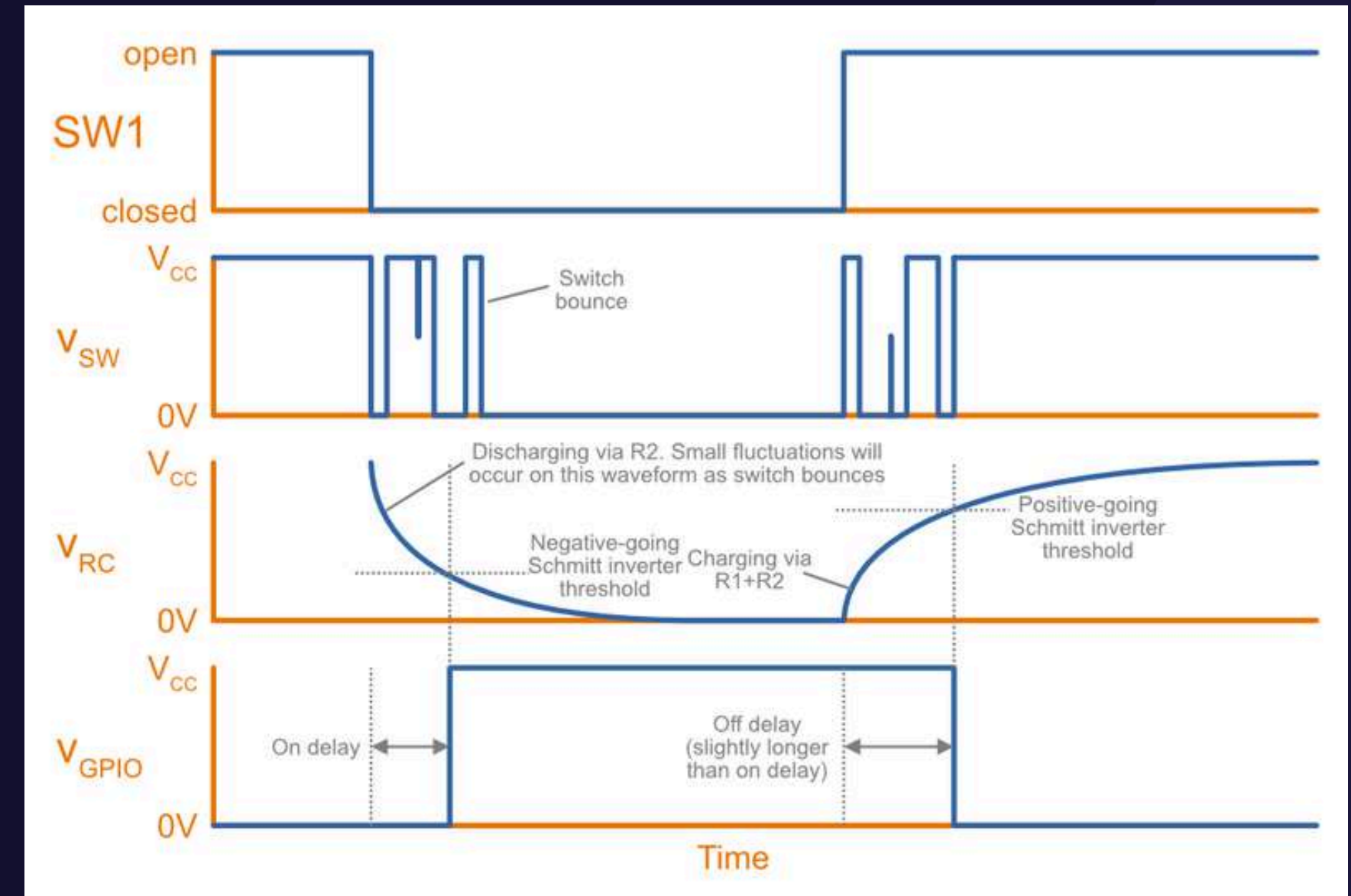
CAPACITOR - APPLICATION2 - DEBOUNCING

For Example: Debouncing

- Removing unwanted input noise from buttons, switches, or other user input.
- preventing extra activations or slow functions from triggering too often.



Debouncing cap in parallel to a button



*Note: in this case, SW1 open means logic 1

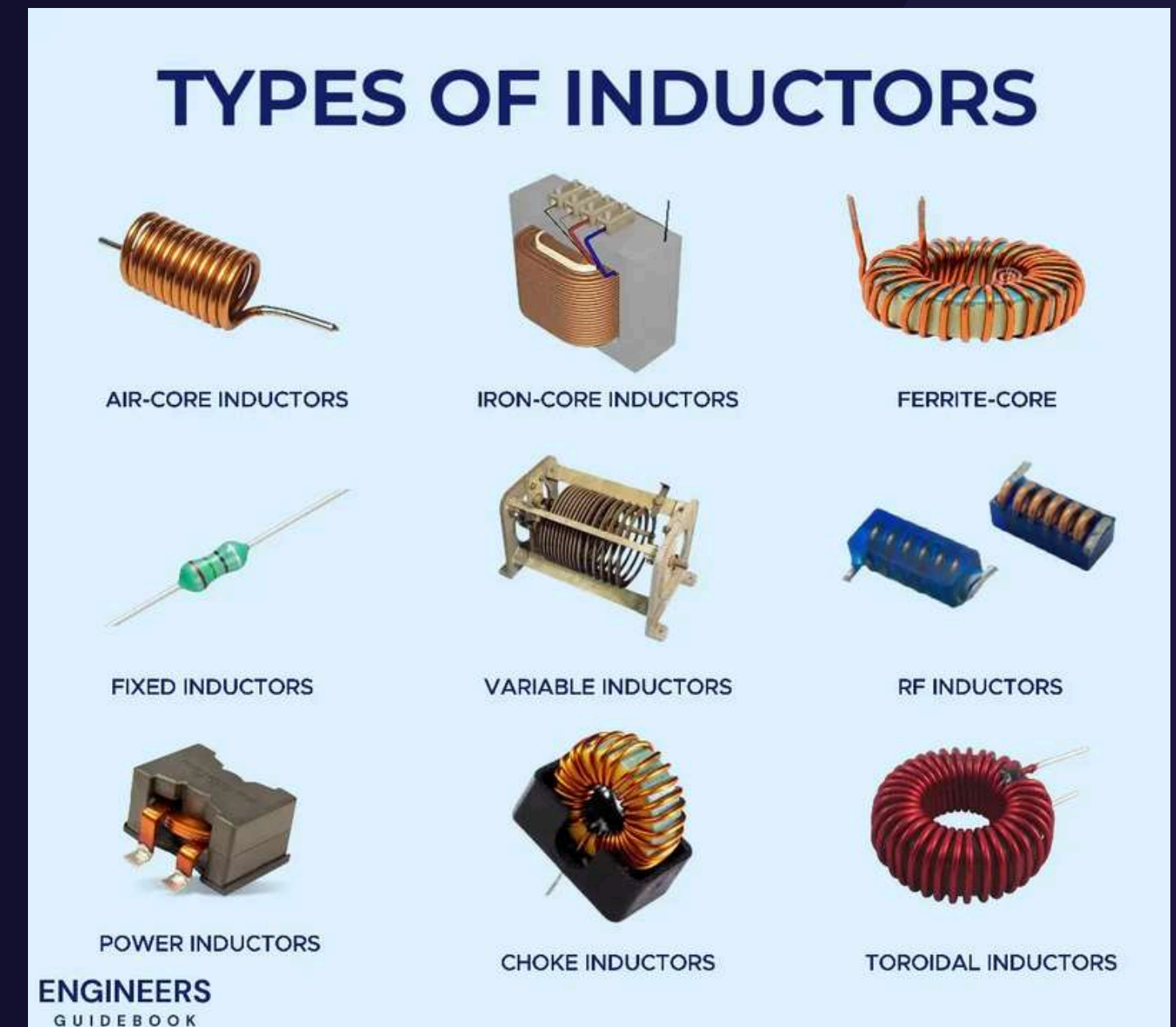
BASICS⁴ OF CIRCUIT

INDUCTOR

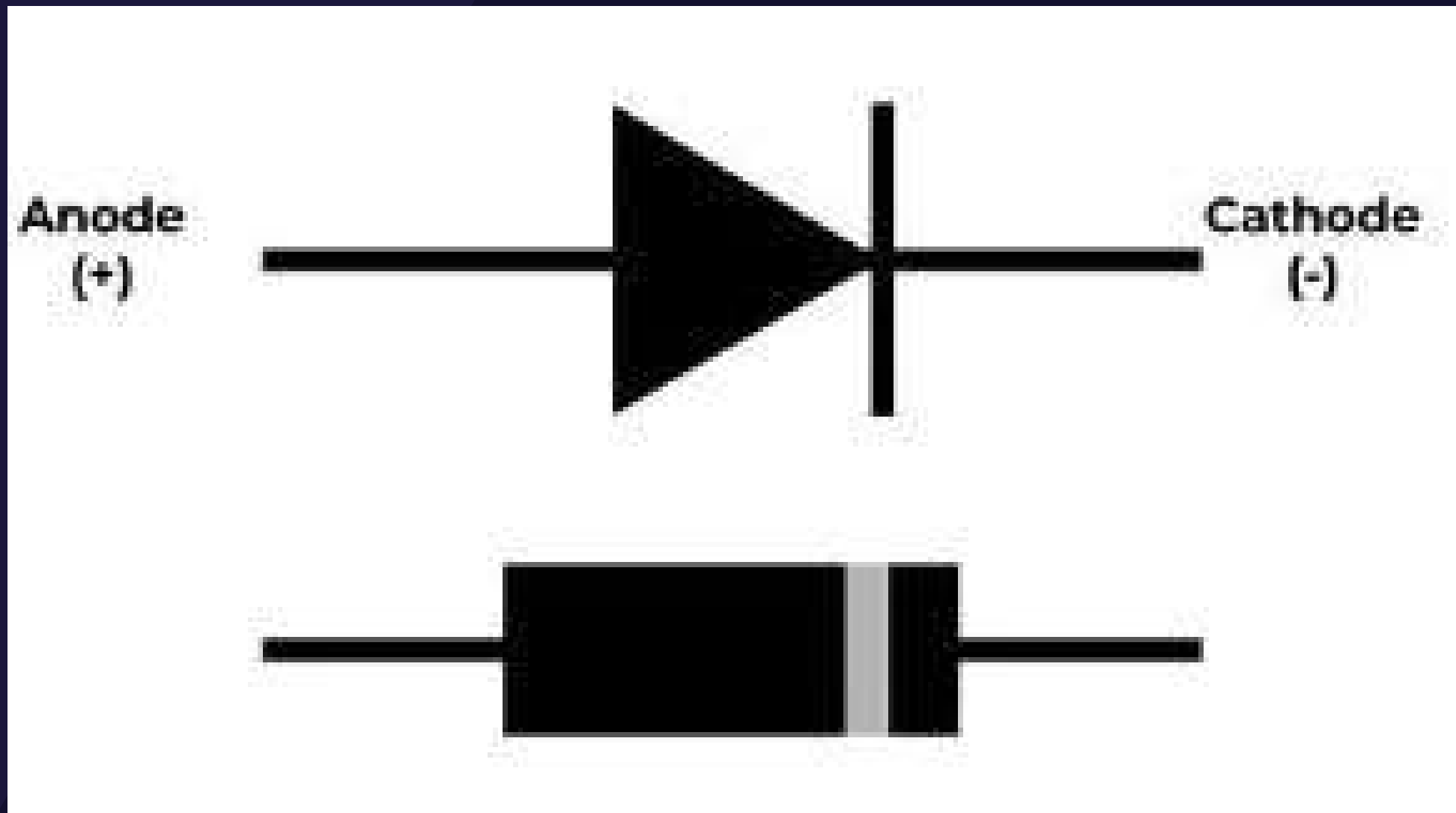
Voltage = Inductance * Rate of change of current
→ Oppose change in current

• Inductor

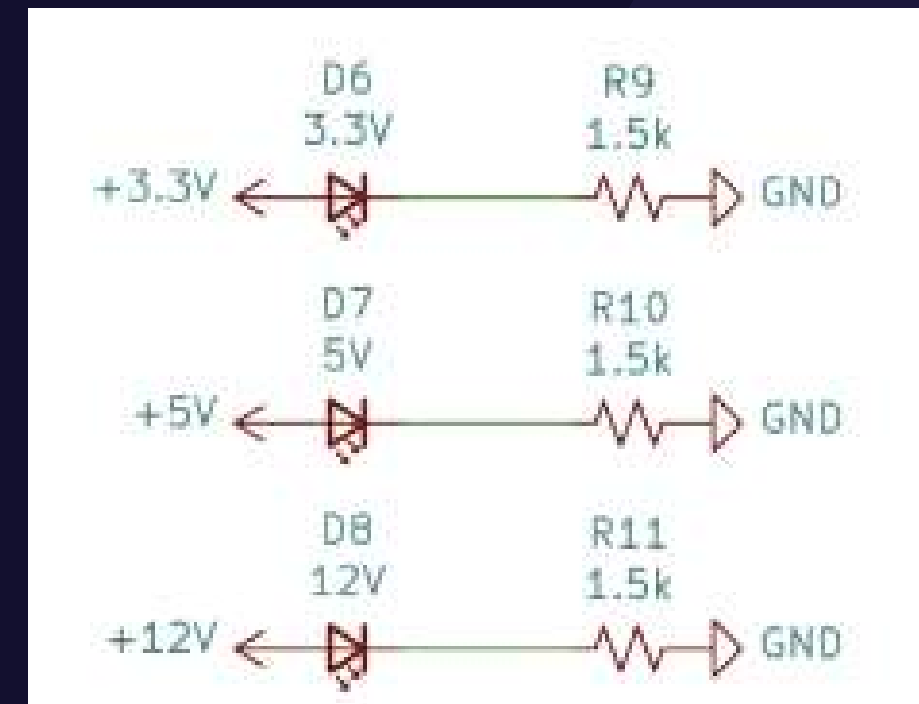
- Stores energy in a magnetic field and opposes changes in current.
- Typical Applications
 - Energy storage in switching power supplies (buck, boost converters).
 - Filtering/chokes: block high-frequency noise while passing DC or low frequencies.
 - Tuned circuits with capacitors for radios, oscillators, and filters.



DIODE



- Prevent reverse voltage
- LEDs



XXXX

ITEM 5

DESIGN FLOW (POWER)

XXXX

GENERAL FLOW TO DESIGN A PCB

1. Know what functions are necessary
2. Discuss with Mech & Software
3. Draw a routing draft
4. Find ICs and read the datasheet
5. Compare IC
6. Schematic & PCB Design
7. Send to manufacturer
8. Debug

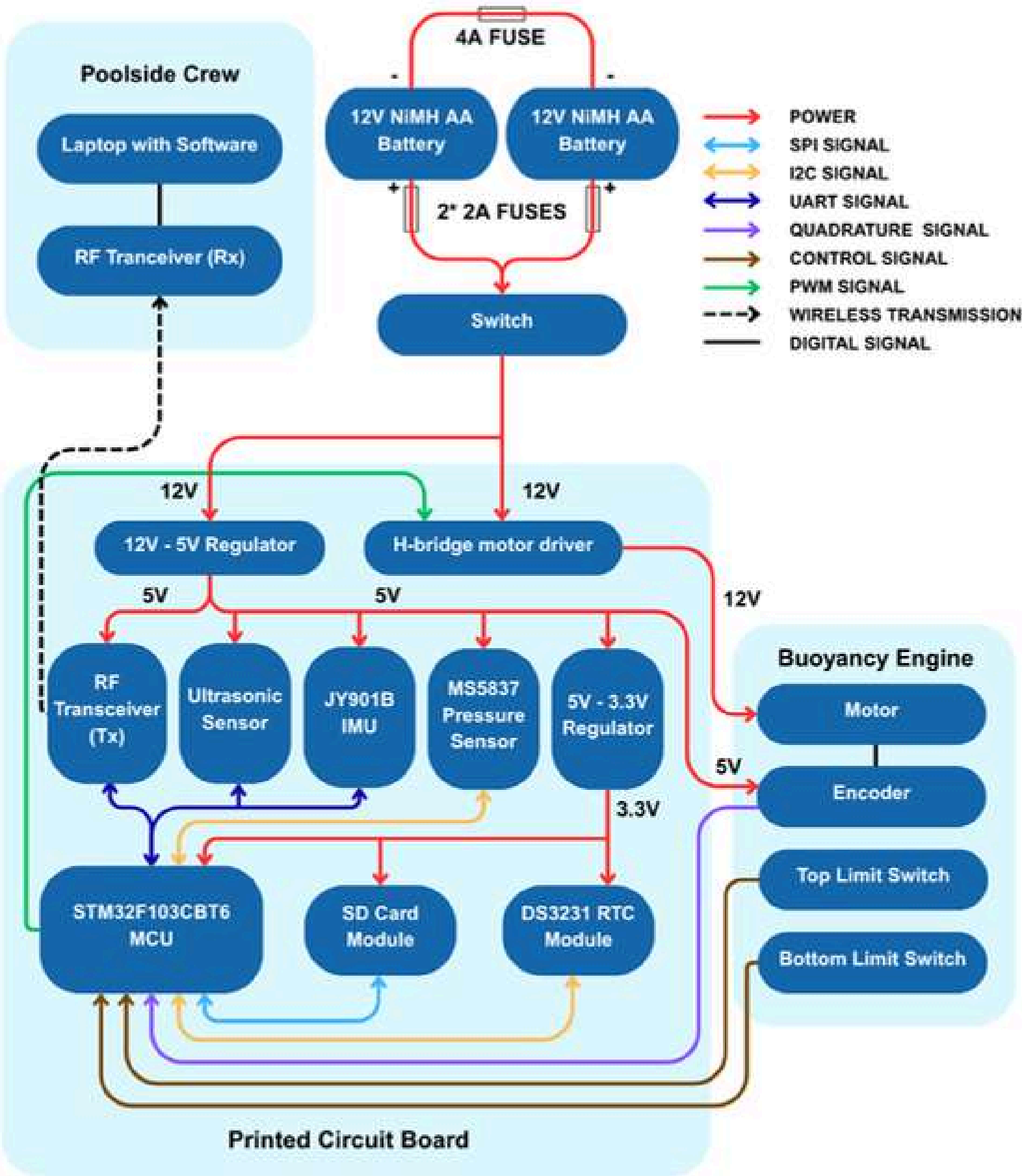


DESIGN⁵ FLOW

HOW TO CONNECT ALL THESE?

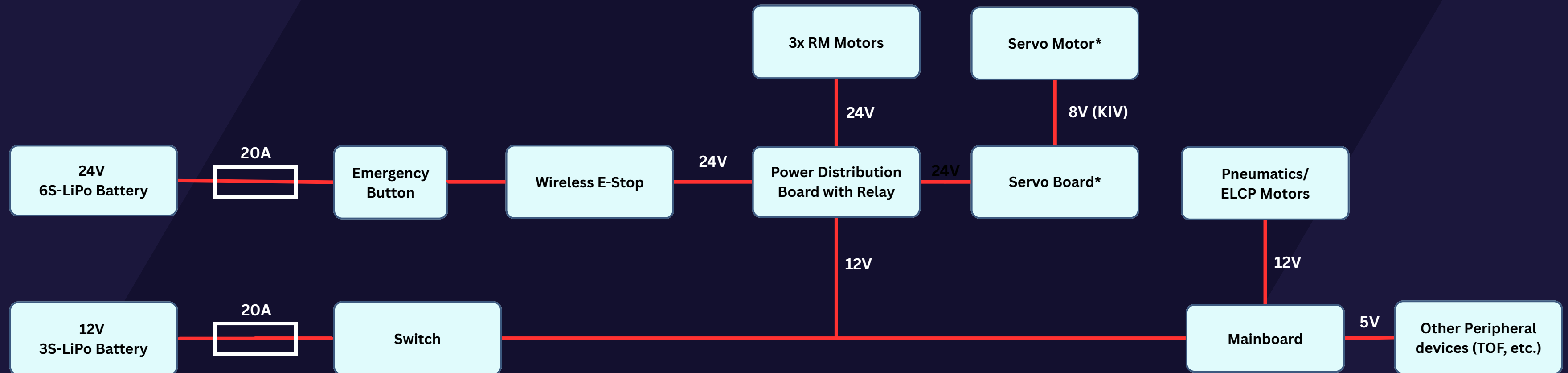


DESIGN⁵
FLOW



DESIGN⁵ FLOW

EXAMPLE



Recommendation: start to think from power input (where it all begins)



DESIGN⁵ FLOW

POWER INPUT - BATTERY SELECTION

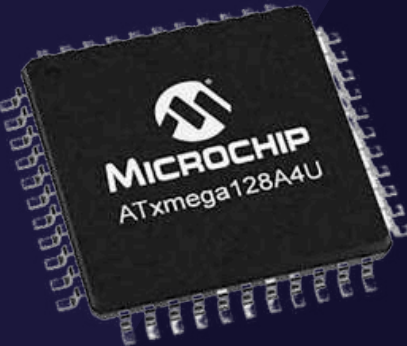
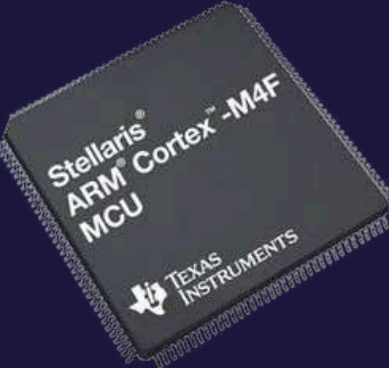
Imagine each component needs an independent battery:

- **Battery voltage selection**
- * **In most cases:**
 - Use one battery for the entire system
 - Pick the maximum voltage among all components
 - Gradually use different power regulators to step down the voltage
- **Reasons:**
 - Directly plugging batteries into sensitive components without regulations will GG them
 - Usually, higher voltage is preferred
 - Usually, stepping down is preferred over stepping up voltage

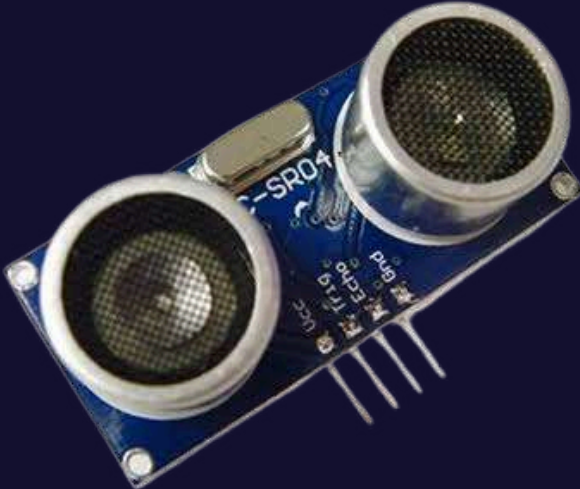


DESIGN⁵
FLOW

POWER INPUT - BATTERY SELECTION



3.3V



5V



24V



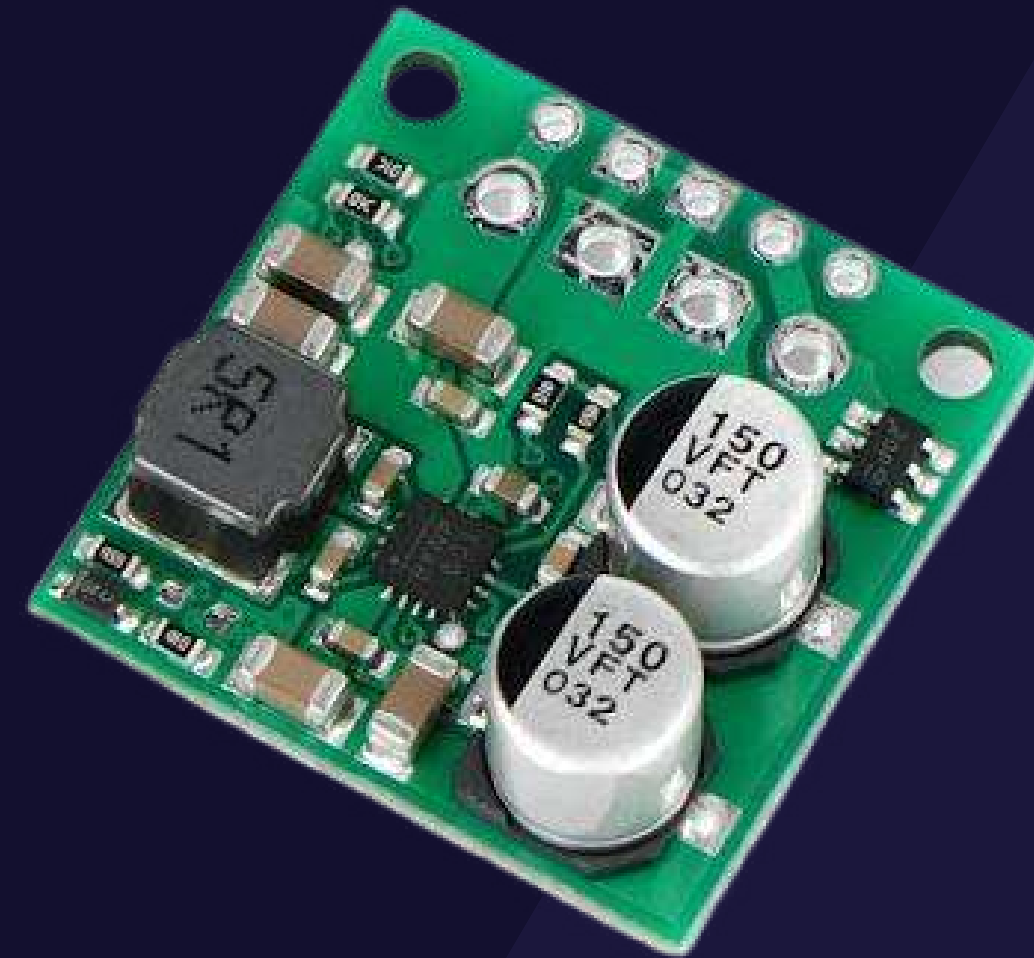
24V



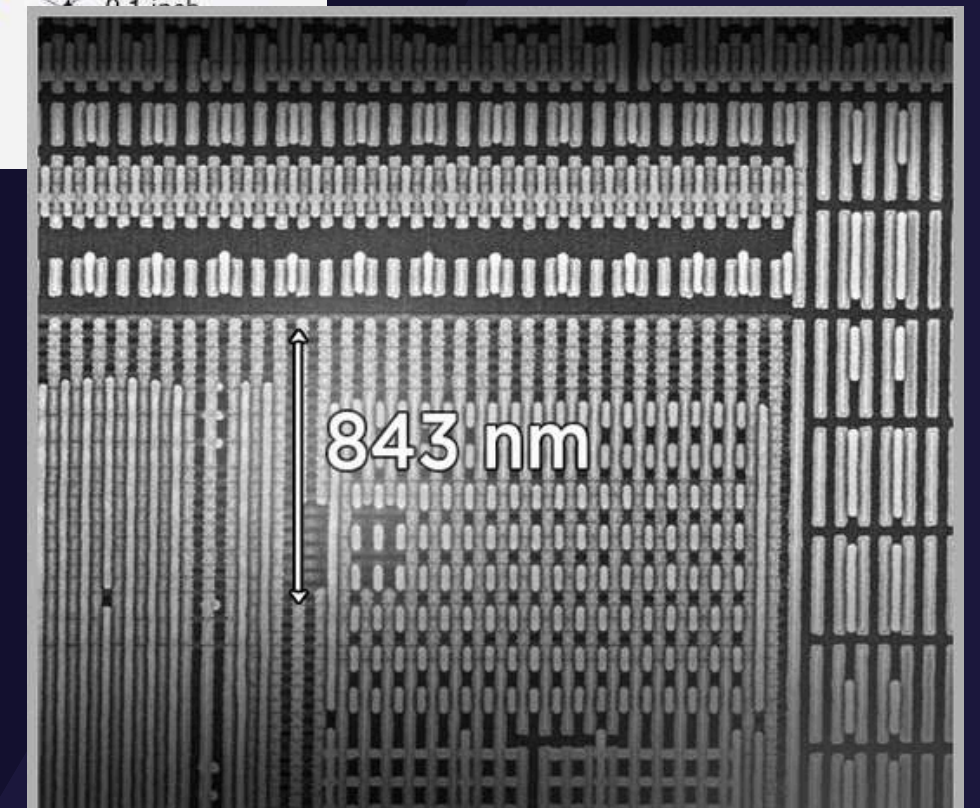
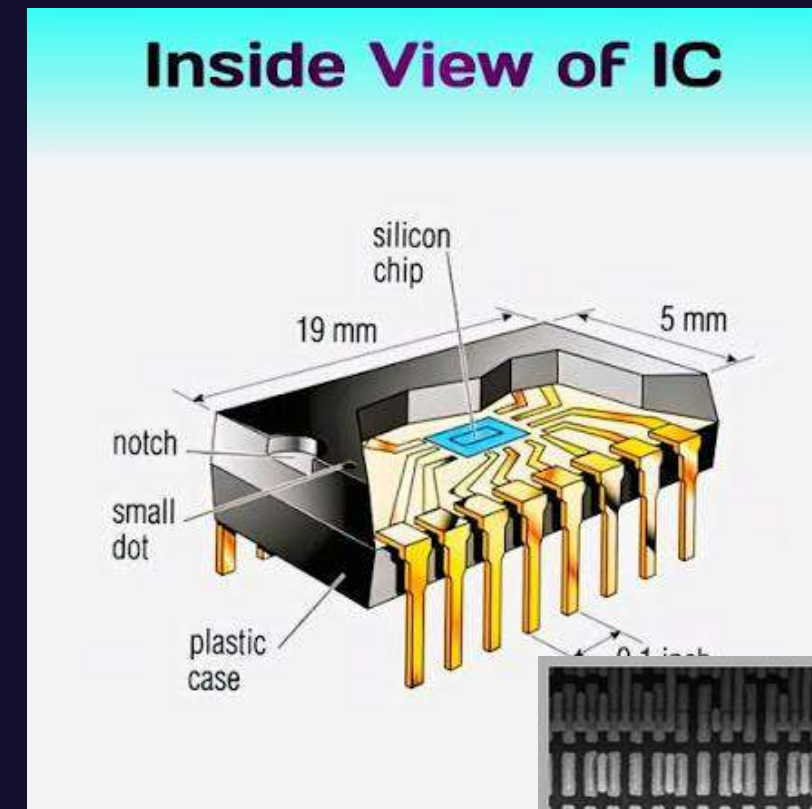
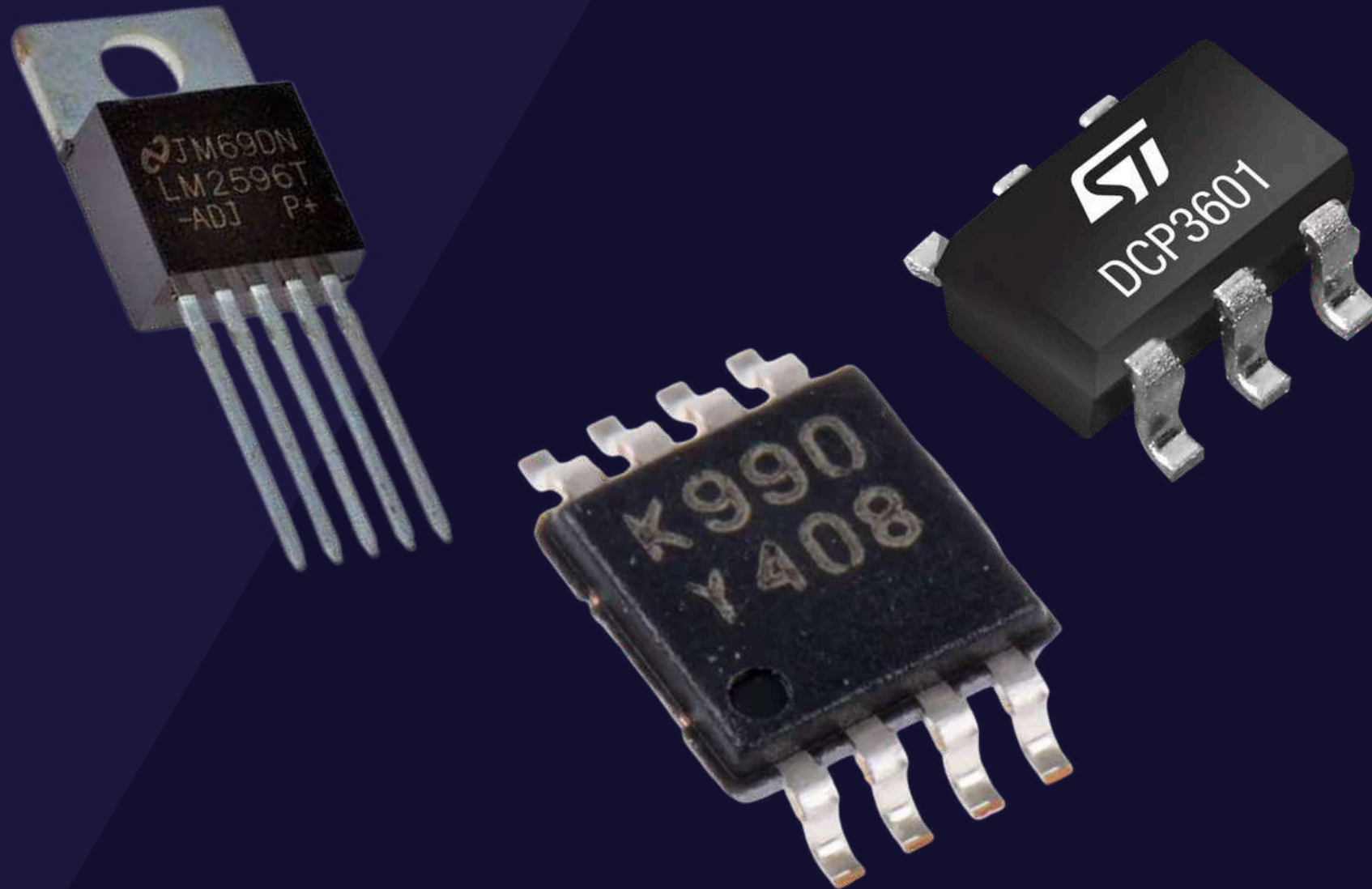
POWER STEPPING DOWN

- Power Regulators are used for stepping down voltage

Example power regulators modules



BLACK BOXES - INTEGRATED CIRCUITS

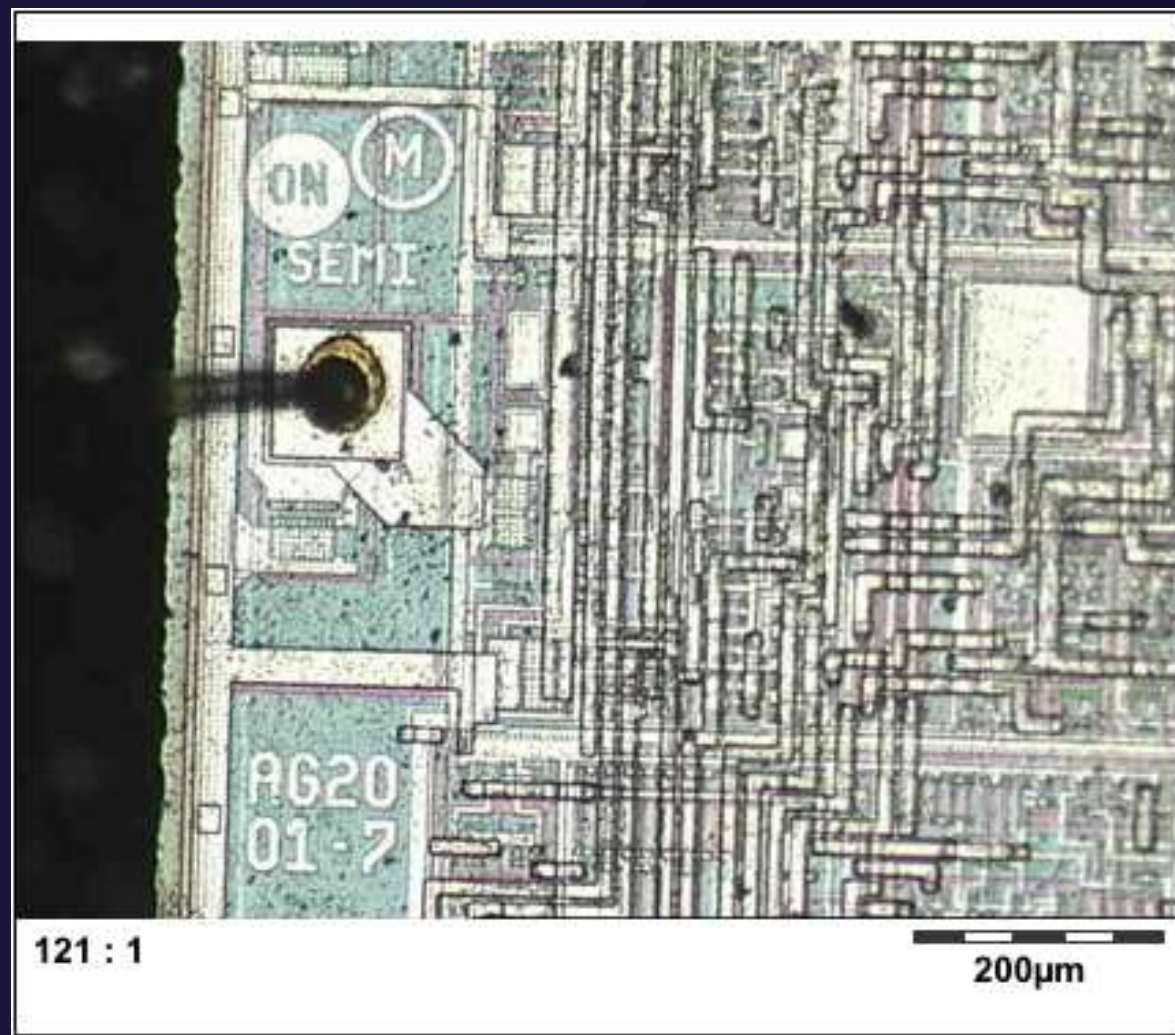


Arrays of transistors & logic gates
to achieve specific function



BLACK BOXES - INTEGRATED CIRCUITS

Would NOT be covered in the Robotics Team



ELIC 2000 - Introduction to Integrated Circuits and Systems (8 units)		
Section	Date & Time	Room
L1 (2000)	TuTh 09:00AM - 10:30AM	Lecture Theatre
T1 (2000)	TuTh 02:30PM - 04:00PM	Res 1007, L1
T2 (2000)	Fr 04:30PM - 06:00PM	Res 2006, L1
LAP (2000)	TuTh 02:30PM - 04:00PM	Res 1007, L1
LAP (2000)	Wed 02:30PM - 04:00PM	Res 1015, L1
LAP (2000)	Wed 04:30PM - 06:00PM	Res 1015, L1
LAP (2000)	Wed 06:30PM - 08:00PM	Res 1015, L1

ELIC 2010 - CMOS VLSI Design (2 units)		
Section	Date & Time	Room
L1 (2010)	Wed 04:30PM - 06:00PM	Res 1015, L1
LAP (2010)	TuTh 02:30PM - 04:00PM	Res 1007, L1

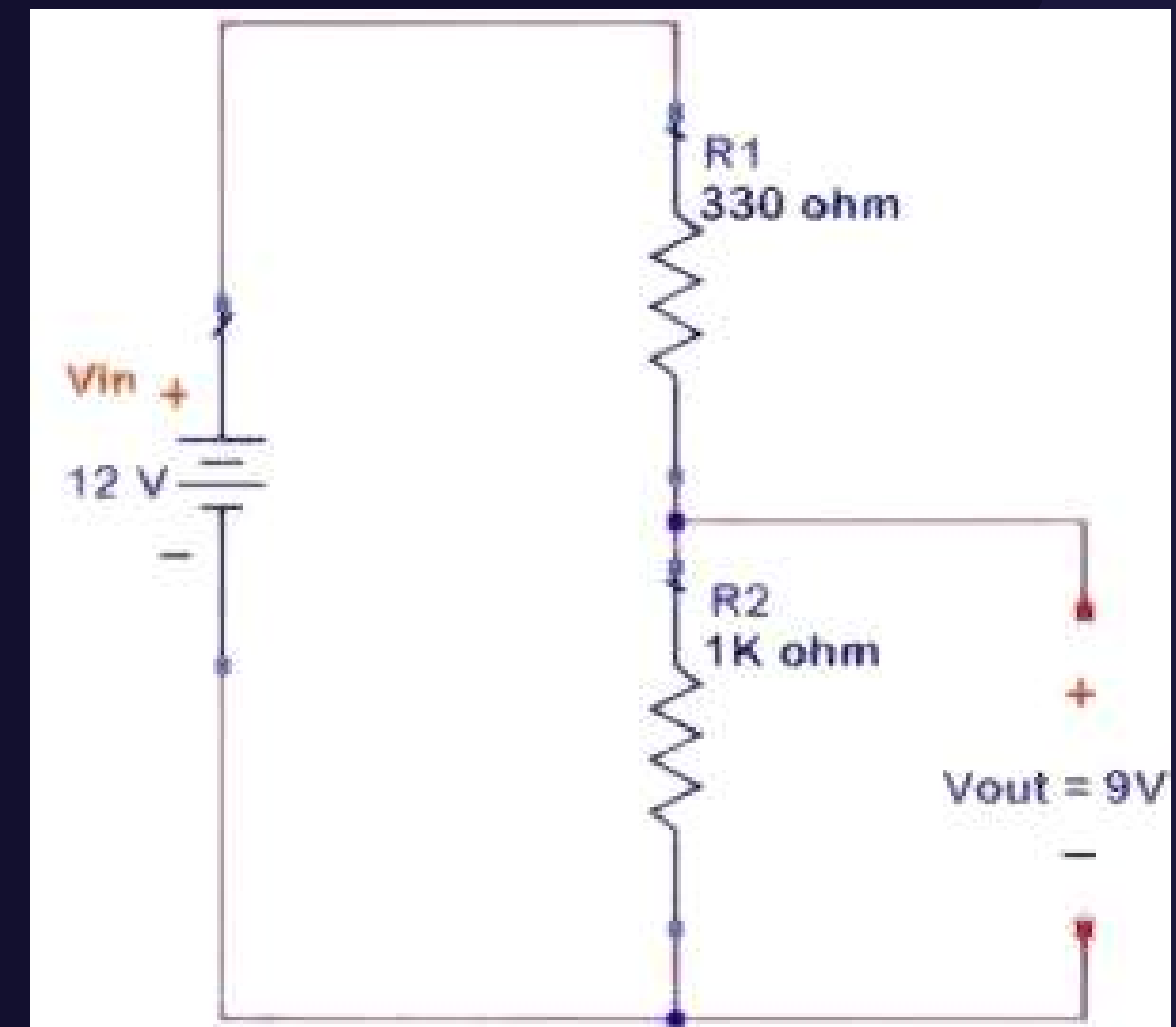
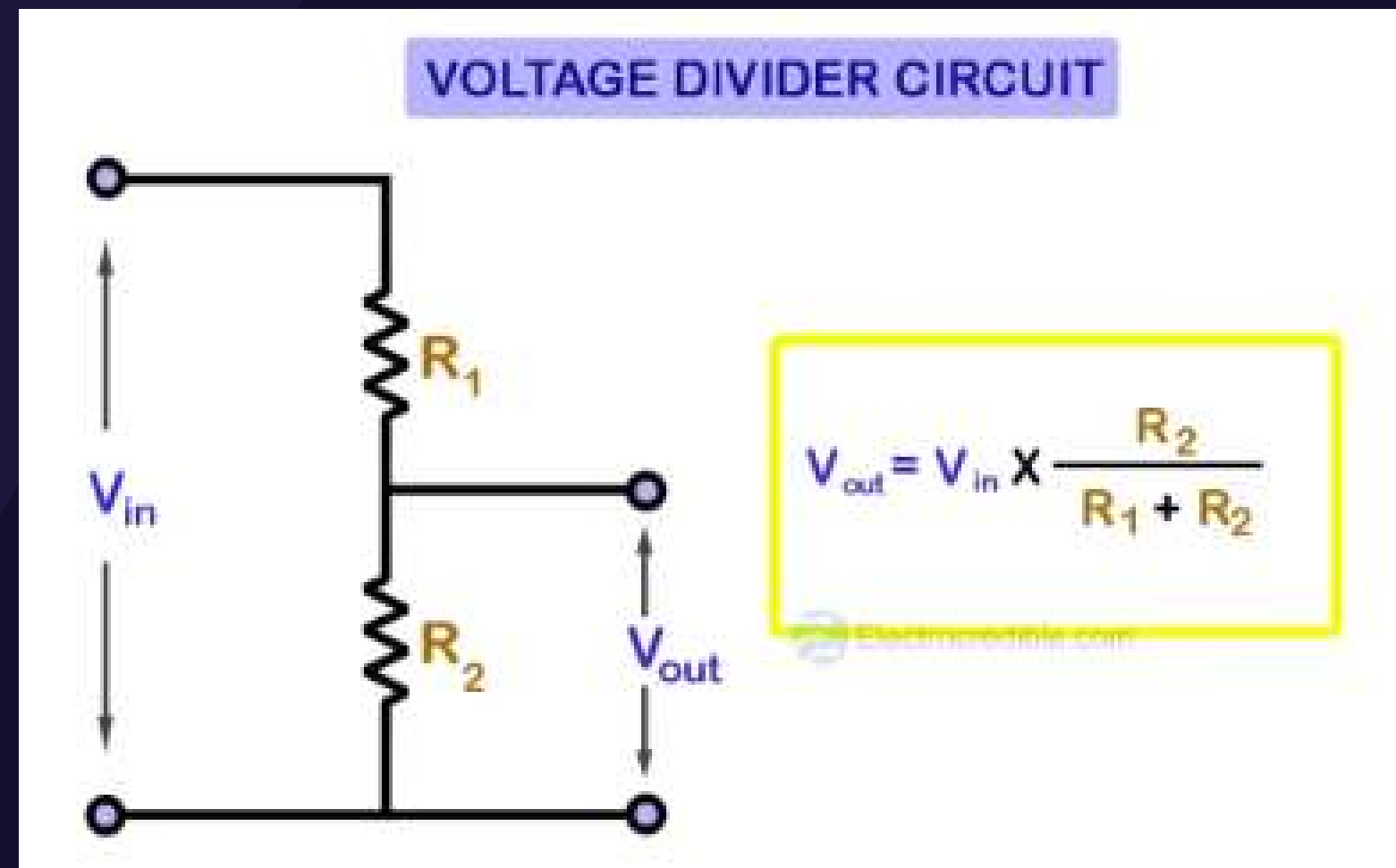
ELIC 2000 - Integrated Circuit Devices (4 units)		
Section	Date & Time	Room
L1 (2000)	TuTh 09:00AM - 10:30AM	Res 1007, L1
LAP (2000)	TuTh 02:30PM - 04:00PM	Res 1015, L1
LAP (2000)	Wed 02:30PM - 04:00PM	Res 1015, L1
LAP (2000)	Wed 04:30PM - 06:00PM	Res 1015, L1
LAP (2000)	Wed 06:30PM - 08:00PM	Res 1015, L1

We only need to know
how to apply them

(Already many to
learn, trust me)

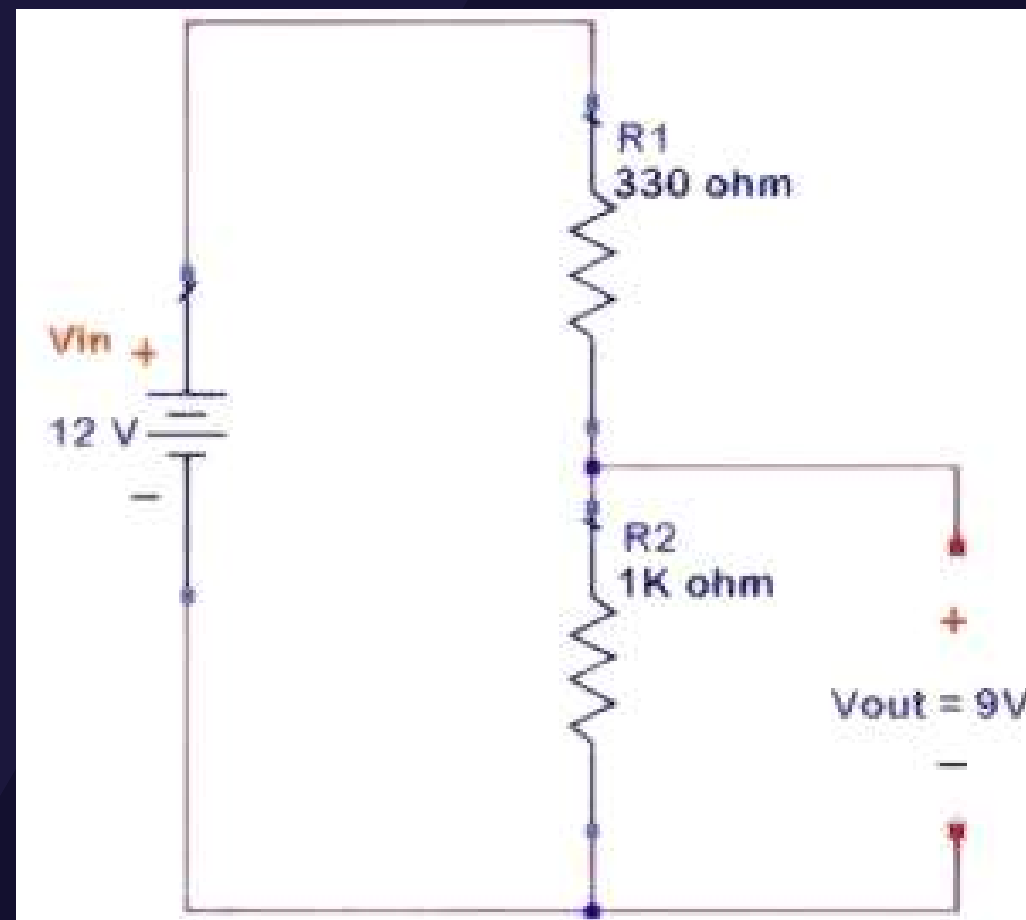
POWER REGULATORS - INCENTIVE

What if simple resistor voltage circuit is used to step down voltage?



POWER REGULATORS - INCENTIVE

What if simple resistor voltage circuit is used to step down voltage?



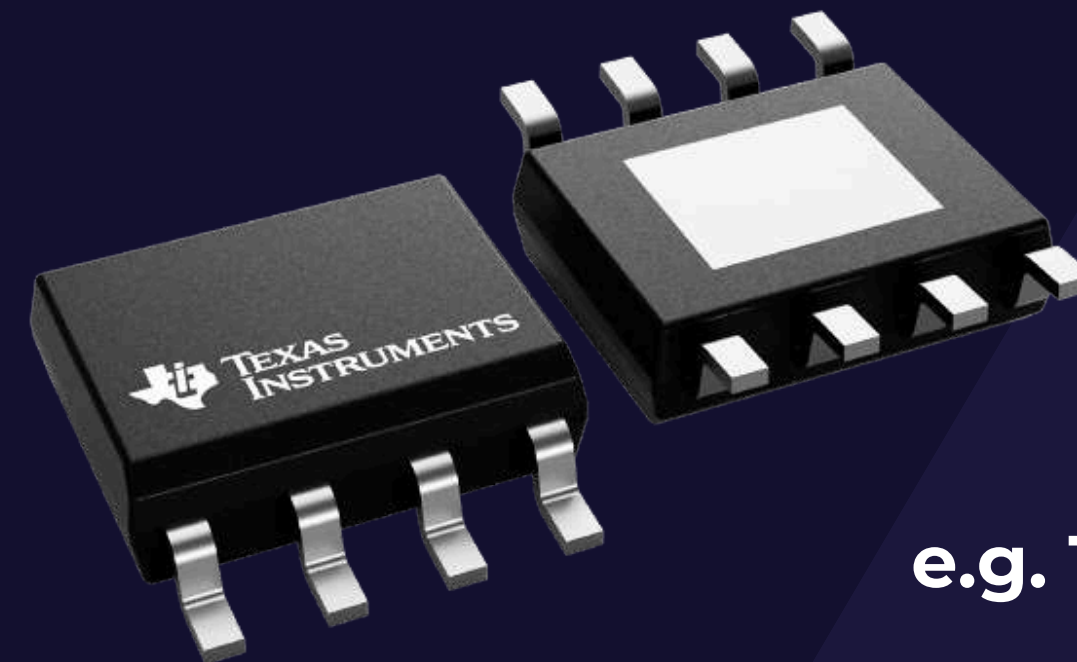
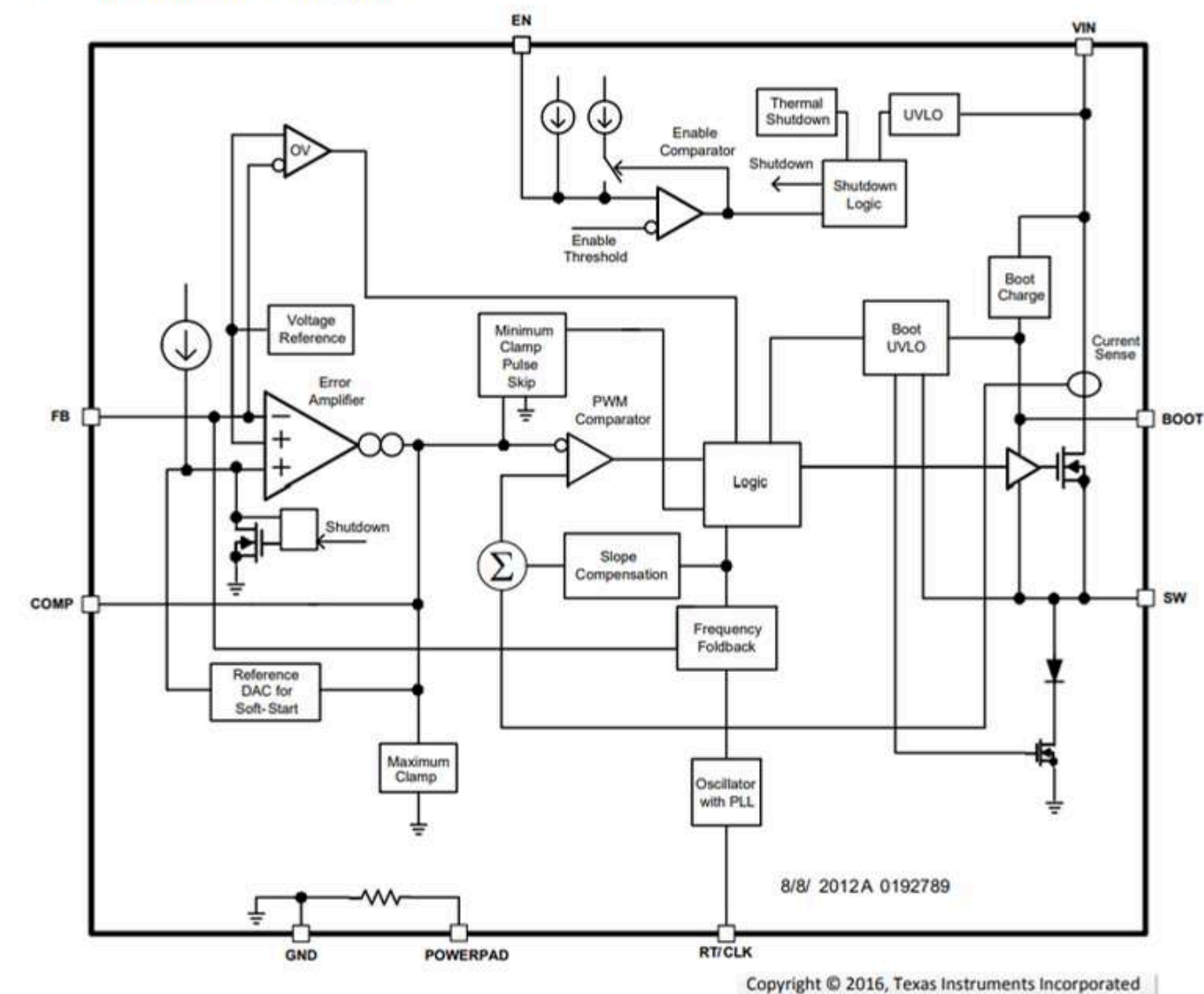
- Output voltage depends on input voltage
- Output voltage depends on load current
- Large % of power loss via heat (in resistors)

DESIGN⁵ FLOW

POWER REGULATORS - INCENTIVE

What if regulator ICs are used to step down voltage?

7.2 Functional Block Diagram



e.g. TPS54360

- Output voltage is independent of input voltage
- Output voltage is independent of load current
- Less power loss via heat → higher efficiency



*You don't have to understand all these, please don't go

POWER REGULATORS - HOW TO USE?

1. **Determine the electrical requirements (input & output voltage range, input & output current, IC package)**
2. **Choose the IC**
 - a. **Ask AI**
 - b. **Go to WEBENCH[®] POWER DESIGNER**
3. **Find the datasheet of the IC**
4. **Read through (though most people don't)**
5. **Find the typical application diagram**
6. **Copy and paste into your schematic**
7. **Adjust values of some peripheral components**

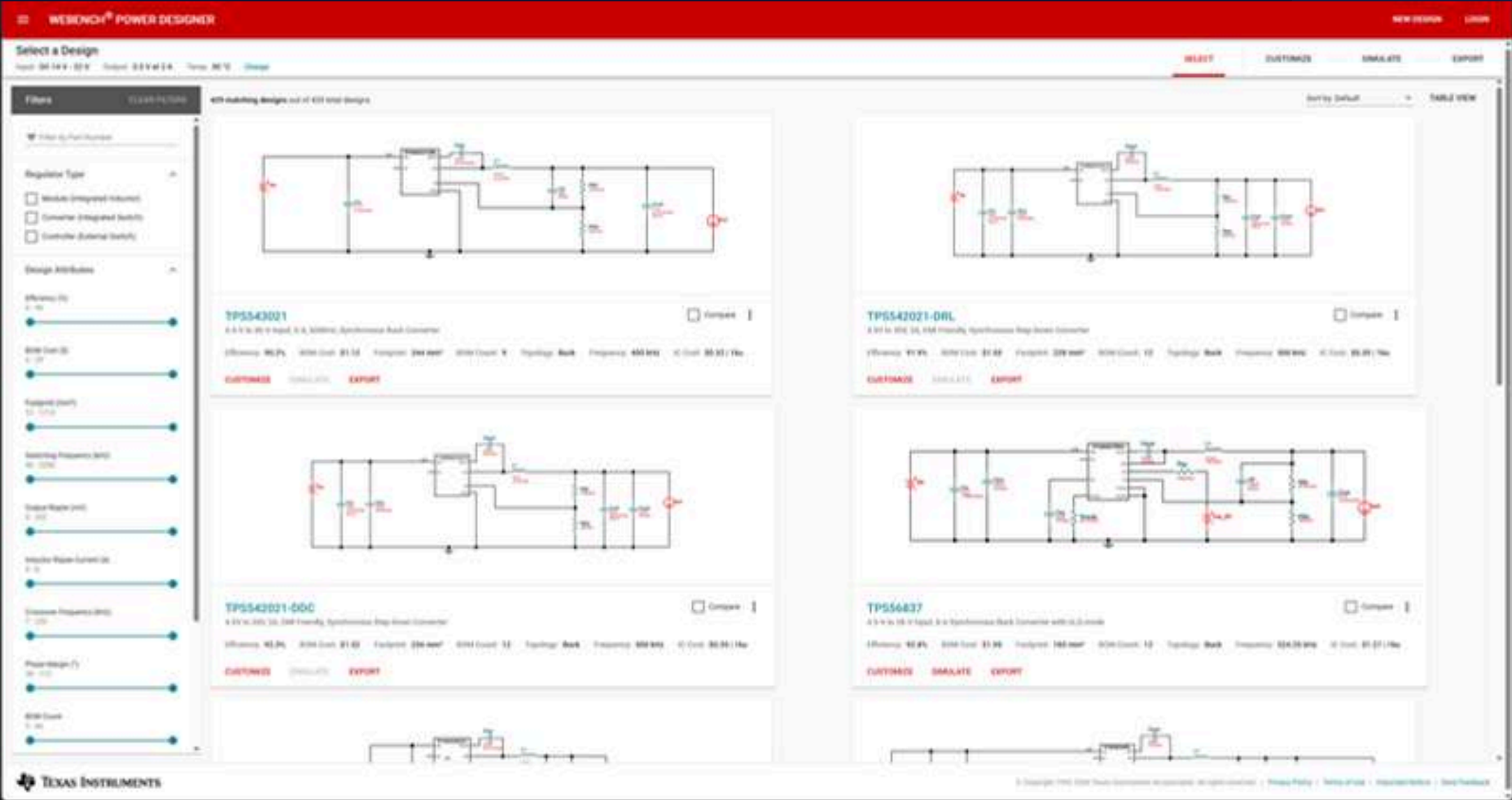
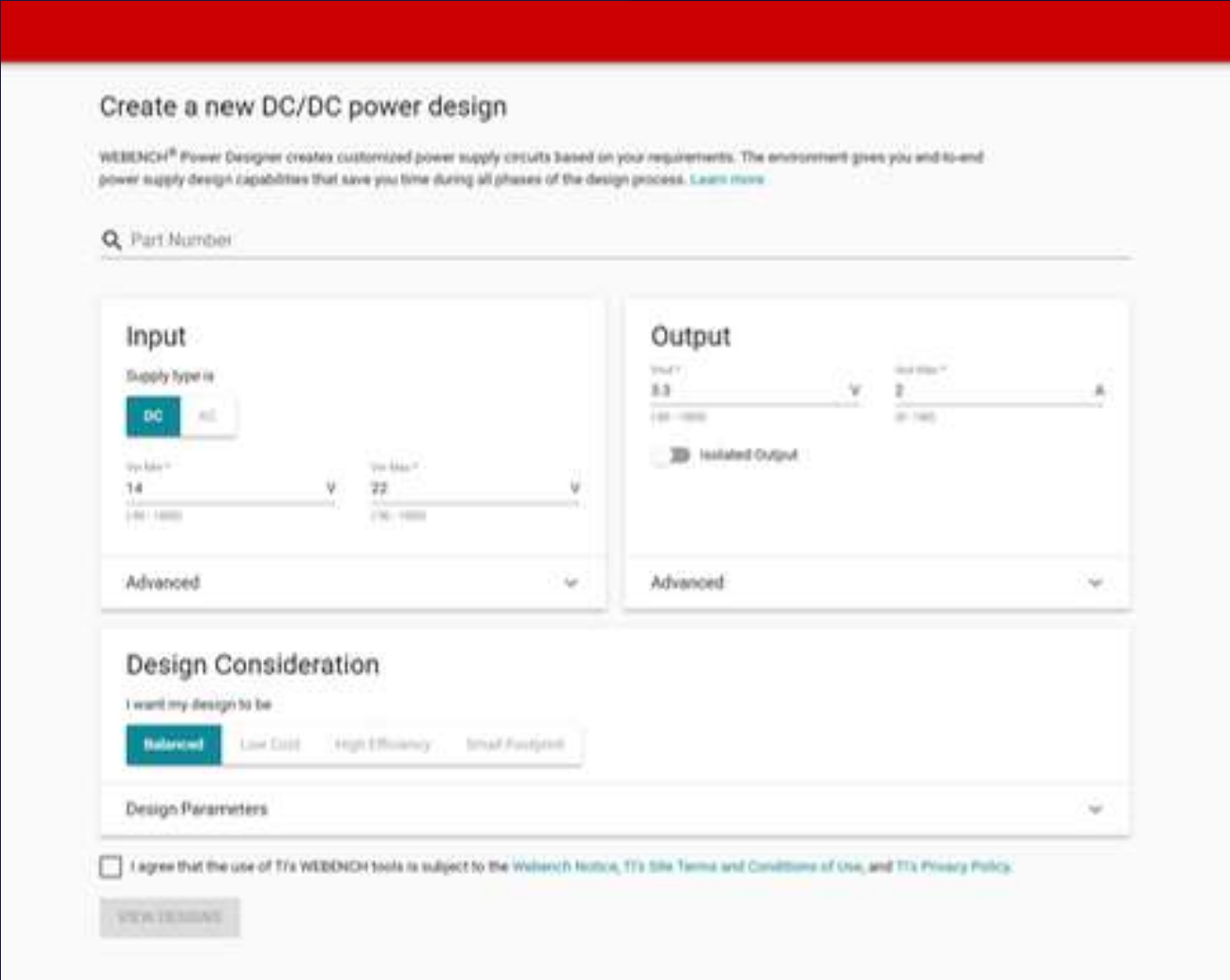
Personal Exp:
Hardware at the start is quite mechanical and boring, until you really read through the intricacies of the datasheet (you will suffer for sure tho)



***This is bascially your HW1 content. Will demonstrate if time allows

DESIGN⁵ FLOW

WEBENCH[®] POWER DESIGNER



Product Folder

Order Now

Technical Documents

Tools & Software

Support & Community

Reference Design

TEXAS
INSTRUMENTS

TPS54360

SLVS884G – AUGUST 2012 – REVISED JUNE 2018

TPS54360 60-V Input, 3.5-A, Step-Down DC/DC Converter With Eco-Mode™

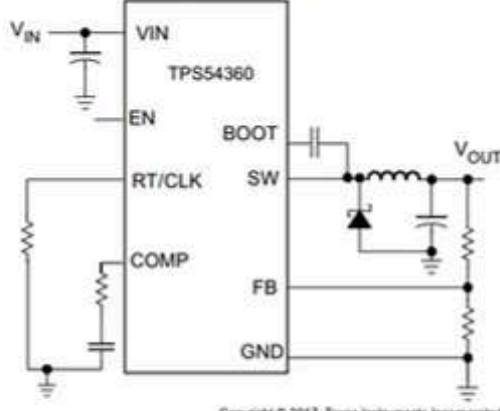
1 Features

- 4.5-V to 60-V (65-V Abs Max) Input Range
- 3.5-A Continuous Current, 4.5-A Minimum Peak Inductor Current Limit
- Current Mode Control DC/DC Converter
- 92-mΩ High-Side MOSFET
- High Efficiency at Light Loads with Pulse Skipping Eco-mode™
- Low Dropout at Light Loads with Integrated BOOT Recharge FET
- 146-μA Operating Quiescent Current
- 2-μA Shutdown Current
- 100-kHz to 2.5-MHz Fixed Switching Frequency
- Synchronizes to External Clock
- Adjustable UVLO Voltage and Hysteresis
- Internal Soft Start
- Accurate Cycle-by-Cycle Current Limit
- Thermal, Overvoltage, and Frequency Foldback Protection
- 0.8 V 1% Internal Voltage Reference
- 8-Terminal HSOIC with PowerPAD™ Package
- -40°C to 150°C T_J Operating Range
- Create a Custom Design using the TPS54360 with the [WEBENCH® Power Designer](#)

2 Applications

12-V, 24-V and 48-V Industrial, Automotive and Communications Power Systems

Simplified Schematic



3 Description

The TPS54360 is a 60-V, 3.5-A, step-down regulator with an integrated high side MOSFET. The device survives load dump pulses up to 65 V per ISO 7637. Current mode control provides simple external compensation and flexible component selection. A low ripple pulse skip mode reduces the no load supply current to 146 μA. Shutdown supply current is reduced to 2 μA when the enable pin is pulled low.

Undervoltage lockout is internally set at 4.3 V but can be increased using the enable pin. The output voltage start-up ramp is internally controlled to provide a controlled start-up and eliminate overshoot.

A wide switching frequency range allows either efficiency or external component size to be optimized. Frequency foldback and thermal shutdown protect internal and external components during an overload condition.

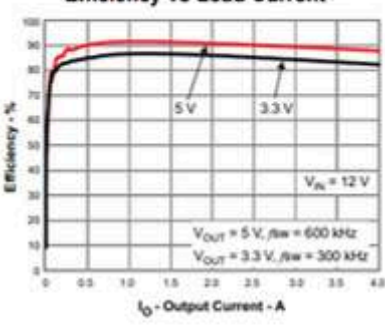
The TPS54360 is available in an 8-terminal thermally enhanced HSOIC PowerPAD™ package.

Device Information

ORDER NUMBER	PACKAGE	BODY SIZE
TPS54360	HSOIC (8)	4.89 mm × 3.90 mm

(1) For all available packages, see the orderable addendum at the end of the data sheet.

Efficiency vs Load Current



An IMPORTANT NOTICE at the end of this data sheet addresses availability, warranty, changes, use in safety-critical applications, intellectual property matters and other important disclaimers. PRODUCTION DATA.

8.2 Typical Application

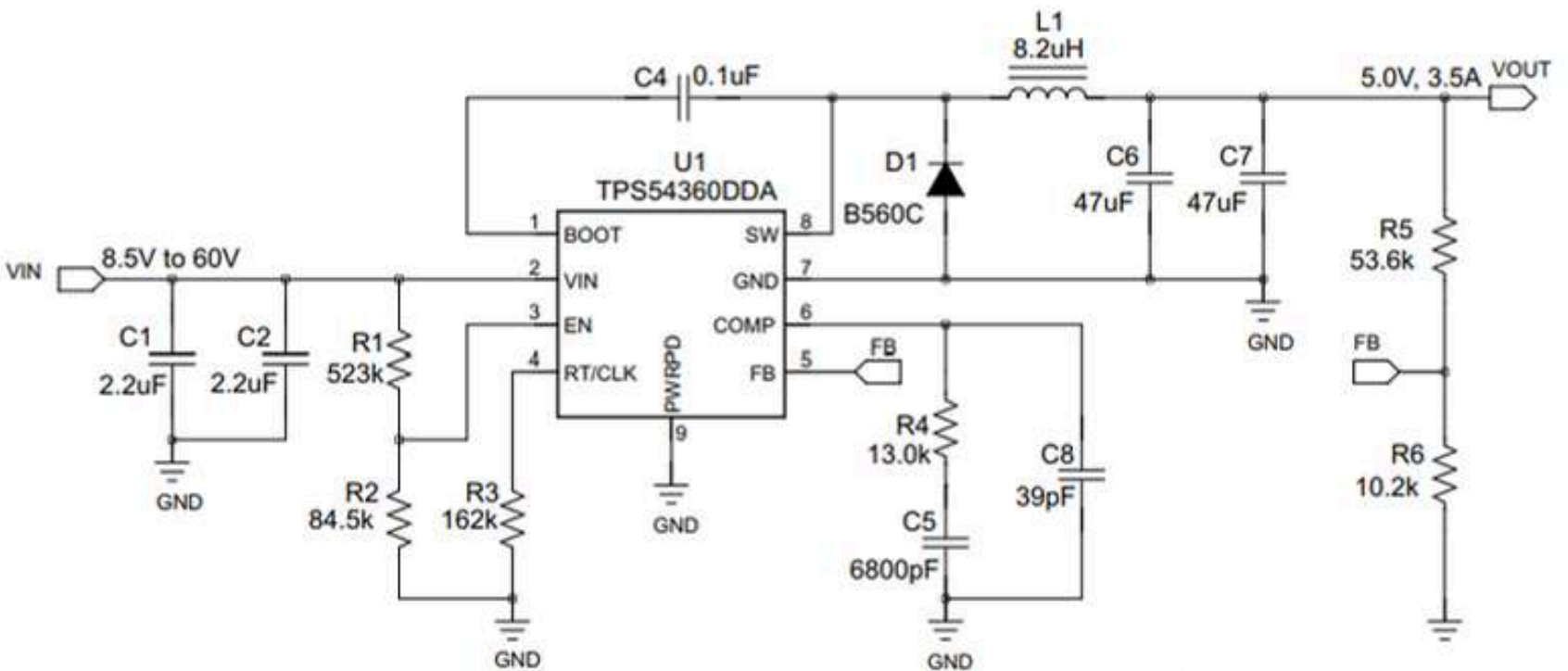


Figure 34. 5 V Output TPS54360 Design Example

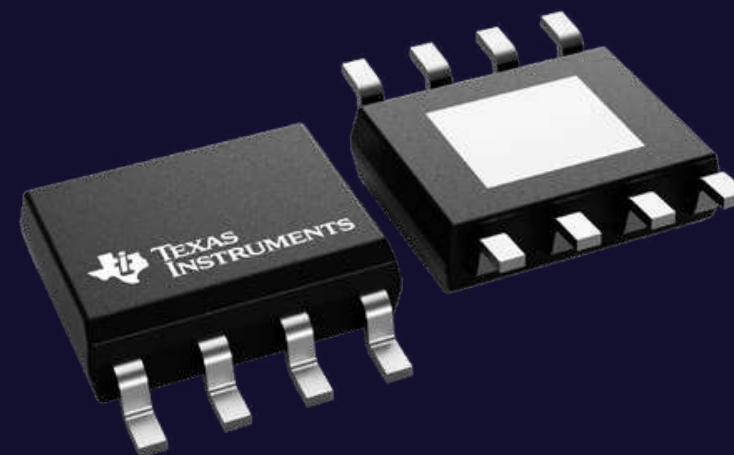
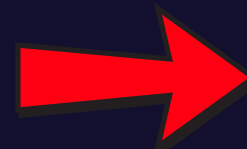
Copyright © 2017, Texas Instruments Incorporated

BACK TO STEPPING DOWN VOLTAGE

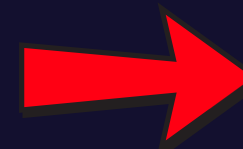
To conclude,



24V



5V



P.S. The size is not in ratio

DESIGN⁵
FLOW

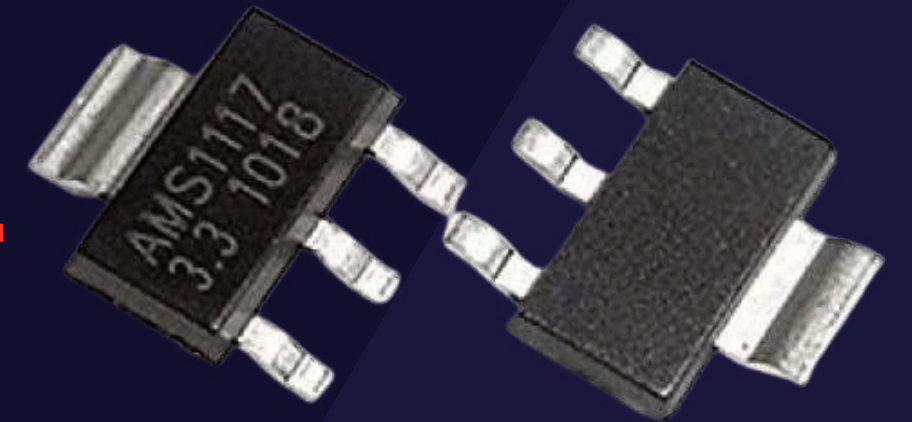
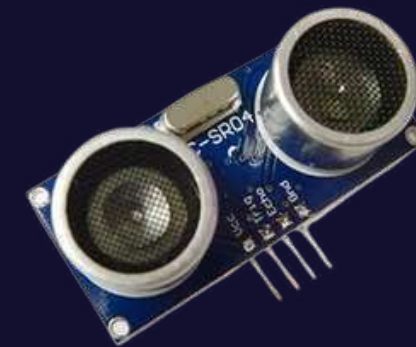
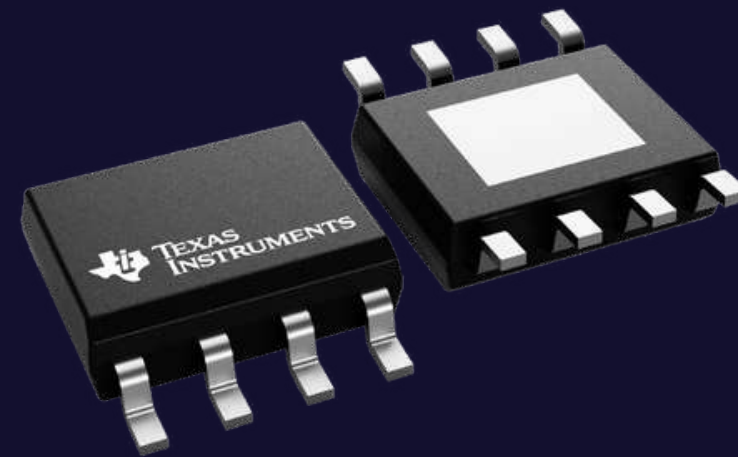
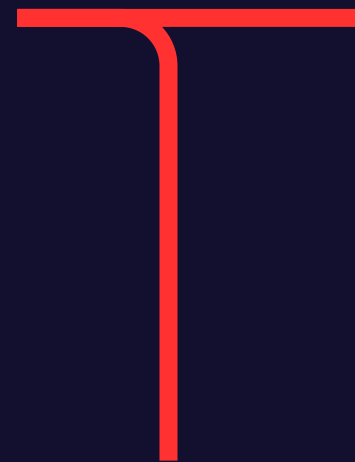
POWER CONNECTIONS

To conclude,

24V

5V

3.3V

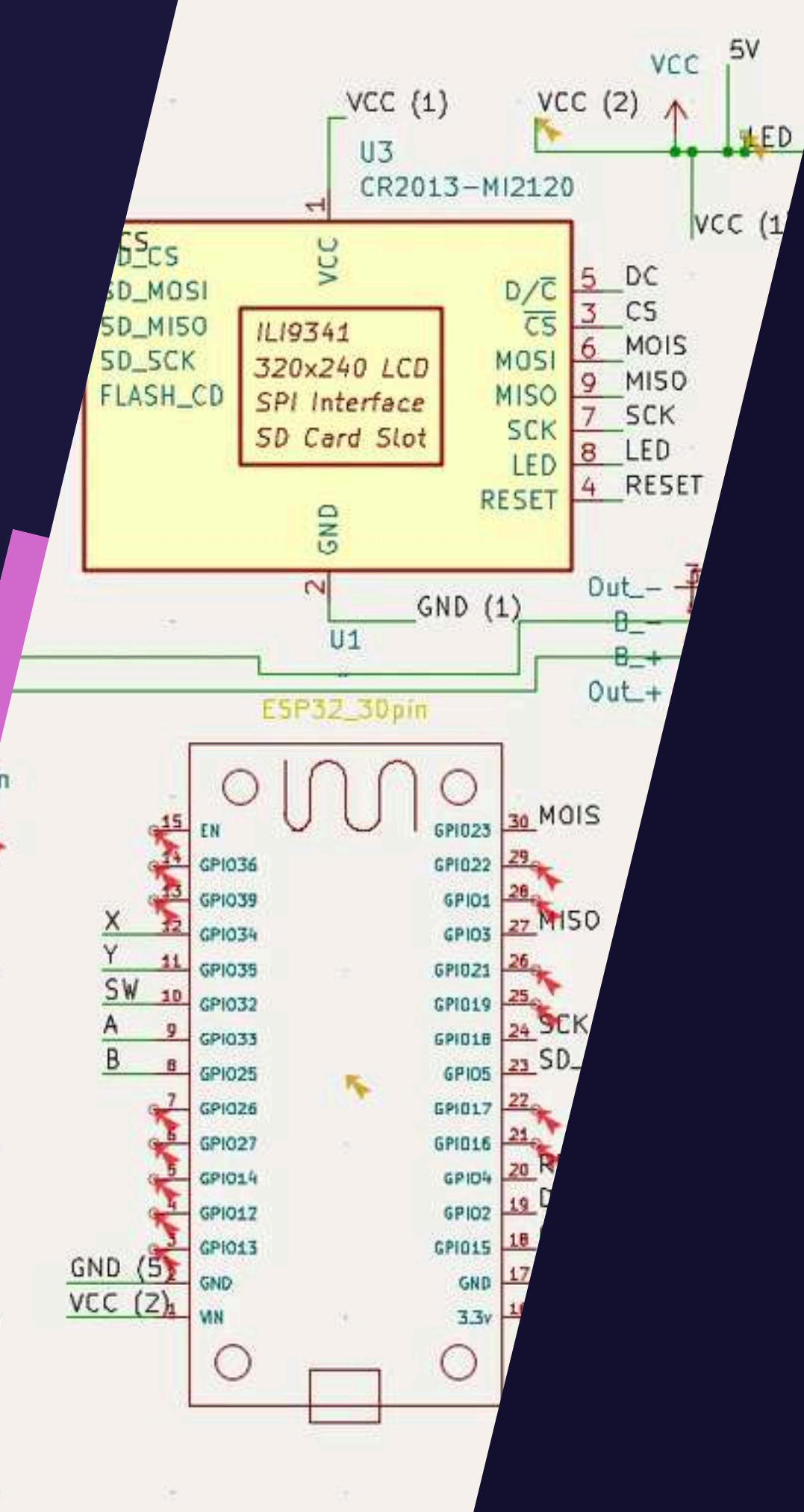


XXXX

ITEM 6

IC IMPLEMENTATION & KICAD (HW) GUIDELINES

XXXX



HW1 CONTENT

1. Search for the required IC (just google)
2. Find the datasheet of the IC
3. Read through the datasheet (though most people don't)
4. Find the typical application diagram
5. Copy and paste into your schematic
6. Adjust values of some peripheral components

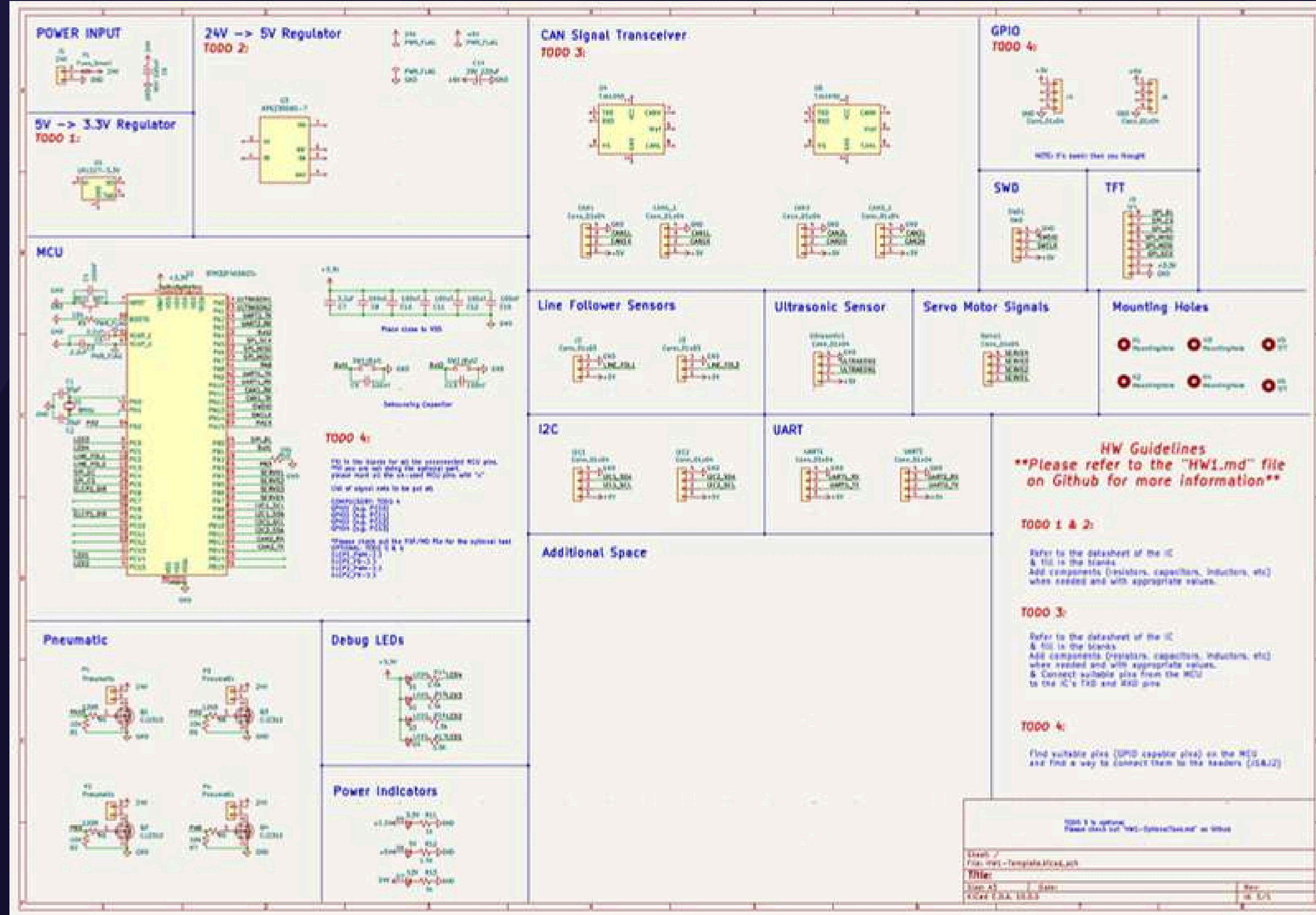
Purpose:

- To familiarize with operations in KiCAD (schematics)
- To familiarize with datasheet of ICs



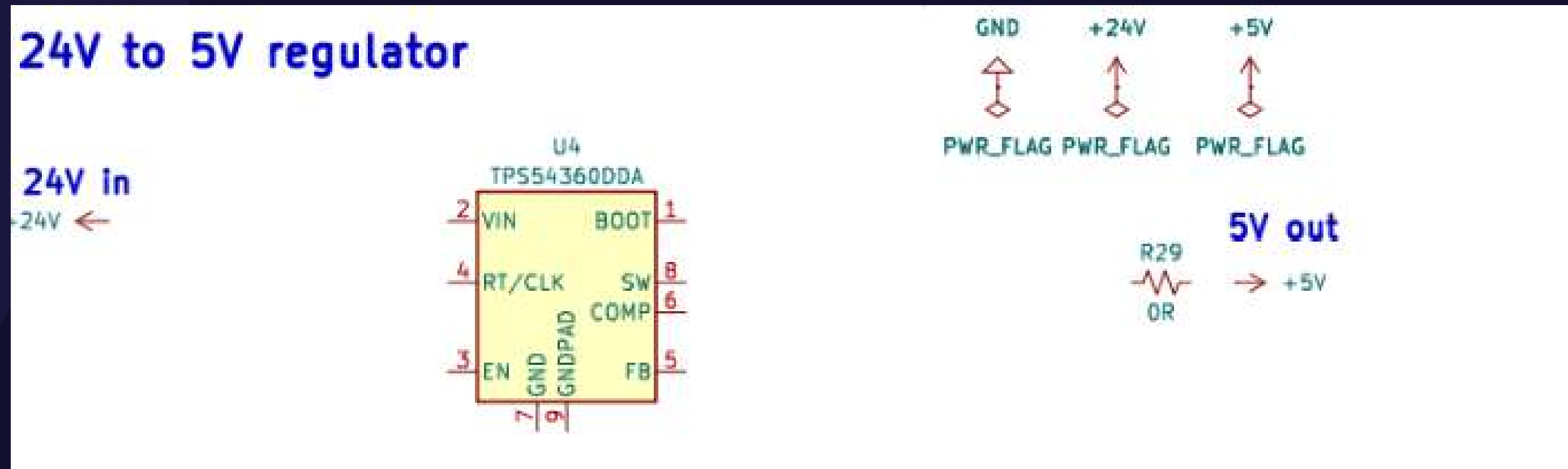
HW 6

GUIDELINES



HW⁶ GUIDELINES

HW1 EXAMPLE



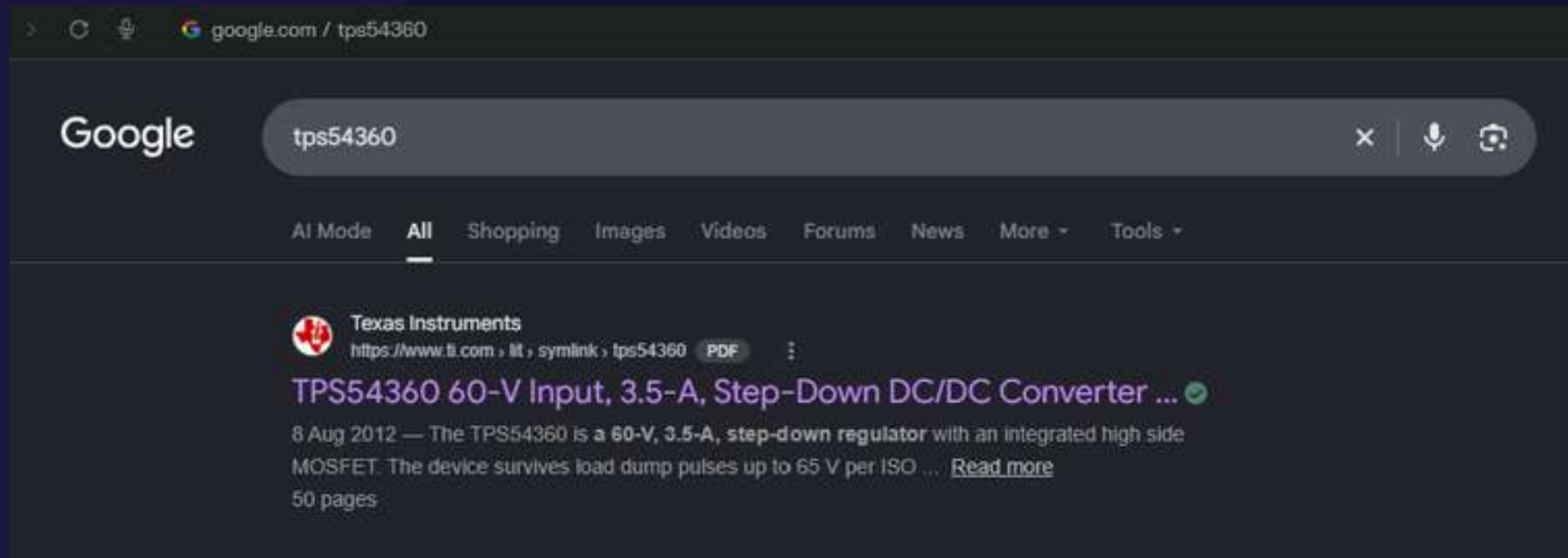
Note: The power flags are just to indicate power used by this IC, optional but surely contribute to a more organised design of schematic

HW⁶ GUIDELINES

HW1 EXAMPLE

2. Find the datasheet of the IC

1. Search for the required IC (just google)





Product Folder

Order Now

Technical Documents

Tools & Software

Support & Community

Reference Design

TPS54360

SLSV54360 – AUGUST 2012 – REVISED JUNE 2018

TPS54360 60-V Input, 3.5-A, Step-Down DC/DC Converter With Eco-Mode™

1 Features

- 4.5-V to 60-V (65-V Abs Max) Input Range
- 3.5-A Continuous Current, 4.5-A Minimum Peak Inductor Current Limit
- Current Mode Control DC/DC Converter
- 92-mΩ High-Side MOSFET
- High Efficiency at Light Loads with Pulse Skipping Eco-mode™
- Low Dropout at Light Loads with Integrated BOOT Recharge FET
- 146-μA Operating Quiescent Current
- 2-μA Shutdown Current
- 100-kHz to 2.5-MHz Fixed Switching Frequency
- Synchronizes to External Clock
- Adjustable UVLO Voltage and Hysteresis
- Internal Soft Start
- Accurate Cycle-by-Cycle Current Limit
- Thermal, Overvoltage, and Frequency Foldback Protection
- 0.8 V 1% Internal Voltage Reference
- 8-Terminal HSOIC with PowerPAD™ Package
- 40°C to 150°C T_J Operating Range
- Create a Custom Design using the TPS54360 with the [WEBENCH® Power Designer](#)

3 Description

The TPS54360 is a 60-V, 3.5-A, step-down regulator with an integrated high side MOSFET. The device survives load dump pulses up to 65 V per ISO 7637. Current mode control provides simple external compensation and flexible component selection. A low ripple pulse skip mode reduces the no load supply current to 146 μA. Shutdown supply current is reduced to 2 μA when the enable pin is pulled low.

Undervoltage lockout is internally set at 4.3 V but can be increased using the enable pin. The output voltage start-up ramp is internally controlled to provide a controlled start-up and eliminate overshoot.

A wide switching frequency range allows either efficiency or external component size to be optimized. Frequency foldback and thermal shutdown protect internal and external components during an overload condition.

The TPS54360 is available in an 8-terminal thermally enhanced HSOIC PowerPAD™ package.

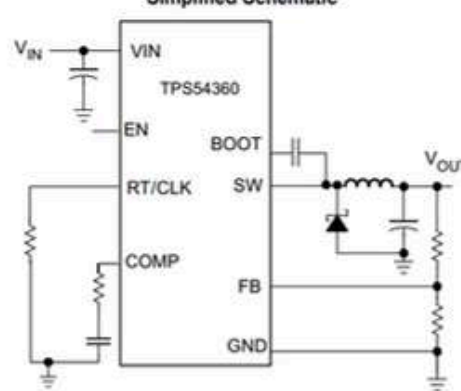
Device Information		
ORDER NUMBER	PACKAGE	BODY SIZE
TPS54360	HSOIC (8)	4.89 mm × 3.90 mm

(1) For all available packages, see the orderable addendum at the end of the data sheet.

2 Applications

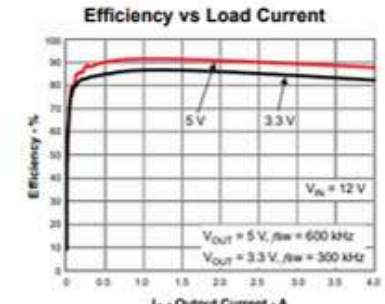
12-V, 24-V and 48-V Industrial, Automotive and Communications Power Systems

Simplified Schematic



The schematic shows the TPS54360 IC with its pins connected to a 60V input (VIN), an enable pin (EN), a bootstrap capacitor (BOOT), a switching MOSFET (SW), a diode, an output capacitor (COUT), and a feedback network (FB, COMP, RT/CLK). The output is VOUT.

Efficiency vs Load Current



The graph shows Efficiency (%) on the y-axis (0 to 100) versus I_O - Output Current - A on the x-axis (0 to 4.0). Two curves are shown for V_{OUT} = 5 V, f_{sw} = 600 kHz and V_{OUT} = 3.3 V, f_{sw} = 300 kHz. The input voltage V_{IN} is 12 V.

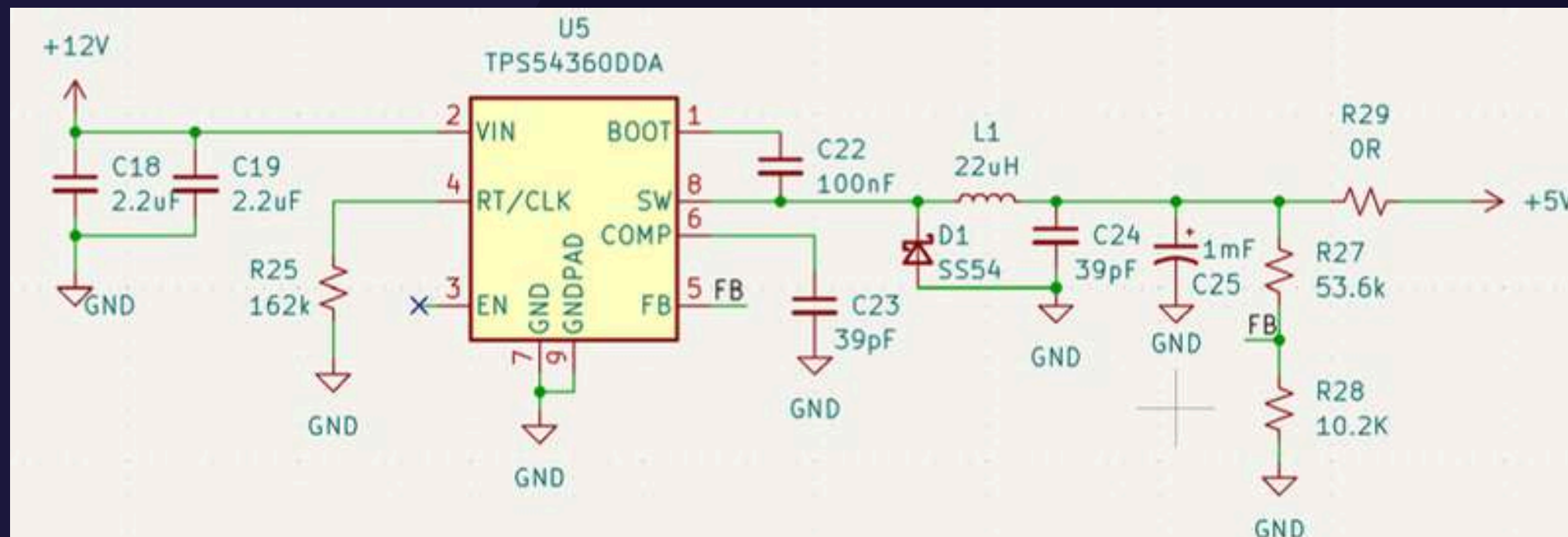
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HW⁶ GUIDELINES

HW1 EXAMPLE

5. Copy and paste into your schematic
6. Adjust values of some peripheral components



8.2.2.9 Output Voltage and Feedback Resistors Selection

The voltage divider of R5 and R6 sets the output voltage. For the example design, 10.2 kΩ was selected for R6. Using Equation 3, R5 is calculated as 53.5 kΩ. The nearest standard 1% resistor is 53.6 kΩ. Due to the input current of the FB terminal, the current flowing through the feedback network should be greater than 1 μA to maintain the output voltage accuracy. This requirement is satisfied if the value of R6 is less than 800 kΩ. Choosing higher resistor values decreases quiescent current and improves efficiency at low output currents but may also introduce noise immunity problems.

$$R_{HS} = R_{LS} \times \frac{V_{OUT} - 0.8 \text{ V}}{0.8 \text{ V}} = 10.2 \text{ k}\Omega \times \left(\frac{5 \text{ V} - 0.8 \text{ V}}{0.8 \text{ V}} \right) = 53.5 \text{ k}\Omega$$

(42)



FEEDBACK RESISTORS

8.2.2.9 Output Voltage and Feedback Resistors Selection

The voltage divider of R5 and R6 sets the output voltage. For the example design, 10.2 kΩ was selected for R6. Using Equation 3, R5 is calculated as 53.5 kΩ. The nearest standard 1% resistor is 53.6 kΩ. Due to the input current of the FB terminal, the current flowing through the feedback network should be greater than 1 μA to maintain the output voltage accuracy. This requirement is satisfied if the value of R6 is less than 800 kΩ. Choosing higher resistor values decreases quiescent current and improves efficiency at low output currents but may also introduce noise immunity problems.

$$R_{HS} = R_{LS} \times \frac{V_{OUT} - 0.8 V}{0.8 V} = 10.2 \text{ k}\Omega \times \left(\frac{5 V - 0.8 V}{0.8 V} \right) = 53.5 \text{ k}\Omega \quad (42)$$

So $R_{HS} / R_{LS} = (V_{out} - 0.8V) / 0.8V$,
intrinsically it's just a ratio of resistance,
why large resistance, 10.2k ohm is selected for R6?

FEEDBACK RESISTORS

Signal Tracks

vs

Power Tracks

Could be same voltage

Definitely different current values

“mA”

“A”

Short ans: the FB pin is a signal pin, we don't want huge current to go through it



****SUPPLEMENTARY****

HW⁶ GUIDELINES

NO WORRIES, LET'S DO IT TGR



SHORT CUTS FOR KICAD

Mouse Right Click

Mouse Left Click

**Mouse Left Hold - from
left to right(green)**

**Mouse Left Hold - from
right to left (blue)**

Mouse Scroll up & down

Toggle around the schematics

Select

**Only selects objects fully inside the
rectangle.**

**Selects objects that are fully or partially
inside the rectangle**

Zoom in & out

SHORT CUTS FOR KICAD

A

Place symbol mode

R

Rotate

W

Wiring mode

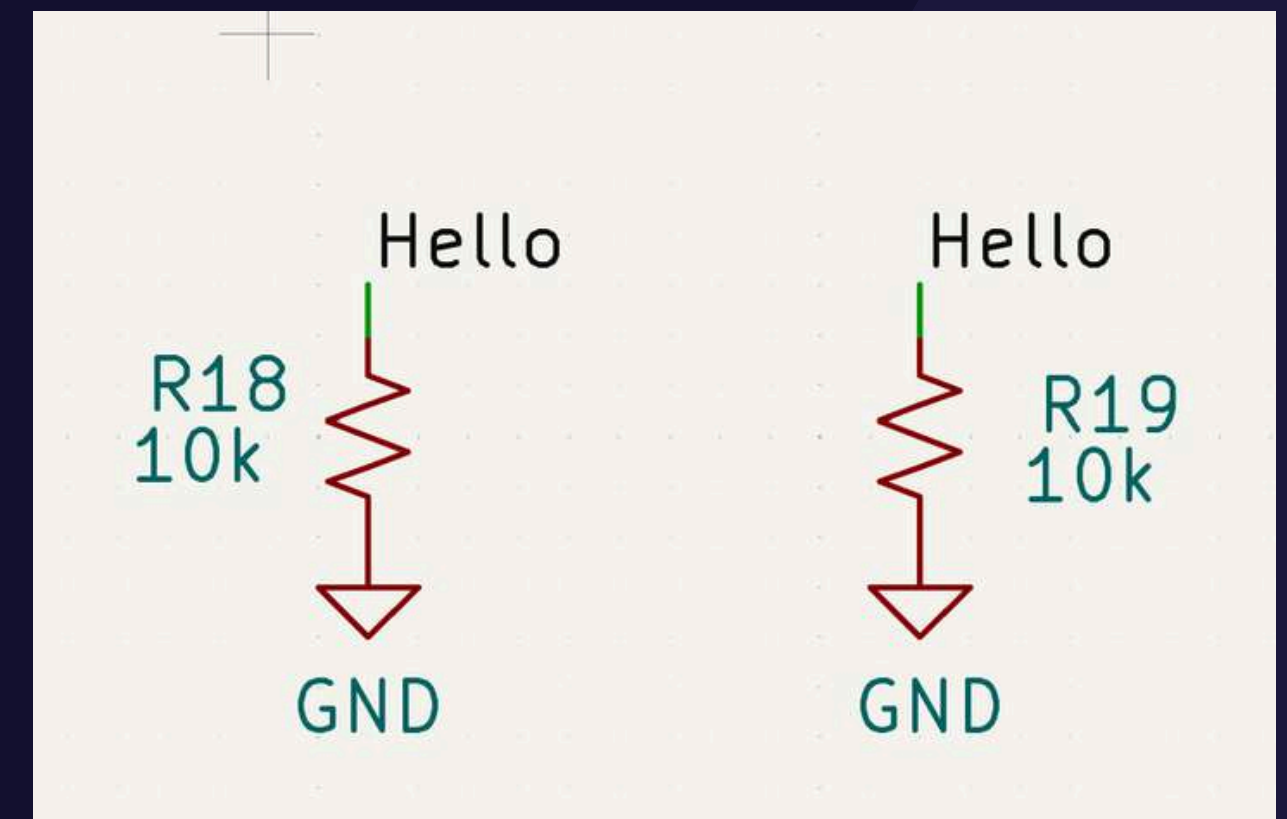
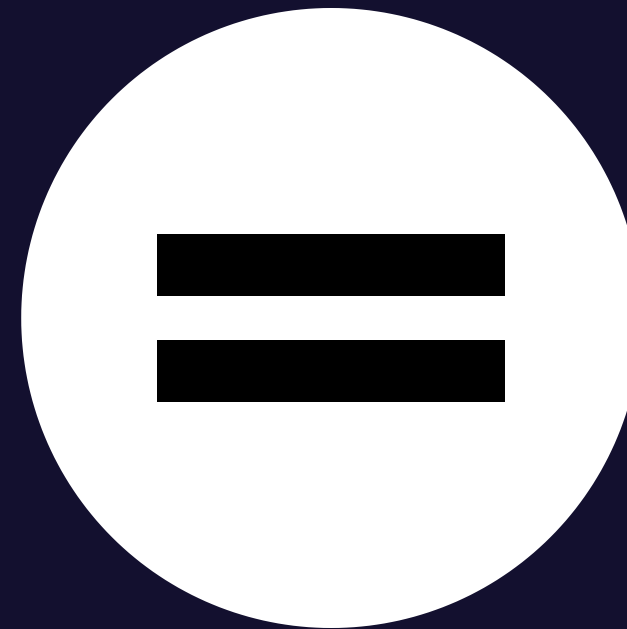
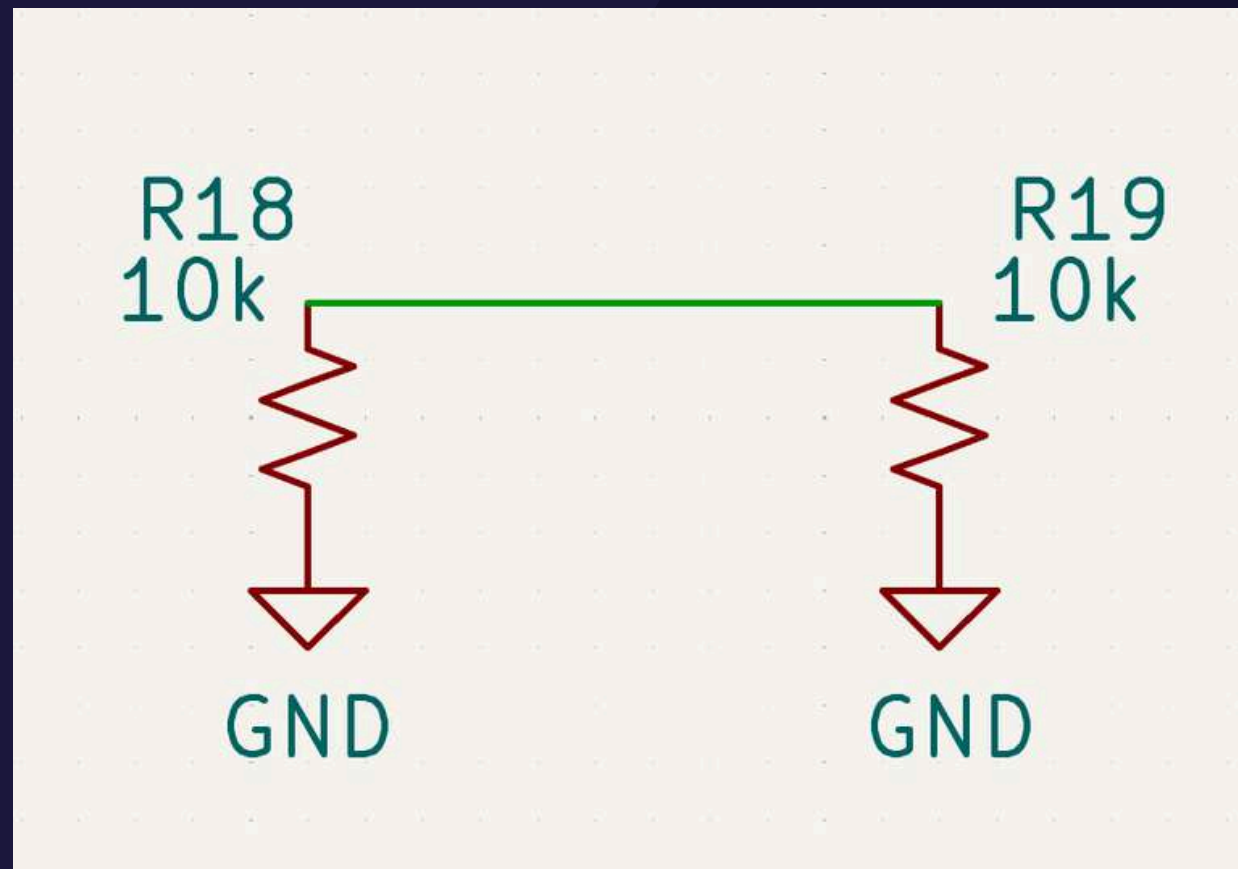
L

Placing net label mode

ESC

Quit existing mode

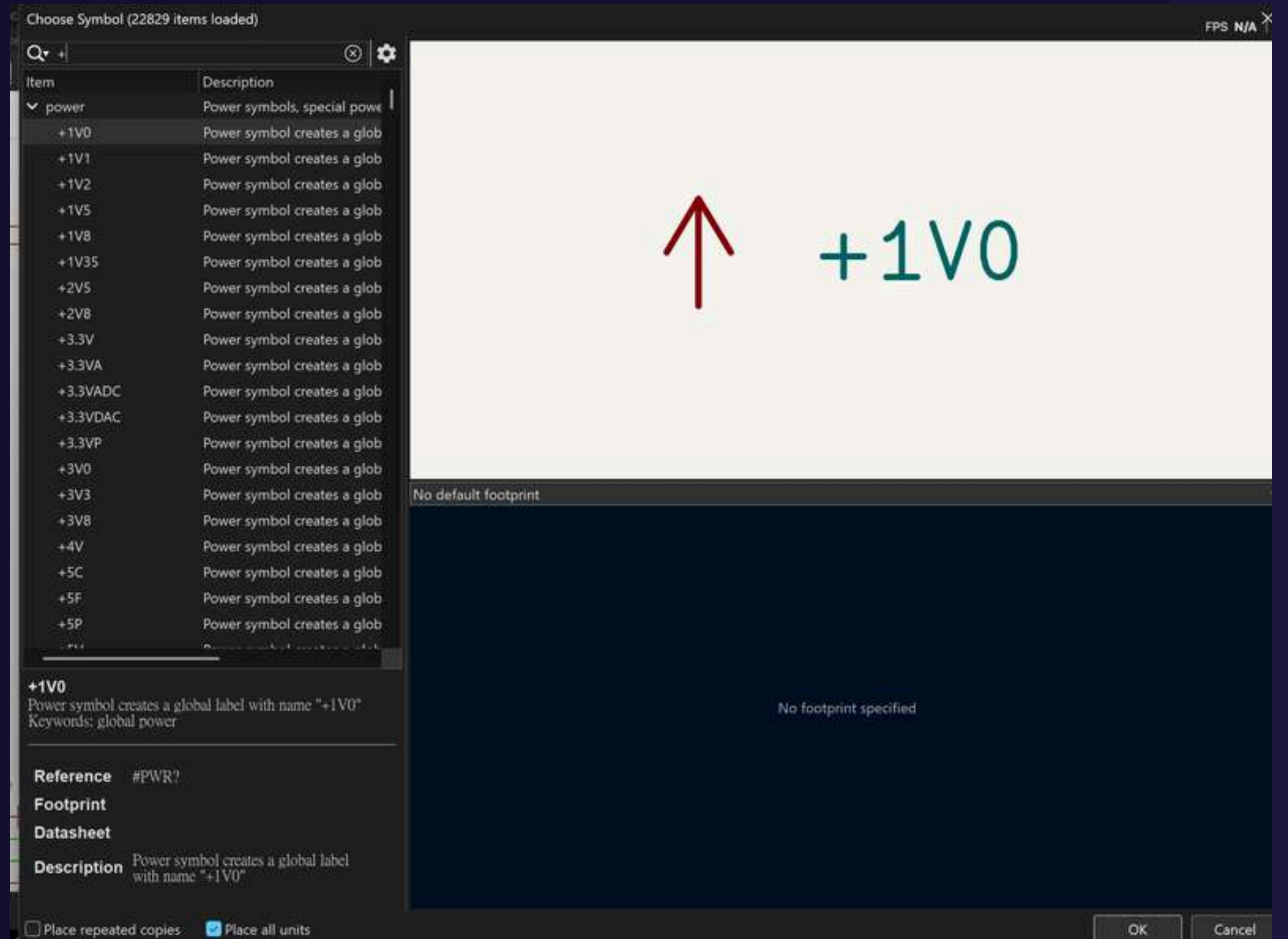
MORE ABOUT KICAD - NET LABELS



HW 6 GUIDELINES

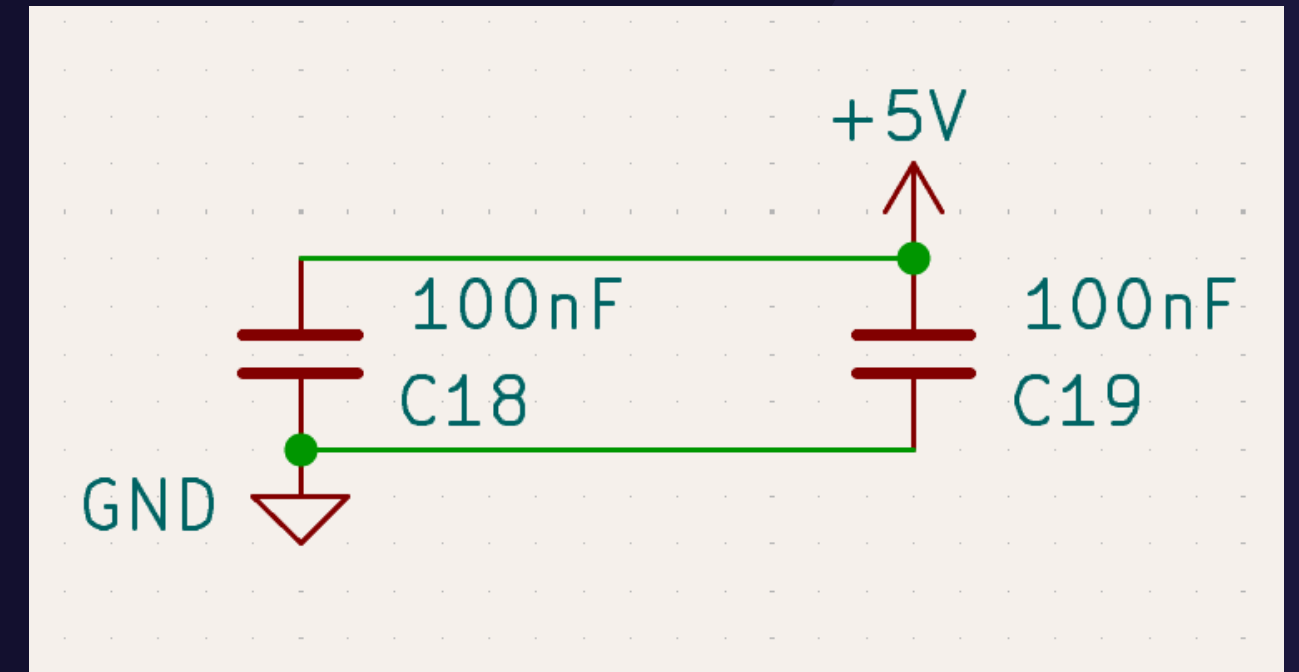
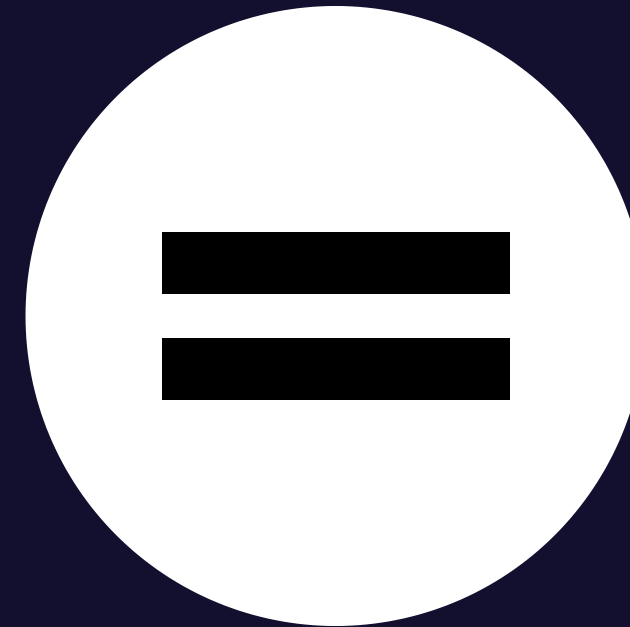
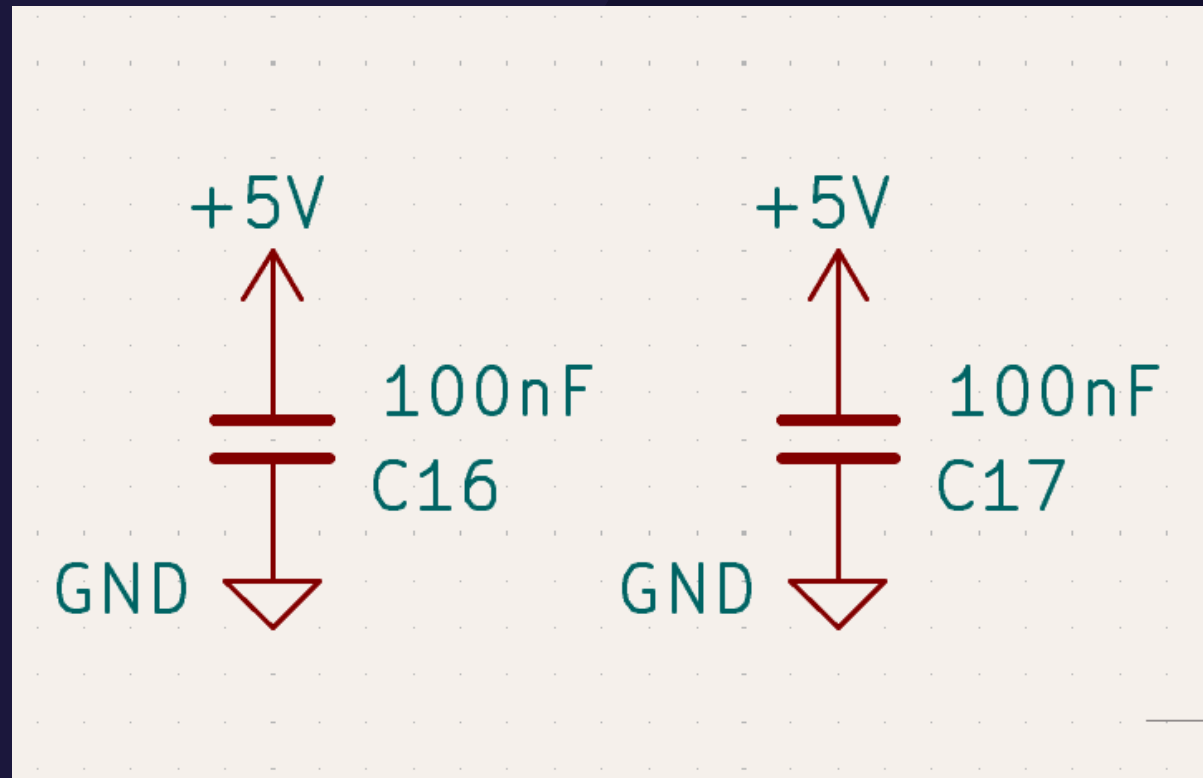
MORE ABOUT KICAD - POWER NET LABELS

In symbol
(Press A & type
“+” or “GND”)

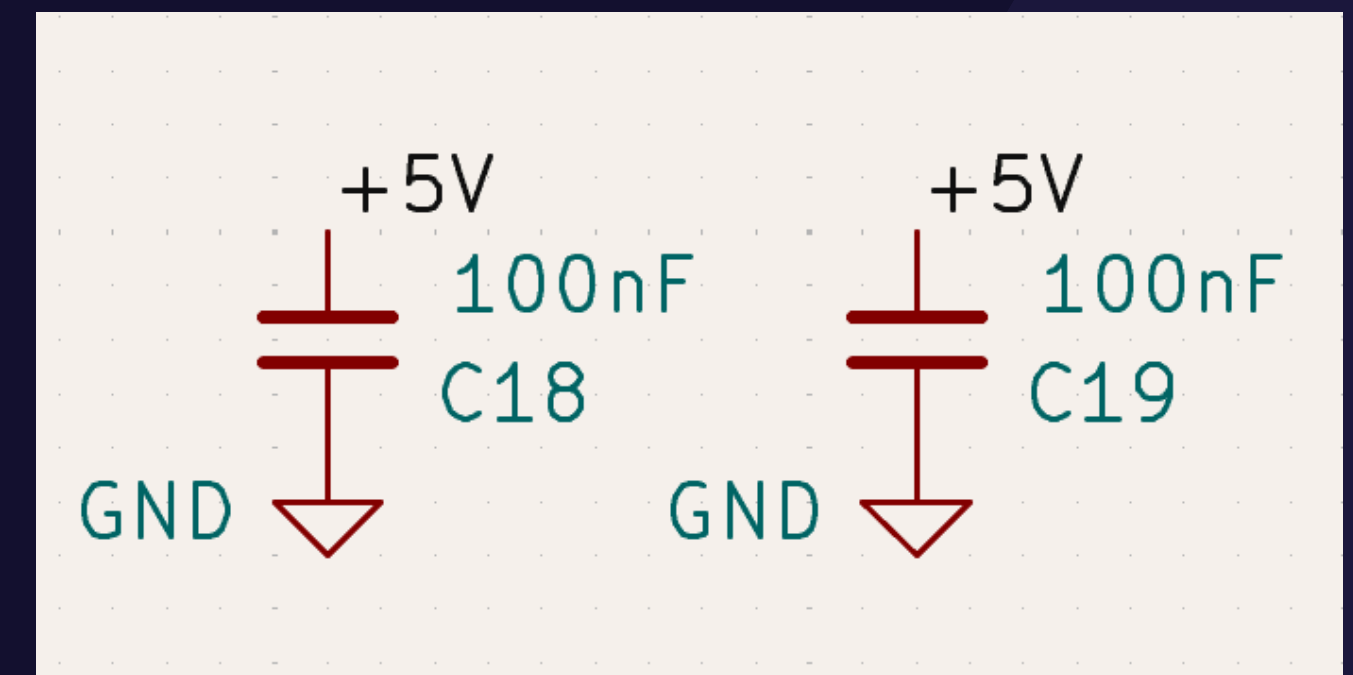
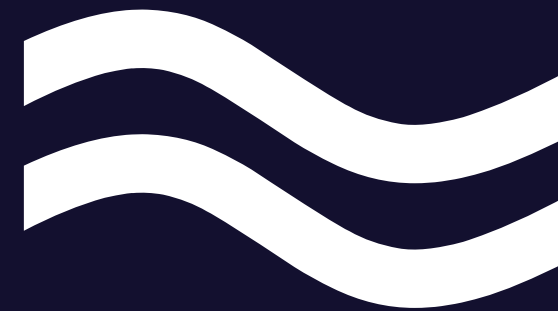
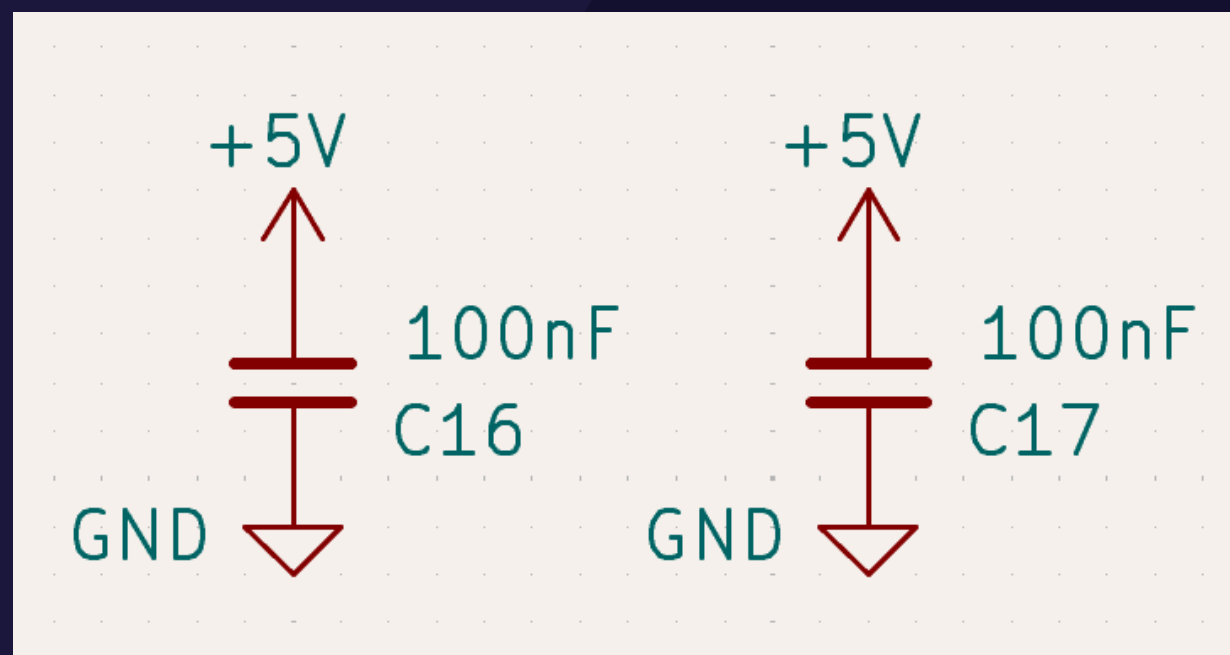


HW ⁶ GUIDELINES

MORE ABOUT KICAD - POWER NET LABELS

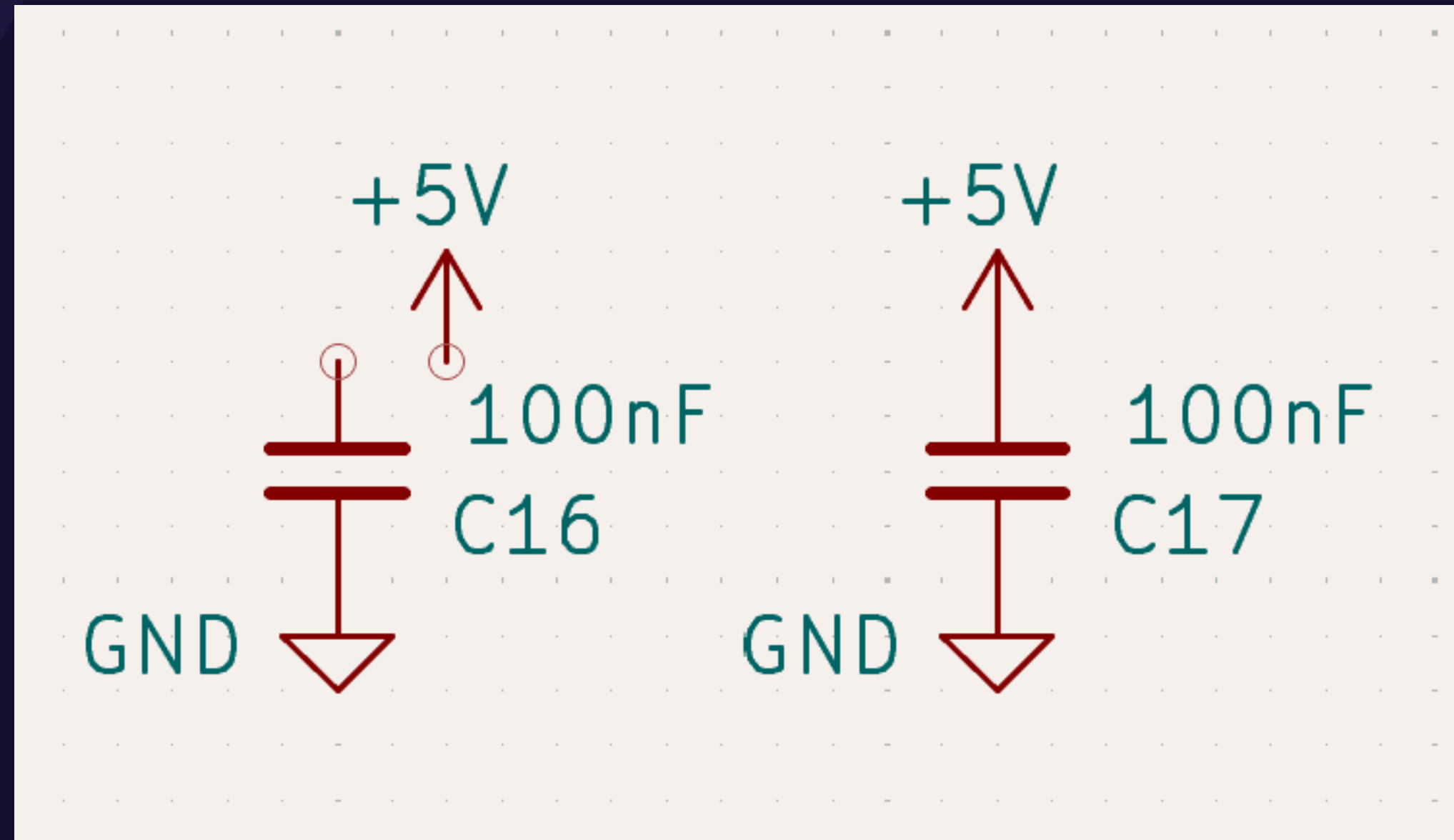


MORE ABOUT KICAD - NET LABELS



Electrically the same
but best practice is to vividly show power V.S. signal connections

MORE ABOUT KICAD - POWER NET LABELS



circle: no connection

MORE ABOUT KICAD - POWER NET LABELS



If you want to indicate sth on schem without electrical implications

××××

MORE ABOUT KICAD - POWER NET LABELS



If you want to use border lines for more readable schematics:



HW⁶ GUIDELINES

LET'S TRY TGR WITH TPS54360



HW⁶
GUIDELINES

2026-Hardware-Tutorial / Homework / Datasheets / 

Add file ▾ 

 YuanL-013 Update HW-1 

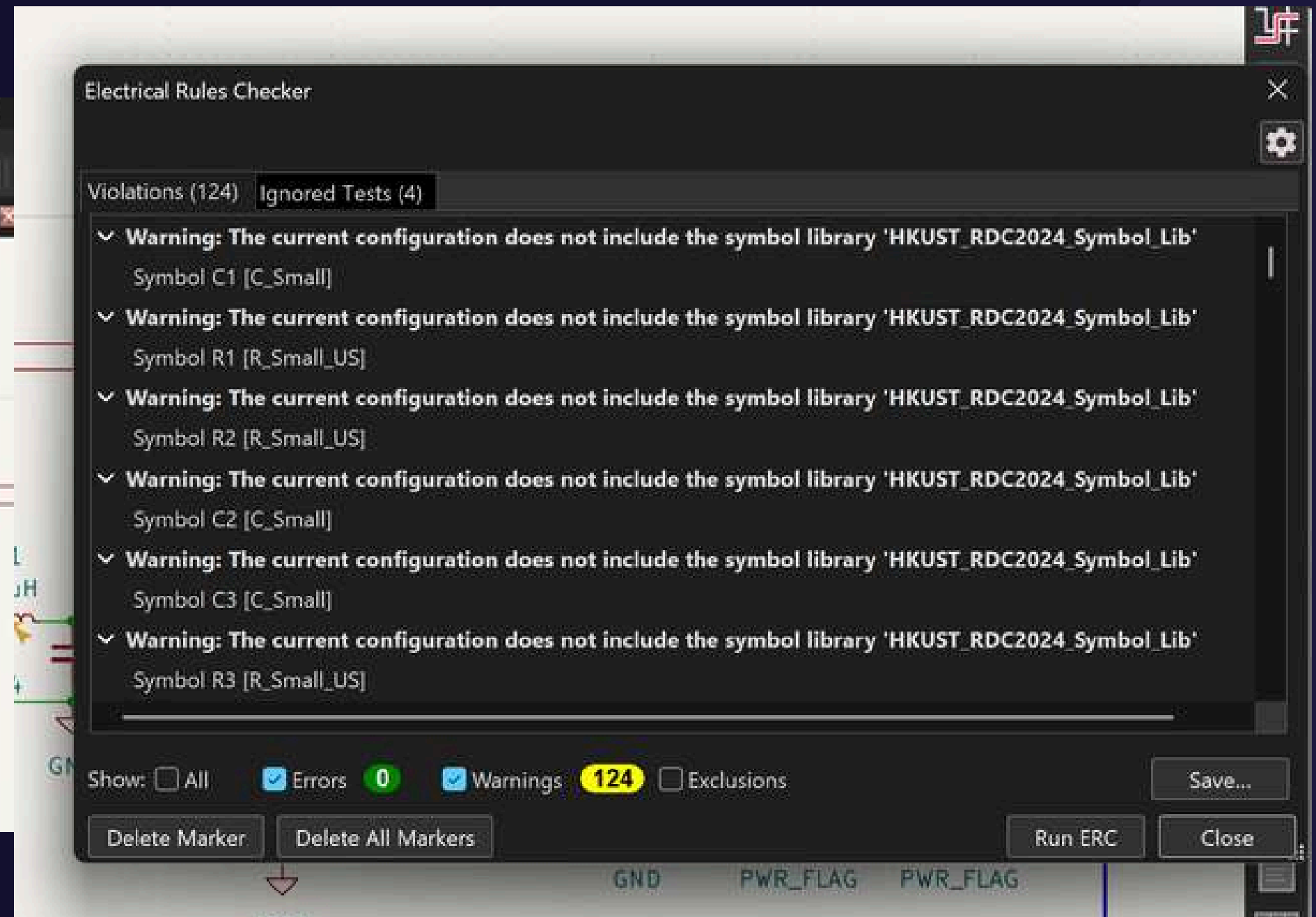
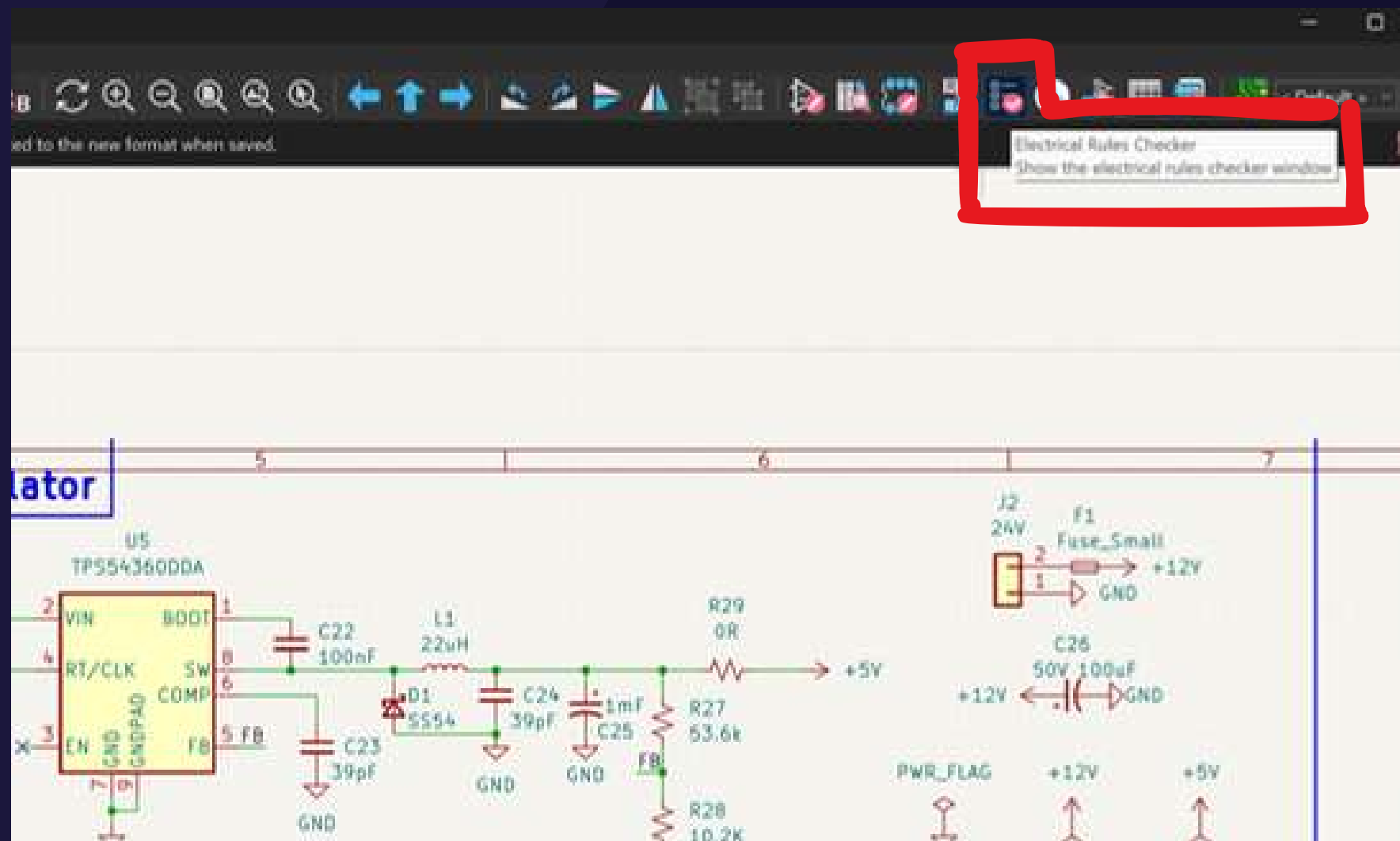
f13f646 · 1 hour ago  History

Name	Last commit message	Last commit date
 ..		
 AP63300-AP63301.pdf	Update HW-1	1 hour ago
 LM1117.PDF	Update HW-1	1 hour ago
 STM32f405RG.pdf	Update HW-1	1 hour ago
 TJA1050.pdf	Update HW-1	1 hour ago
 TP554360.pdf	Update HW-1	1 hour ago
 TXS0102.pdf	Update HW-1	1 hour ago



HW ⁶ GUIDELINES

Before Submission, PLEASE UNDERGO ERC



Please make sure there's no error before you submit it! If you are unsure, please consult seniors



XXXX

ITEM 7

EXTERNAL RESOURCES

XXXX

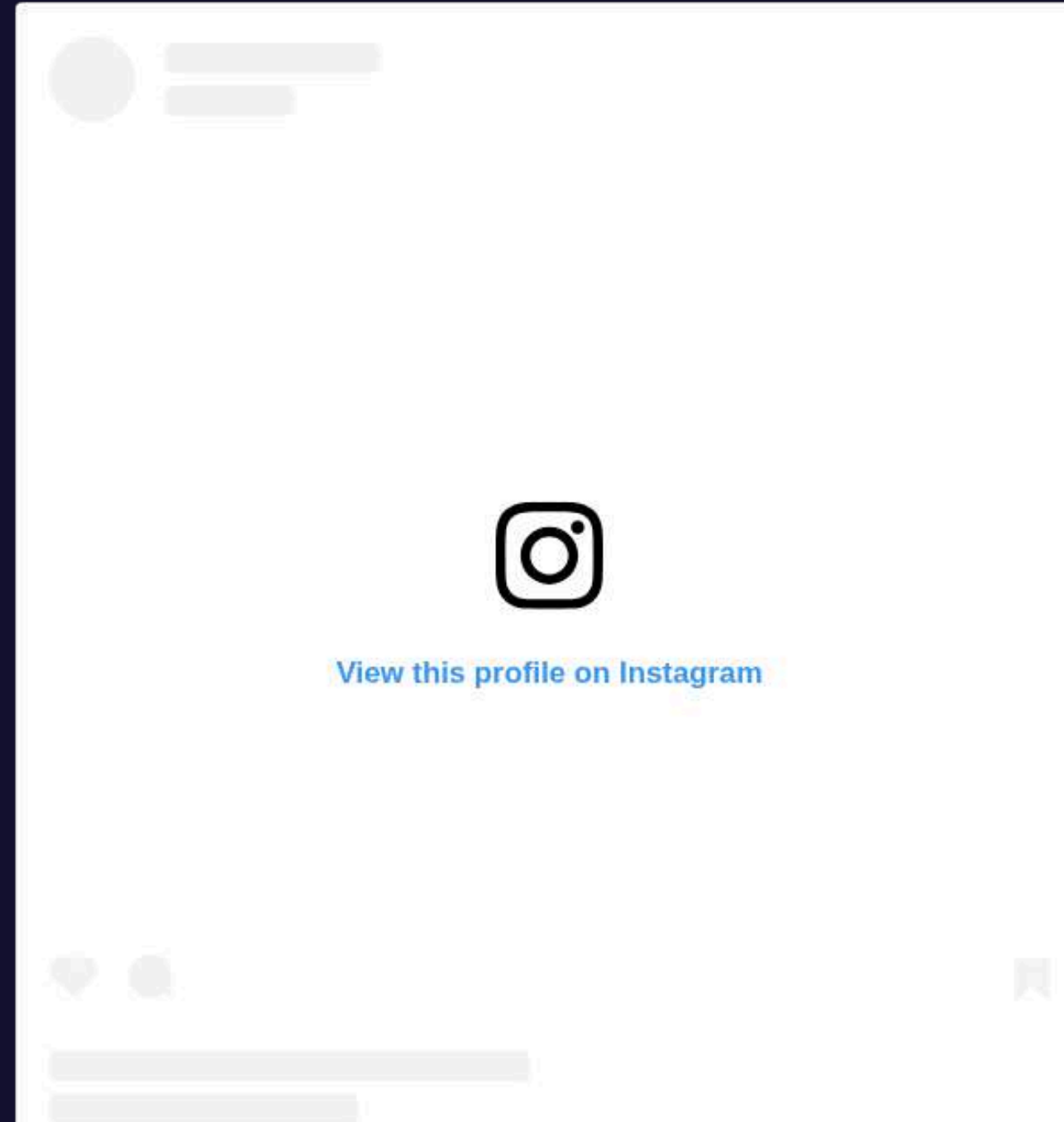


RESOURCES

- **Component search engine:**
 - <https://hken.rs-online.com/web/>
 - <https://www.mouser.hk/>
 - <https://www.szlcsc.com/>
- **Cheap(Free) Labour**
 - **Your seniors**
 - **Feel free to ask questions anytime!**

EXTERNAL⁷
RESROUCES

RESOURCES



XXXX

ITEM 8

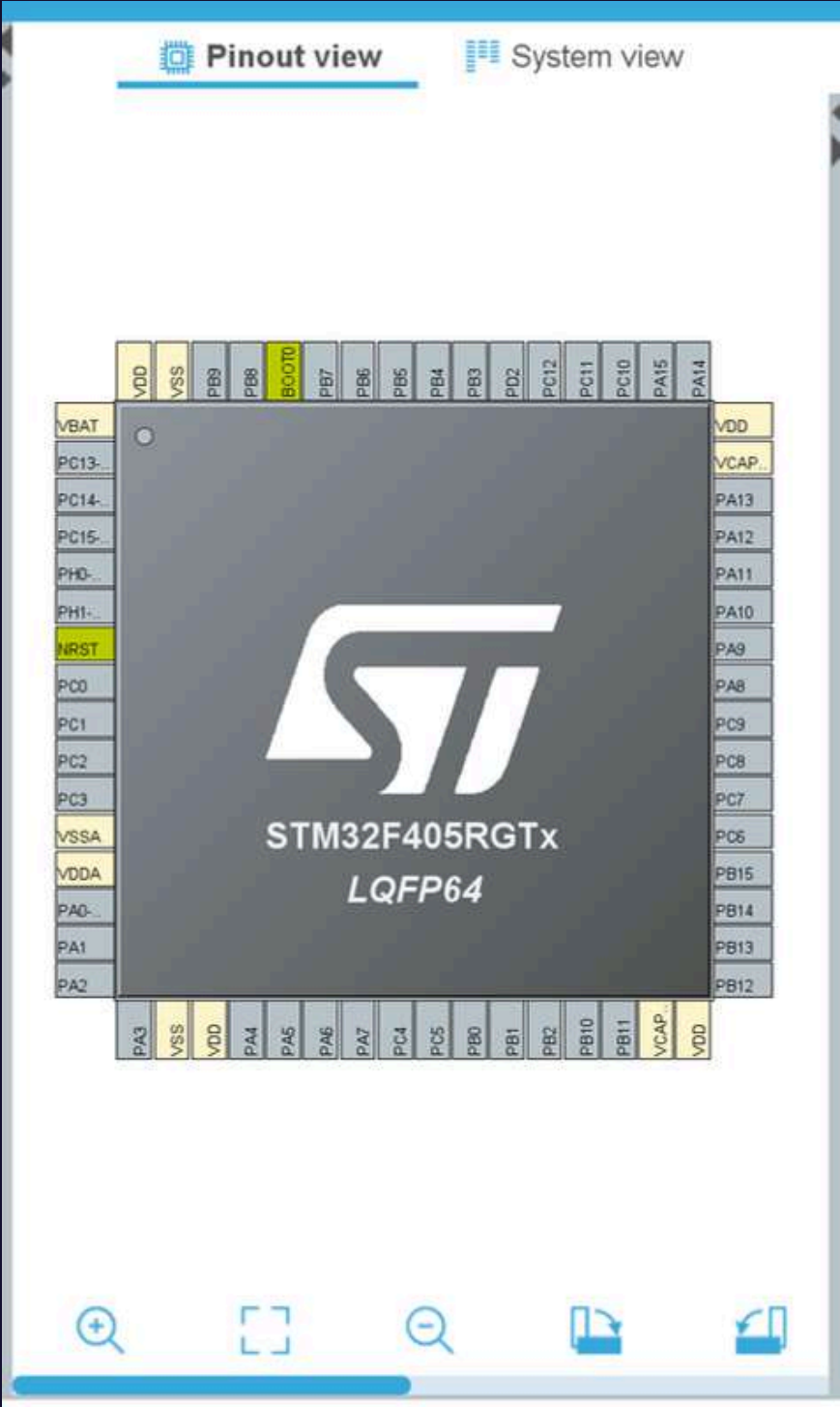
MCU CONTROL SIGNALS

XXXX



CONTROL ⁸
COMM SIGNALS

CUBE MX



CONTROL SIGNALS - GPIO

General Purpose Input/Output (GPIO)

- a flexible pin on a microchip or development board
 - act as an input
 - an output
 - or both under software control
- Simple logic 1 (3.3V for STM32) and 0 (GND)

Application: Simple ON & OFF



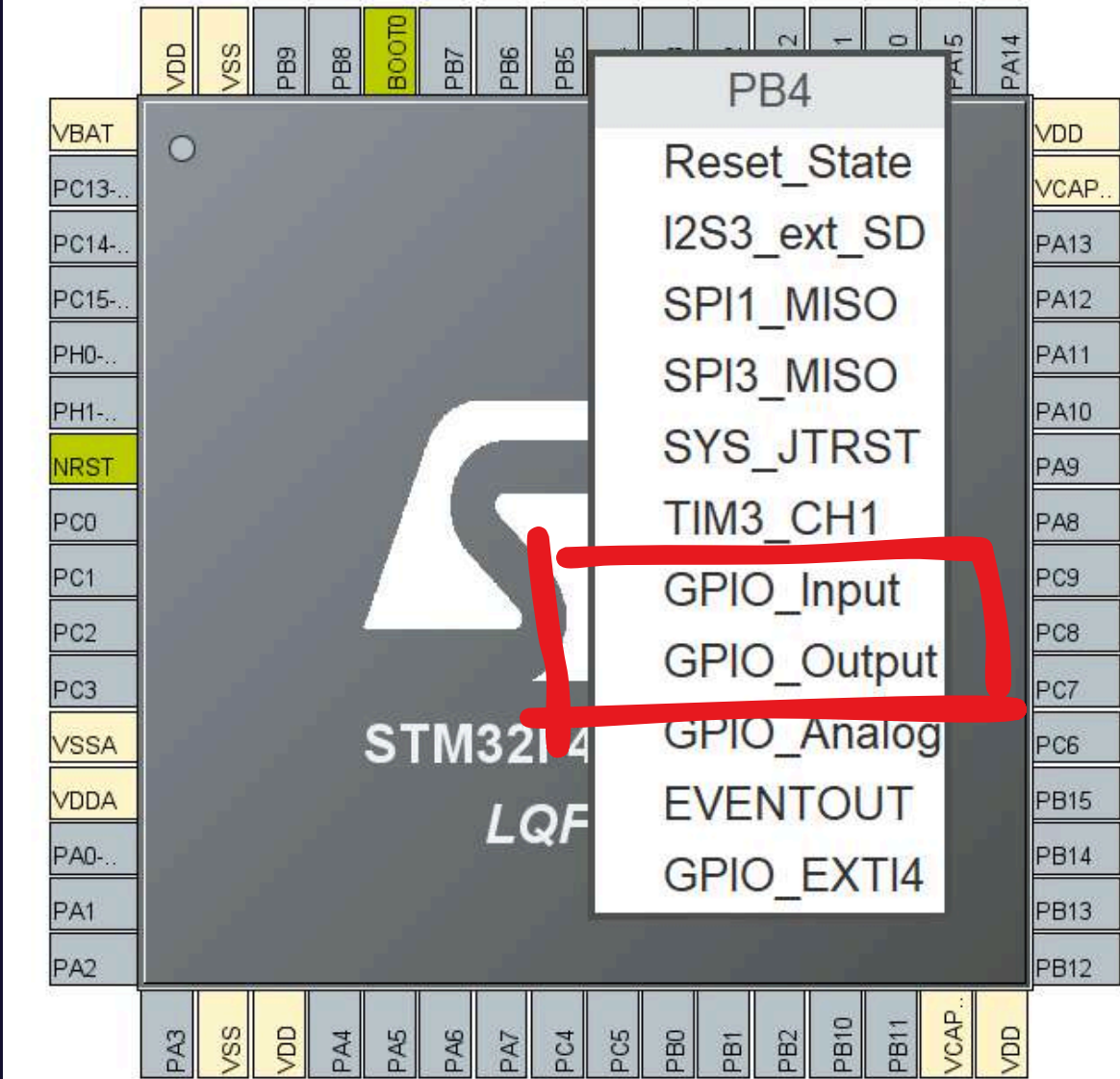
Brief AO3400 explanation:

If Gate (G) is applied a control signal of logic HIGH, Drain(D) and Source(S) would be connected

CONTROL SIGNALS - GPIO

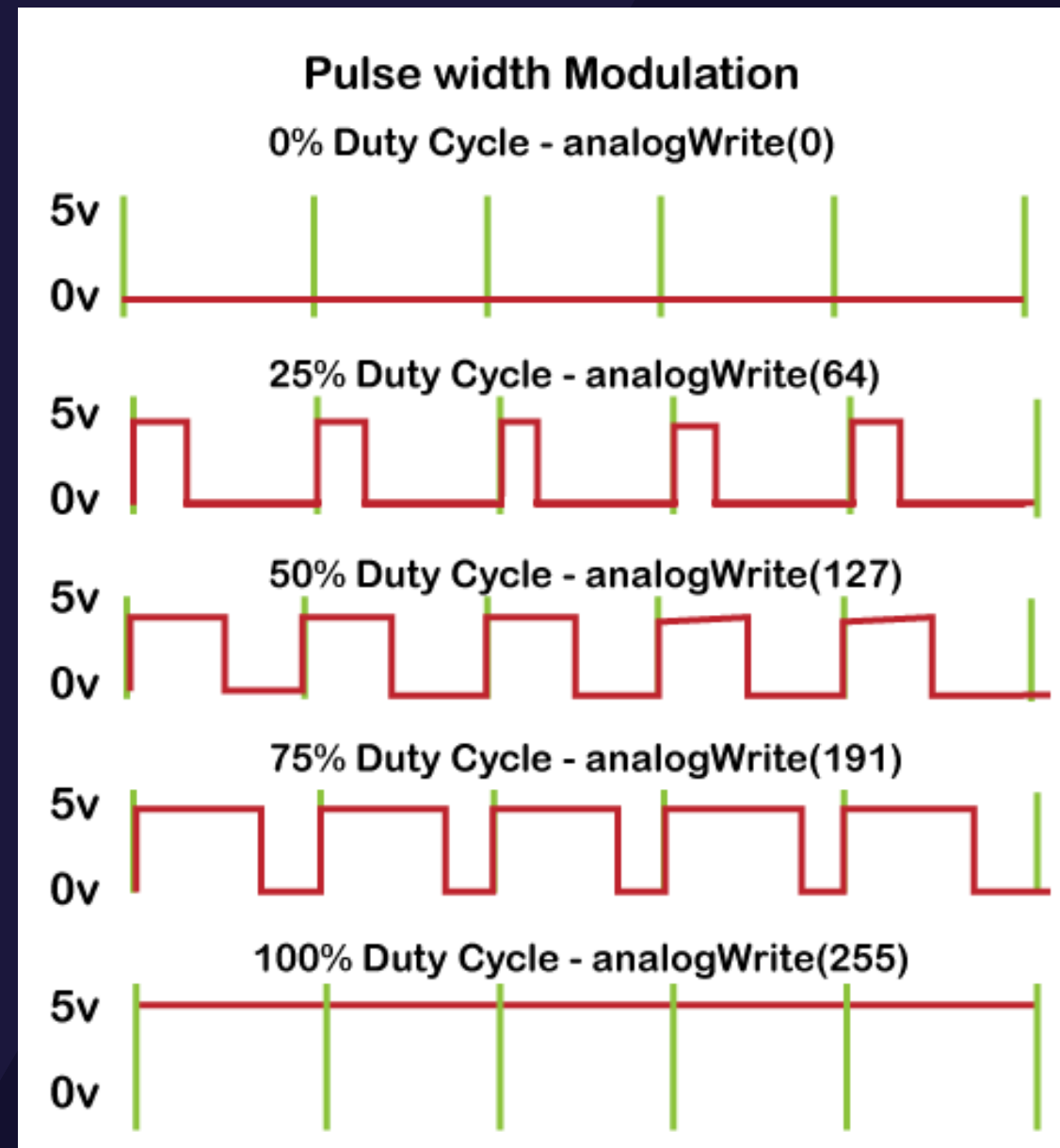
Table 7. STM32F40xxx pin and ball definitions⁽¹⁾

Pin number						Pin name (function after reset) ⁽²⁾	Pin type	I/O structure	Notes	Alternate functions	Additional functions
LQFP64	WLCSP90	LQFP100	LQFP144	UFBGA176	LQFP176						
-	-	1	1	A2	1	PE2	I/O	FT	-	TRACECLK/ FSMC_A23 / ETH_MII_TXD3 / EVENTOUT	-



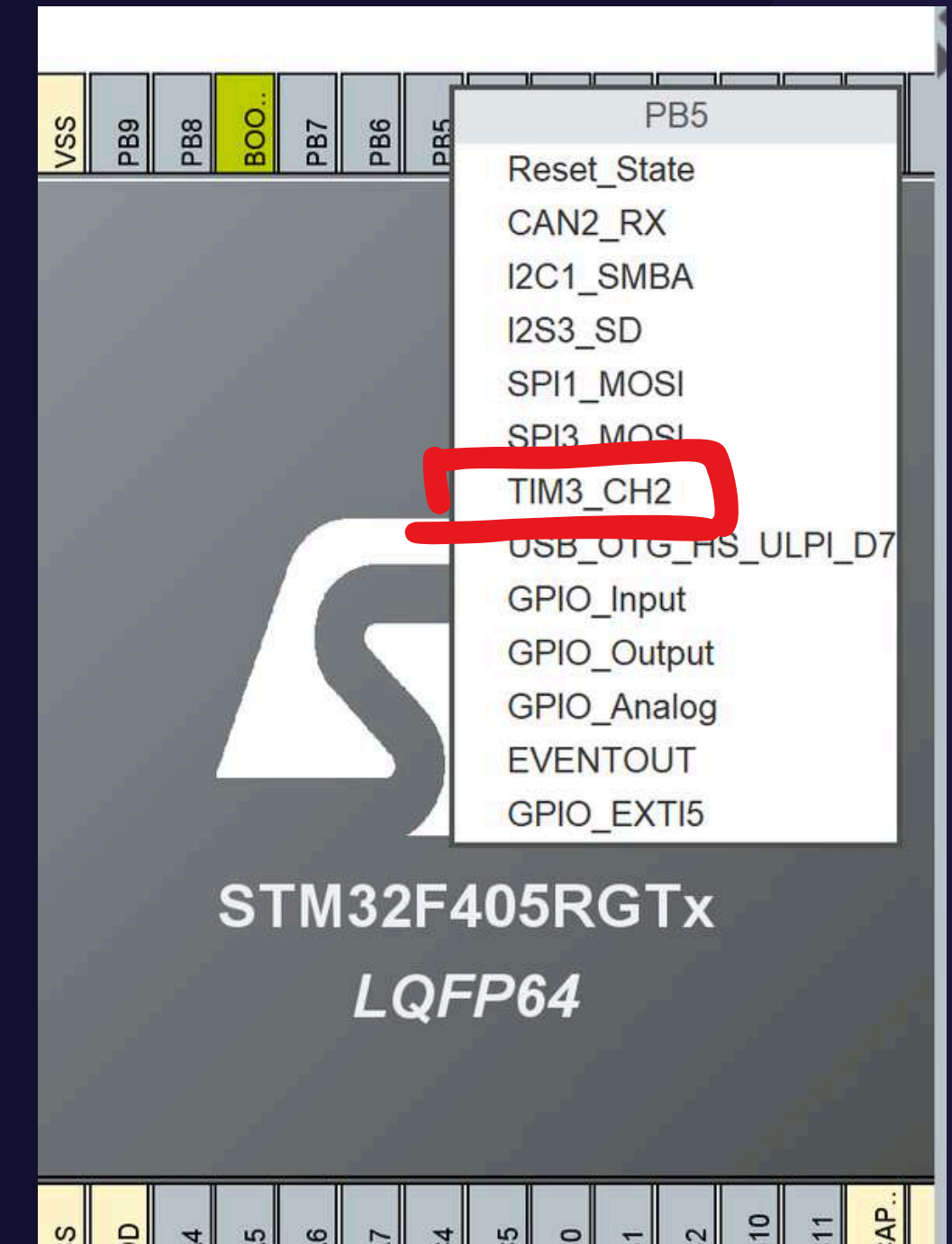
CONTROL ⁸ COMM SIGNALS

CONTROL SIGNALS - PWM



Note: Logic of STM32 is 3.3V as 100% rather than 5V

Application:
Controlling the strength of the force applied by the motor (at a % of its maximum force)

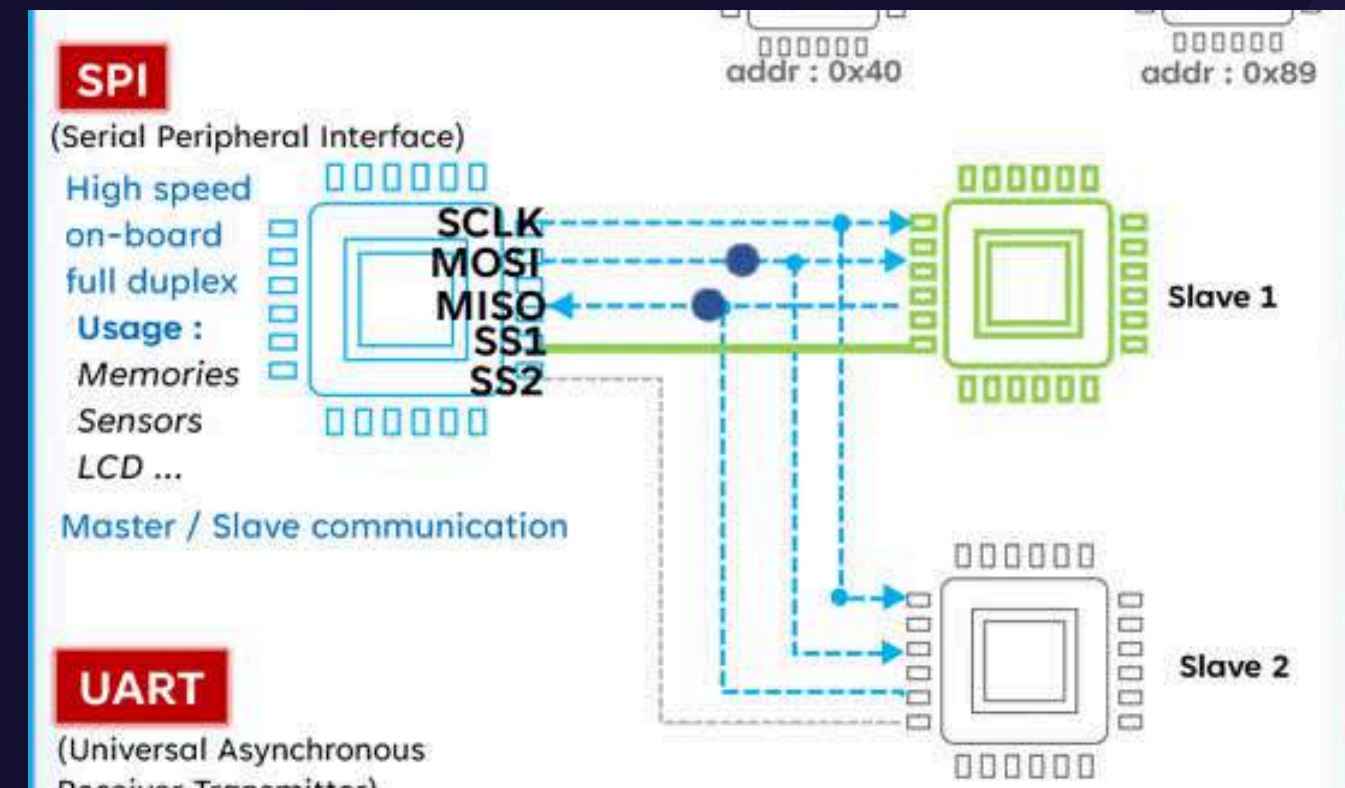
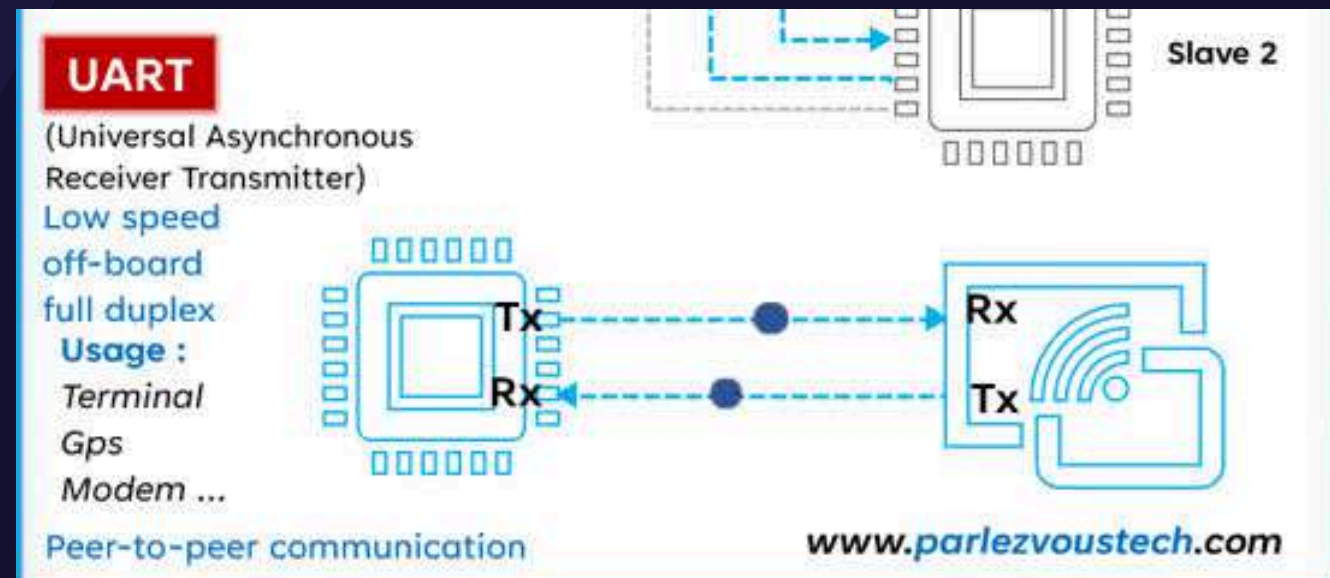
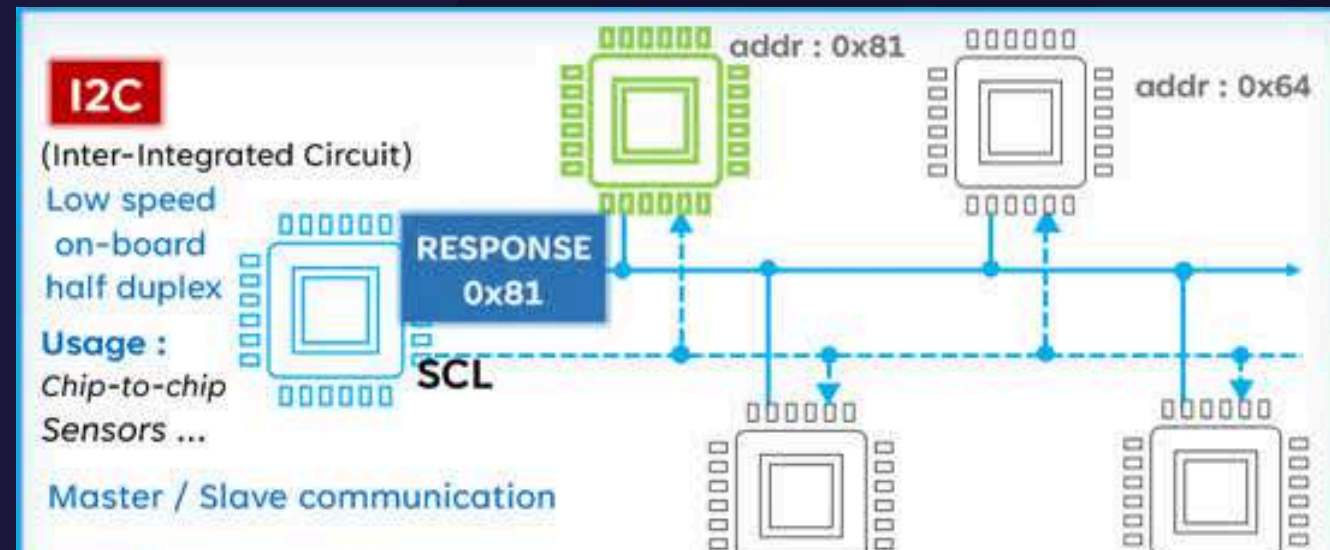


MCU Pin: TIMx_CHx



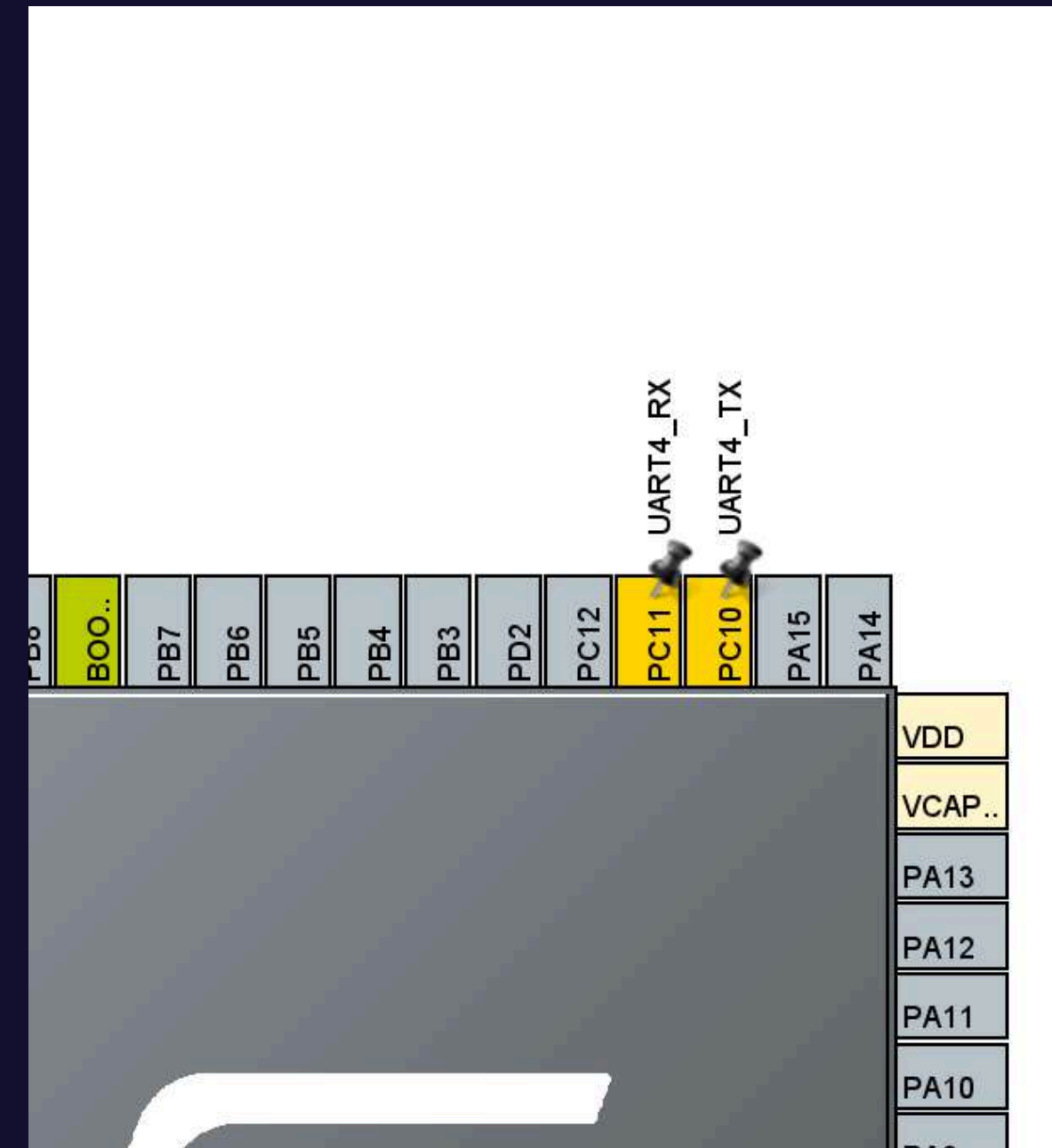
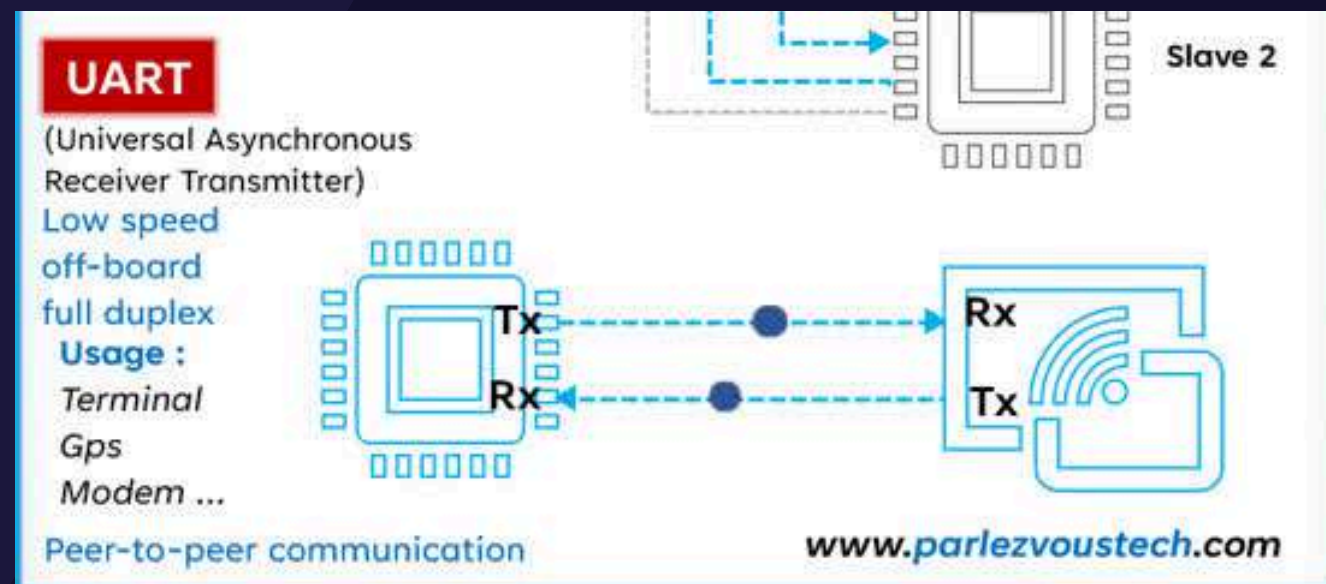
CONTROL 8 COMM SIGNALS

COMMUNICATION PROTOCOLS



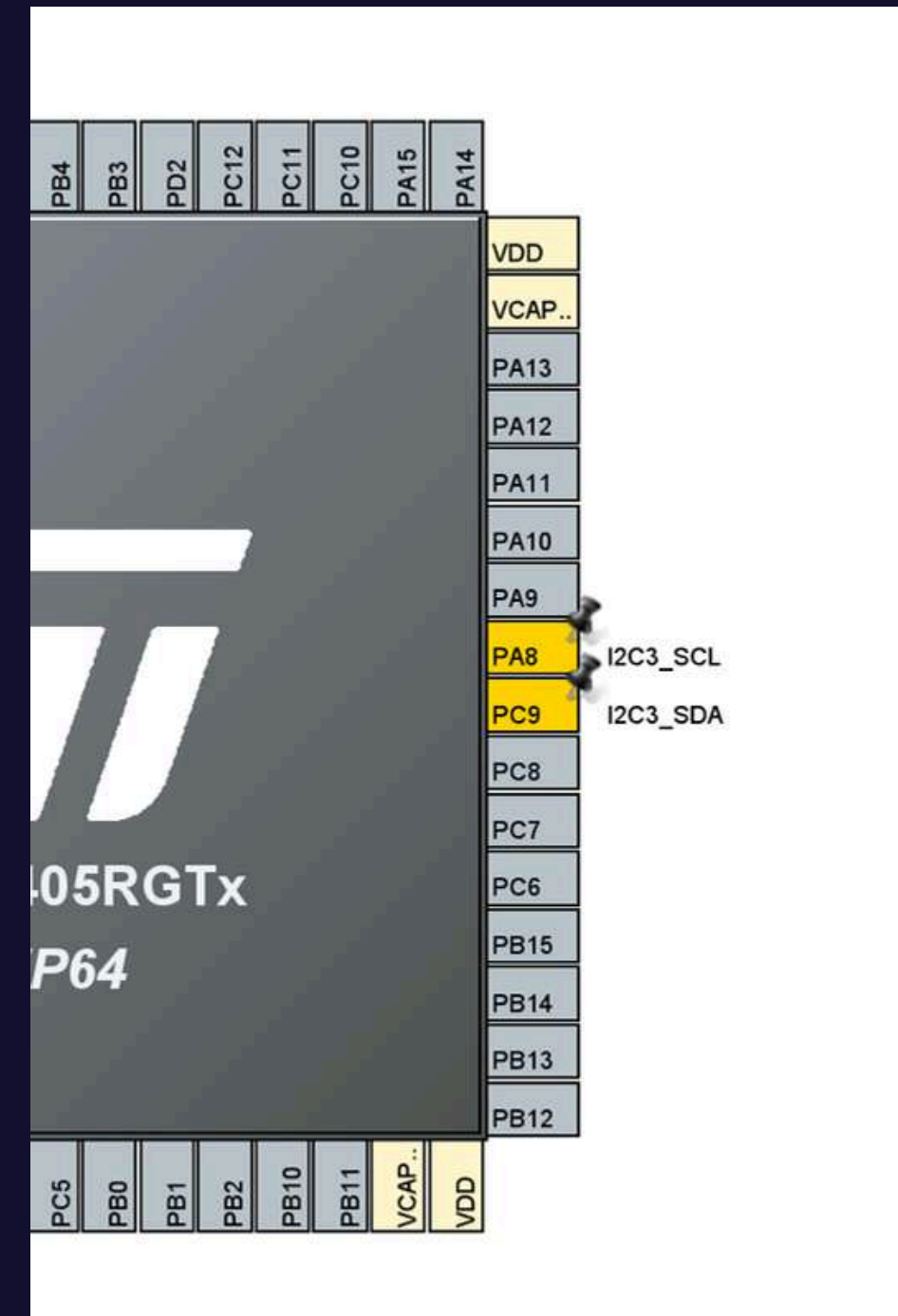
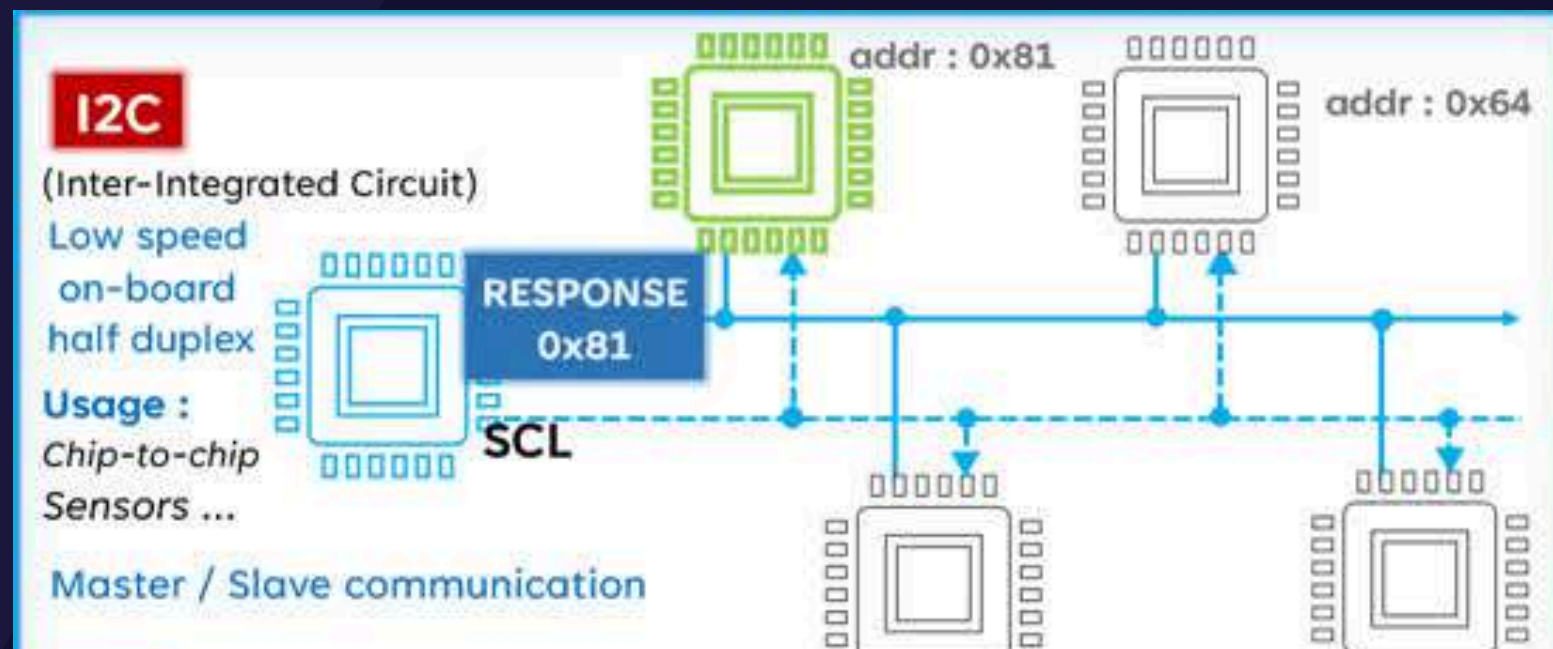
CONTROL 8 COMM SIGNALS

COMMUNICATION PROTOCOLS



CONTROL 8 COMM SIGNALS

COMMUNICATION PROTOCOLS - I2C



CONTROL 8 COMM SIGNALS

COMMUNICATION PROTOCOLS				
AT A GLANCE COMPARISON				
PROTOCOL	WIRES	SPEED	MAX DEVICES	BEST USED FOR
I²C Inter-Integrated Circuit	2 SDA, SCL	 SLOW / MED Up to 3.4 Mbps	127 (7-bit Addressing)	Local sensors & OLED displays
SPI Serial Peripheral Interface	4+ MOSI, MISO, SCK, CS (+)	 VERY FAST Up to 100+ Mbps	Limited by pins (No fixed limit)	SD Cards & TFT screens
UART Universal Asynchronous Receiver / Transmitter	2 TX, RX	 MEDIUM Up to 1 Mbps	Only 1 (Point-to-Point)	PC Debugging, GPS, Bluetooth
CAN Controller Area Network	2 CAN_H, CAN_L	 MEDIUM Up to 1 Mbps	~110 (Depending on bus load & length)	Automotive & high-noise environments
MODBUS Industrial Communication Protocol	2 A(+), B(-) (RS485)	 SLOW Up to 115.2 kbps	247 (On RS485 Bus)	Industrial PLCs, Solar, HVAC
1-WIRE Dallas / Maxim Protocol	1 DATA (DQ) + GND	 SLOW Up to 16.3 kbps	UNLIMITED (Practical limit depends on cable length)	Temperature sensors, IoT, Asset tracking
CHOOSE WISELY Match the protocol to your speed, distance & device requirements.	DESIGN BETTER Optimized communication leads to stable & efficient systems.	BUILT FOR INDUSTRY From simple sensors to complex systems, there's a protocol for every need.	RELIABLE CONNECTIONS The right protocol ensures performance, scalability & long-term reliability.	
SAVE THIS CHEAT SHEET				





HOORAY!

THANK YOU

END GAME

